Renewable and Sustainable Energy Reviews 244 (2027) 117497

Contents lists available at ScienceDirect

Renewable and Sustainable Energy Reviews

journal homepage: www.elsevier.com/locate/rser

Synergistic photovoltaic-thermoelectric-phase change material (PV-TEG-PCM) systems for solar energy harvesting: A comprehensive review of thermal management, post-sunset power generation, and multi-energy synergy

Ning Yang [a], Zishuo Zhao [a], Ze Wang [a], Shipeng Yang [a], Bowen Wang [c], Zhiming Bao [c], Yuhang Zhang [a], Yaoxuan Shang [a], Binjian Nie [d], Hongwei Li [a,*], Lei Xing [b,**]

[a] College of Energy Resource and Power Engineering, Northeast Electric Power University, Jilin, 132012, China
[b] School of Chemistry and Chemical Engineering, University of Surrey, Guildford, GU2 7XH, United Kingdom
[c] National Industry-Education Platform for Energy Storage, Tianjin University, Tianjin, 300350, China
[d] Department of Engineering Sciences, University of Oxford, Oxford, OX1 3PJ, United Kingdom

ARTICLE INFO

Keywords:
Photovoltaics
Thermoelectric generators
Phase change materials
PV-TEG-PCM system
Post-sunset power generation

ABSTRACT

Photovoltaic (PV) technology is a cornerstone of the global energy transition; however, its large-scale deployment is constrained by efficiency losses at elevated temperatures and the intermittent nature of solar irradiation. This review provides a comprehensive and systematic assessment of advanced hybrid systems, with a focus on the photovoltaic-thermoelectric generator-phase change material (PV-TEG-PCM) architecture, aiming to synthesise the currently fragmented knowledge across the three subsystems into a unified system-level framework. The PV-TEG-PCM system enables coupled thermal-electrical energy management, improving energy utilization and enhancing the temporal stability of power output. Phase change materials act as both thermal buffers and energy storage media, reducing PV operating temperature during daytime and releasing stored heat to extend thermoelectric generation beyond daylight hours under favorable thermal conditions. This review evaluates the fundamental principles of PV, TEG, and PCM technologies, and analyses recent advances in materials (e.g., nano-enhanced PCMs and functionally graded thermoelectrics), system configurations, and modelling approaches for performance optimization. Key performance indicators, including efficiency, power density, and reliability, are compared across reported systems. Overall, PV-TEG-PCM systems offer a promising pathway to improve the efficiency, reliability, and temporal flexibility of solar energy. Future research priorities include advanced materials development, intelligent control strategies, and integrated techno-economic and lifecycle assessments.

Nomenclature and Abbreviations

V_{oc}	open-circuit voltage, V	I_{sc}	short-circuit current, A
I	thermoelectric-generator output current, A	P_{max}	maximum output power, W
P_{PV}	photovoltaic output power, W	P_{TEG}	thermoelectric-generator output power, W
P_{aux}	auxiliary power consumption, W	T_j	junction temperature, K
T_h	TEG hot-side temperature, K	T_c	TEG cold-side temperature, K
T	average absolute temperature of the TEG, K	T_{PV}	photovoltaic-module temperature, K
T_{amb}	ambient temperature, K	T_{ref}	reference-system temperature, K
T_{test}	test-system temperature, K	ΔT	hot-to-cold-side temperature difference of the TEG, K
ΔT_{PV}	PV-temperature reduction relative to the reference system, K	α	effective Seebeck coefficient, V K^{-1}

(continued on next column)

(continued)

(continued on next page)

* Corresponding author.
** Corresponding author.
E-mail addresses: lihongwei@neepu.edu.cn (H. Li), l.xing@surrey.ac.uk (L. Xing).

https://doi.org/10.1016/j.rser.2026.117497
Received 13 June 2026; Received in revised form 4 September 2026; Accepted 7 September 2026
Available online 10 September 2026
1364-0321/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

(*continued*)

σ	electrical conductivity, S m^{-1}	k	thermal conductivity, W m^{-1} K^{-1}
K	thermal conductance of the TEG, W K^{-1}	R_i	internal electrical resistance of the TEG, Ω
R_L	external load resistance, Ω	$R_{PV\text{-}TEG}$	thermal resistance of the PV-to-TEG interface, K W^{-1}
R_{TEG}	thermal resistance of the TEG module, K W^{-1}	$R_{TEG\text{-}cold}$	thermal resistance of the TEG cold-side interface, K W^{-1}
R_{sink}	thermal resistance of the heat sink or cooling boundary, K W^{-1}	Q	heat-transfer rate, W
Q_h	heat absorbed at the TEG hot side, W	Q_c	heat rejected at the TEG cold side, W
Q_u	useful thermal output, W	Q_{PCM}	thermal-energy storage capacity of the PCM, J
G	incident solar irradiance, W m^{-2}	A	system or collector area, m^2
A_{PV}	active photovoltaic area, m^2	m	PCM mass, kg
L	latent heat of phase change, J kg^{-1}	$c_{p,s}$	specific heat capacity of solid PCM, J kg^{-1} K^{-1}
$c_{p,l}$	specific heat capacity of liquid PCM, J $kg^{-1}K^{-1}$	η_{PV}	photovoltaic electrical efficiency, dimensionless
η_{TEG}	thermoelectric conversion efficiency, dimensionless	η_{el}	overall system electrical efficiency, dimensionless
η_{th}	thermal efficiency, dimensionless	η_{total}	total energy efficiency, dimensionless
η_{ex}	exergy efficiency, dimensionless	PD	power density, W m^{-2}
EY	energy yield, Wh or Wh m^{-2}	ZT	dimensionless thermoelectric figure of merit, dimensionless
N_f	predicted number of cycles to failure, dimensionless	D	cumulative fatigue damage, dimensionless
n	number of thermal cycles, dimensionless		

Abbreviations

ABC-PO	artificial-bee-colony-based perturb and observe	ASHP	air-source heat pump
BIPV	building-integrated photovoltaics	BIPV/T	building-integrated photovoltaic/thermal
BIPVT	building-integrated photovoltaic/thermal	CFD	computational fluid dynamics
cPCMs	composite phase change materials	CTE	coefficient of thermal expansion
DOI	digital object identifier	DSC	differential scanning calorimetry
DX-PVT-HP	direct-expansion photovoltaic/thermal heat pump	EVA	ethylene-vinyl acetate
FEA	finite element analysis	GHI	global horizontal irradiation
GSHP	ground-source heat pump	ICC	initial capital cost
IDX-PVT-HP	indirect-expansion photovoltaic/thermal heat pump	IIGA	improved immune genetic algorithm
LCA	life-cycle assessment	LCOE	levelized cost of electricity
MCHP	micro heat pipe	MPC	model predictive control
MPPT	maximum power point tracking	NEPCM	nano-enhanced phase change material
PBP	payback period	PCM	phase change material
PV	photovoltaic	PV/T	photovoltaic/thermal
PVT	photovoltaic/thermal	PV-PCM	photovoltaic-phase change material
PV-TEG	photovoltaic-thermoelectric generator	PV-TEG-PCM	photovoltaic-thermoelectric generator-phase change material
PV-PCM-TEG	photovoltaic-phase change material-thermoelectric generator	PV/T-PCM	photovoltaic/thermal-phase change material
PV/T-TEG	photovoltaic/thermal-thermoelectric generator	PV/T-PCM-TEG	photovoltaic/thermal-phase change material-thermoelectric generator
RMSE	root-mean-square error	SQ1-SQ3	search queries 1-3
STC	standard test conditions	TEG	thermoelectric generator
TES	thermal energy storage	UDF	user-defined function

1. Introduction

Photovoltaic (PV) technology has become a cornerstone of the global transition towards low-carbon energy because of its scalability, modularity, and broad deployment potential. As summarized in Fig. 1, nearly two centuries of PV development can be broadly divided into four stages: Foundation, Silicon Era, Diversification, and Perovskite-Tandem development. The Foundation stage began with Becquerel's discovery of the photovoltaic effect in liquid electrolytes in 1839 [1], followed by the observation of the solid-state photovoltaic effect in selenium in 1876 [2] and the fabrication of an early selenium solar cell with an efficiency of approximately 1% in 1883 [3]. Einstein's explanation of the photoelectric effect in 1905 established the quantum-mechanical basis of light-matter interaction [4], while the Czochralski crystal-growth method developed in 1918 provided an important foundation for high-quality crystalline-silicon materials [5].

The Silicon Era was initiated by the first practical crystalline-silicon solar cell developed at Bell Laboratories in 1954 [6], followed by the use of PV power on the Vanguard 1 satellite in 1958 [7]. Subsequent developments included amorphous-silicon technology [8], dye-sensitised solar cells [9], continuous efficiency improvements in crystalline-silicon devices [10], and the commercialization of CdTe thin-film modules [11].

PV research then entered a period of material and device diversification. Perovskite solar cells reached a laboratory-scale efficiency of 9.7% in 2012 [12], followed by advances in near-infrared-transparent perovskite devices [13], high-performance n-type silicon cells [14], spectrum-splitting concepts [15], and SnO_2-based interface engineering [16]. These developments established the technological basis for high-efficiency tandem architecture.

The emergence of efficient perovskite cells marked the transition from broad material diversification to the current Perovskite-Tandem stage. The most recent Perovskite-Tandem stage has been marked by monolithic perovskite/silicon tandem cells exceeding 29% efficiency [17], improvements enabled by self-assembled monolayer interface engineering [18], continuing advances in CdTe photovoltaics [19], ultrathin flexible tandem devices [20], and the development of large-area perovskite/silicon tandem cells approaching commercially relevant dimensions [21]. As indicated by the upper trajectory in Fig. 1, the long-term direction of PV development has progressively shifted towards higher efficiency, lower cost, and improved operational stability. Nevertheless, increasing device efficiency alone does not eliminate temperature-induced performance losses or solar intermittency. Recent studies have therefore placed growing emphasis on passive, active, and hybrid thermal-management strategies [22], together with the integration of high-efficiency PV devices with thermal storage and thermoelectric energy-harvesting technologies [23]. These unresolved thermal and temporal limitations motivate the integration of PV conversion with thermoelectric recovery and phase-change thermal storage.

PV-TEG-PCM systems integrate photovoltaic conversion, thermoelectric waste-heat recovery, and latent-heat storage to address two fundamental limitations of conventional PV systems: temperature-induced efficiency degradation and solar intermittency. Rather than providing uninterrupted electricity generation, these hybrid systems offer the potential to extend solar-derived energy availability by regulating PV temperature, recovering part of the rejected heat, and enabling supplementary thermoelectric generation beyond daylight hours. However, the actual performance depends strongly on PCM properties, thermal losses, cold-side heat rejection, and operating conditions, requiring a systematic evaluation of the coupled thermal-electrical mechanisms.

Existing reviews have mainly focused on individual subsystems or binary hybrid configurations. PV-PCM reviews have emphasized PCM selection, thermal regulation, and conductivity enhancement [24–26], while PV/T and PV/T-PCM reviews have mainly addressed heat recovery and latent-heat storage strategies [27,28]. Similarly, PV-TEG studies

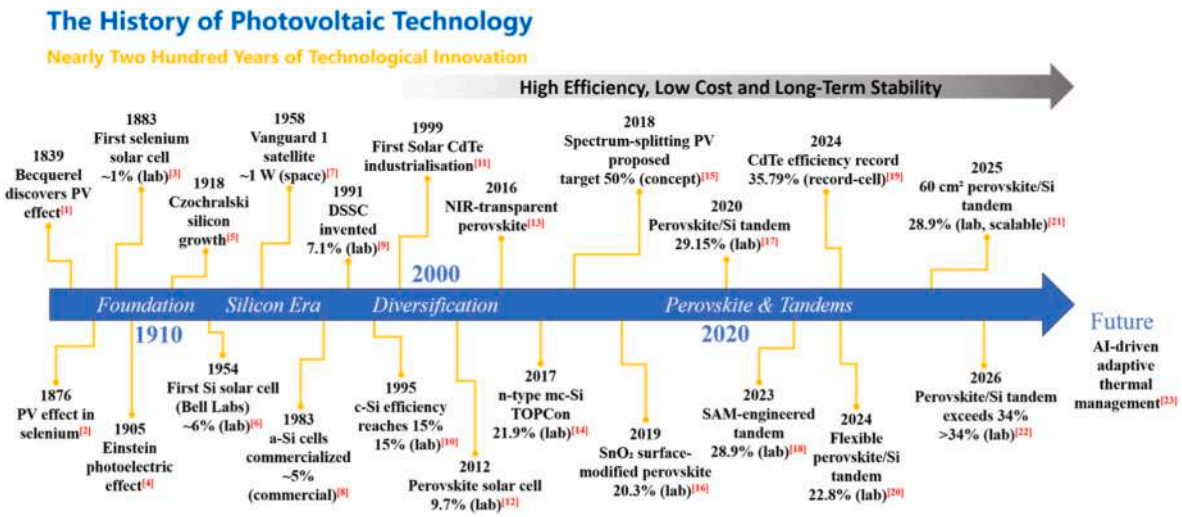


Fig. 1. Evolution and major transitions of photovoltaic technology.

have primarily investigated daytime waste-heat recovery, with limited consideration of PCM-assisted thermal storage and extended-duration thermoelectric generation [29]. Although recent reviews have discussed PCM-assisted and thermoelectric-assisted PV technologies within broader classification frameworks [30], a comprehensive system-level analysis of fully coupled PV-TEG-PCM architecture remains limited.

Accordingly, this review establishes a unified framework for analyzing PV-TEG-PCM systems by integrating material development, operating mechanisms, modelling methodologies, and application scenarios. The main contributions are fourfold: (1) the recent progress of PV, PCM, and thermoelectric materials is systematically synthesized, including nano-enhanced PCMs, composite thermal-storage materials, and advanced thermoelectric materials; (2) coupled operating behaviours involving daytime PV cooling, PCM charging/discharging, thermoelectric recovery, and post-sunset power generation are critically evaluated; (3) modelling approaches, including CFD, FEA, reduced-order models, and optimization strategies, are compared in terms of applicability and predictive capability; and (4) application scenarios and practical deployment constraints are assessed, with particular attention to reliability, techno-economic feasibility, lifecycle impacts, and outdoor validation.

2. Methodology for literature review

To improve the transparency, reproducibility, and consistency of the literature review, a structured methodology was adopted for literature identification, selection, thematic classification, and comparative analysis. The review focused on research related to photovoltaic (PV) thermal management, thermoelectric generators (TEGs), phase change materials (PCMs), and integrated PV-TEG-PCM systems. The methodology comprised four successive stages: (i) literature identification, (ii) literature selection and dataset construction, (iii) thematic coding and quantitative analysis, and (iv) standardized performance comparison.

Three academic databases were used for literature retrieval: ScienceDirect, Web of Science, and Socolar. ScienceDirect was selected because of its extensive coverage of peer-reviewed journals in energy engineering, thermal sciences, materials science, and renewable-energy systems. Web of Science was included because of its comprehensive citation-indexing capability and broad coverage of high-impact interdisciplinary publications. Socolar was employed only as a supplementary retrieval source to improve search completeness, whereas bibliographic verification and eligibility assessment were performed using the corresponding peer-reviewed publication records.

The literature search was last updated on 23 July 2026. To ensure a fixed and reproducible search window, only publications available up to 30 June 2026 were considered for inclusion in the systematic dataset. The quantitative analysis focused primarily on studies published between 2018 and 2026 in order to capture recent advances in PV thermal management, thermoelectric waste-heat recovery, PCM thermal storage, and integrated PV-TEG-PCM technologies. Earlier seminal studies were retained only when they provided essential theoretical foundations, historical developments, governing mechanisms, or technological milestones. These references were used for contextual discussion but were included in the quantitative analysis only when they satisfied the predefined eligibility criteria.

The literature retrieval was performed using three complementary Boolean search strategies. To improve retrieval accuracy and reduce ambiguity, explicit Boolean operators and parentheses were adopted throughout the search process. Search expressions were adapted to the syntax requirements of each database. Whenever possible, searches were restricted to the title, abstract, and author-keyword fields. When identical search fields were unavailable, the closest equivalent topic-search function provided by the database was adopted.

Query 1 (SQ1), PV-TEG systems:

("photovoltaic*" OR "PV module*" OR "solar cell*" OR "solar module*") AND ("thermoelectric generator*" OR "thermoelectric module*" OR "thermoelectric device*")

Query 2 (SQ2), PV-PCM and PV/T-PCM systems:

("photovoltaic*" OR "PV module*" OR "solar cell*" OR "solar module*" OR "photovoltaic thermal" OR "PV/T″ OR PVT) AND ("phase change material*" OR PCM OR "latent heat storage")

Query 3 (SQ3), PV-TEG-PCM integrated systems:

("photovoltaic*" OR "PV module*" OR "solar cell*" OR "solar module*" OR "photovoltaic thermal" OR "PV/T″ OR PVT) AND ("thermoelectric generator*" OR "thermoelectric module*" OR "thermoelectric device*") AND ("phase change material*" OR PCM OR "latent heat storage" OR "thermal energy storage")

Only peer-reviewed journal articles and review papers published in English were considered for the systematic-analysis dataset. Highly relevant peer-reviewed conference papers were included only when they provided unique technical information unavailable in journal publications and were not used as the primary basis for quantitative performance comparison. Books, book chapters, editorials, patents, theses, short communications, and non-peer-reviewed reports were excluded from the systematic-analysis dataset.

Following literature identification, all retrieved records were subjected to a structured selection procedure to construct a reliable dataset for subsequent thematic analysis. Because the three Boolean search

strategies were intentionally complementary and partially overlapping, the same publication could be retrieved by multiple search queries and databases. Consequently, literature selection was performed at two levels: query-level preliminary screening and pooled-record eligibility assessment.

Initially, the records retrieved by SQ1-SQ3 were screened independently within each database according to publication period, document type, indexing status, source quality, and topical relevance. This preliminary screening was designed to remove records that were clearly outside the scope of the review while preserving potentially relevant publications for subsequent assessment. The numbers of records retained at each stage of the query-level screening process are summarized in Table 1.

It should be noted that the values reported in Table 1 represent query-specific screening records rather than unique publications. Since the three search strategies intentionally overlapped and the same publication could be retrieved simultaneously by different search queries and databases, the subtotals shown in Table 1 are simple arithmetic sums of the corresponding SQ1-SQ3 records. These values therefore should not be interpreted as the number of unique publications retained after duplicate removal.

After completion of the query-level screening, all retained records were pooled into a single candidate reference set. Duplicate verification and eligibility assessment were subsequently performed jointly at the pooled-record level rather than independently for each search query. Duplicate publications were identified primarily using digital object identifiers (DOIs). When DOI information was unavailable, bibliographic matching based on normalized titles, author names, publication year, and publication source were employed, followed by manual verification of ambiguous cases where necessary. This pooled-record procedure ensured that publications retrieved repeatedly through different databases or Boolean queries were counted only once during the subsequent analysis.

Following duplicate verification, titles and abstracts were evaluated to determine their relevance to the objectives of the present review. Full-text examination was subsequently conducted whenever the bibliographic information and abstracts were insufficient to determine relevance or when detailed technical information was required for thematic classification and performance comparison.

Studies were included in the core systematic-analysis dataset when they satisfied all of the following criteria: (i) direct investigation of PV thermal behavior, PV thermal management, PV-PCM or PV/T-PCM systems, PV-TEG systems, PCM-assisted thermoelectric recovery, integrated PV-TEG-PCM configurations, or material and device studies with explicit relevance to these systems; (ii) provision of technical information related to thermal management, electrical performance, thermoelectric output, PCM thermal storage, heat-transfer enhancement, system optimization, numerical modelling, experimental validation, or extended energy harvesting; (iii) publication in eligible peer-reviewed sources; and (iv) sufficient bibliographic or full-text information to support reliable thematic classification and comparative analysis.

Studies were excluded when they (i) were unrelated to PV thermal regulation, thermoelectric conversion, PCM thermal storage, or integrated PV-TEG-PCM systems; (ii) addressed generic solar-energy applications without technically relevant discussion of PV thermal behavior, thermoelectric conversion, or latent-heat storage; (iii) lacked sufficient technical information for thematic classification or comparative evaluation; (iv) did not satisfy the predefined publication-type requirements; or (v) could not be assessed reliably because of insufficient publication information. Publications excluded from the core quantitative dataset could still be retained as contextual references when they provided useful theoretical, historical, methodological, techno-economic, lifecycle, or material-property information supporting the broader discussion.

The pooled-record verification and eligibility assessment produced the final literature dataset used throughout the remainder of this review. As summarized in Table 2, the final retained reference pool comprised 171 publications, including 121 core studies used for thematic coding, quantitative synthesis, and performance comparison, together with 50 contextual and supplementary references supporting theoretical background, historical development, methodology, optimization strategies, techno-economic assessment, lifecycle analysis, and related discussions. Only the 121 core publications were used in the quantitative thematic analysis presented in the following section.

Following construction of the core systematic-analysis dataset, a

Table 1
Query-specific retrieval and preliminary screening records from the three databases.

Retrieving Database	Search Criteria	SQ1: PV-TEG systems	SQ2: PV-PCM/PV/T-PCM systems	SQ3: PV-TEG-PCM systems	Subtotal
Science Direct	Initial term screening	239	791	36	1066
	Timeliness screening	197	635	34	866
	Source and publication-type screening	124	442	31	597
	Query-level topical pre-screening	35	48	23	106
Web of Science	Initial term screening	156	1173	43	1372
	Timeliness screening	58	343	22	423
	Source and publication-type screening	36	35	12	83
	Query-level topical pre-screening	8	18	5	31
Socolar	Initial term screening	262	749	1030	2041
	Timeliness screening	114	414	418	946
	Source and publication-type screening	46	57	106	209
	Query-level topical pre-screening	9	18	7	34

Note: SQ1-SQ3 correspond to the Boolean search strategies described in Section 2. Owing to intentional overlap among queries and databases, the reported counts are query-specific rather than unique records. All retained records were pooled for duplicate removal and eligibility assessment. Seminal publications were included for background only unless they satisfied the eligibility criteria for quantitative analysis.

Table 2
Composition and analytical use of the retained literature.

Literature set	Definition and analytical role	No. of publications
Final retained reference pool	Publications retained after duplicate verification and eligibility assessment	171
Core systematic-analysis dataset	Directly relevant publications for thematic coding, quantitative synthesis, and performance comparison	121
Contextual and supplementary references	Retained publications for theoretical, historical, methodological, and supporting discussion	50

Note: Only 121 core publications were used for the quantitative analyses presented in Figs. 2 and 3. The remaining 50 publications were retained solely for contextual discussion. Because non-exclusive multi-label coding was adopted, individual publications may contribute to more than one thematic domain or category.

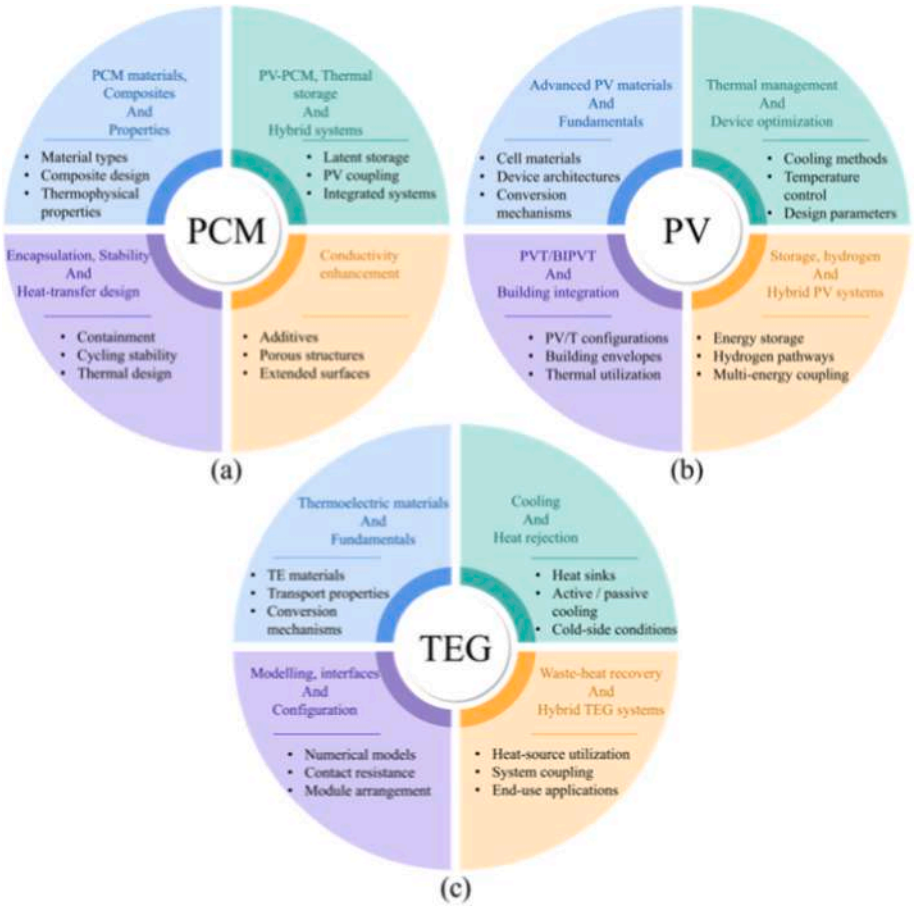


Fig. 2. Thematic framework of the PV, TEG, and PCM literature.

structured thematic-coding procedure was performed to characterize the research landscape of PV, TEG, PCM, and integrated PV-TEG-PCM systems. The objective of this stage was not only to organize the literature into coherent research topics but also to establish a unified analytical framework for the subsequent quantitative synthesis and cross-topic comparison.

The initial coding framework was developed through iterative examination of publication titles, abstracts, author keywords, and, where necessary, full-text contents. Candidate research topics were first identified according to their principal scientific objectives, investigated technologies, and system components. The preliminary thematic categories were subsequently reviewed to eliminate conceptual overlap, reduce thematic fragmentation, and improve consistency across different research directions.

To ensure that the final thematic framework accurately reflected the development of integrated PV-TEG-PCM technologies, conceptually related categories were consolidated according to four principles: (i) similarity of scientific and engineering objectives; (ii) representation of different stages within the materials-device-system research chain; (iii) sufficient thematic independence from neighboring categories; and (iv) direct relevance to the scope of this review, namely PV thermal management, thermoelectric waste-heat recovery, PCM thermal storage, and integrated PV-TEG-PCM systems. Categories containing only a small number of publications were re-examined individually before being either merged or retained to better preserve important emerging research topics.

Following the initial classification and category consolidation, a rule-based verification procedure was applied to improve the reliability and consistency of manual thematic coding. Each coding assignment was examined against the predefined criteria, including the principal scientific objective, investigated technology, system component, and direct relevance to the corresponding thematic domain. Ambiguous or potentially overlapping cases were further checked against the full-text content and reassessed using the same criteria. Through this iterative process of classification, consolidation, and verification, the predefined criteria were applied consistently across the 121 core publications, thereby reducing potential subjectivity in the subsequent quantitative thematic analysis.

This procedure resulted in twelve final thematic categories distributed across three major technological domains, namely PV, TEG, and PCM, as illustrated in Fig. 2. The PV domain comprises advanced PV materials and fundamentals, thermal management and device optimization, PVT/BIPVT and building integration, and storage, hydrogen and hybrid PV systems. The TEG domain comprises thermoelectric materials and fundamentals, cooling and heat rejection, modelling, interfaces and configuration, and waste-heat recovery and hybrid TEG systems. The PCM domain comprises PCM materials, composites and properties, thermal-conductivity enhancement, encapsulation, stability and heat-transfer design, and PV-PCM, thermal storage and hybrid systems. Together, these twelve categories establish a unified framework for organizing the diverse research directions reported in the literature while preserving the relationships among materials, devices, and integrated energy systems.

Because many studies simultaneously investigated multiple technologies or addressed several scientific problems, exclusive classification would not adequately represent their research scope. Therefore, a non-exclusive multi-label coding strategy was adopted. Individual publications could be assigned simultaneously to multiple technological domains and thematic categories whenever justified by their technical contents. To avoid statistical bias, publication frequencies for each

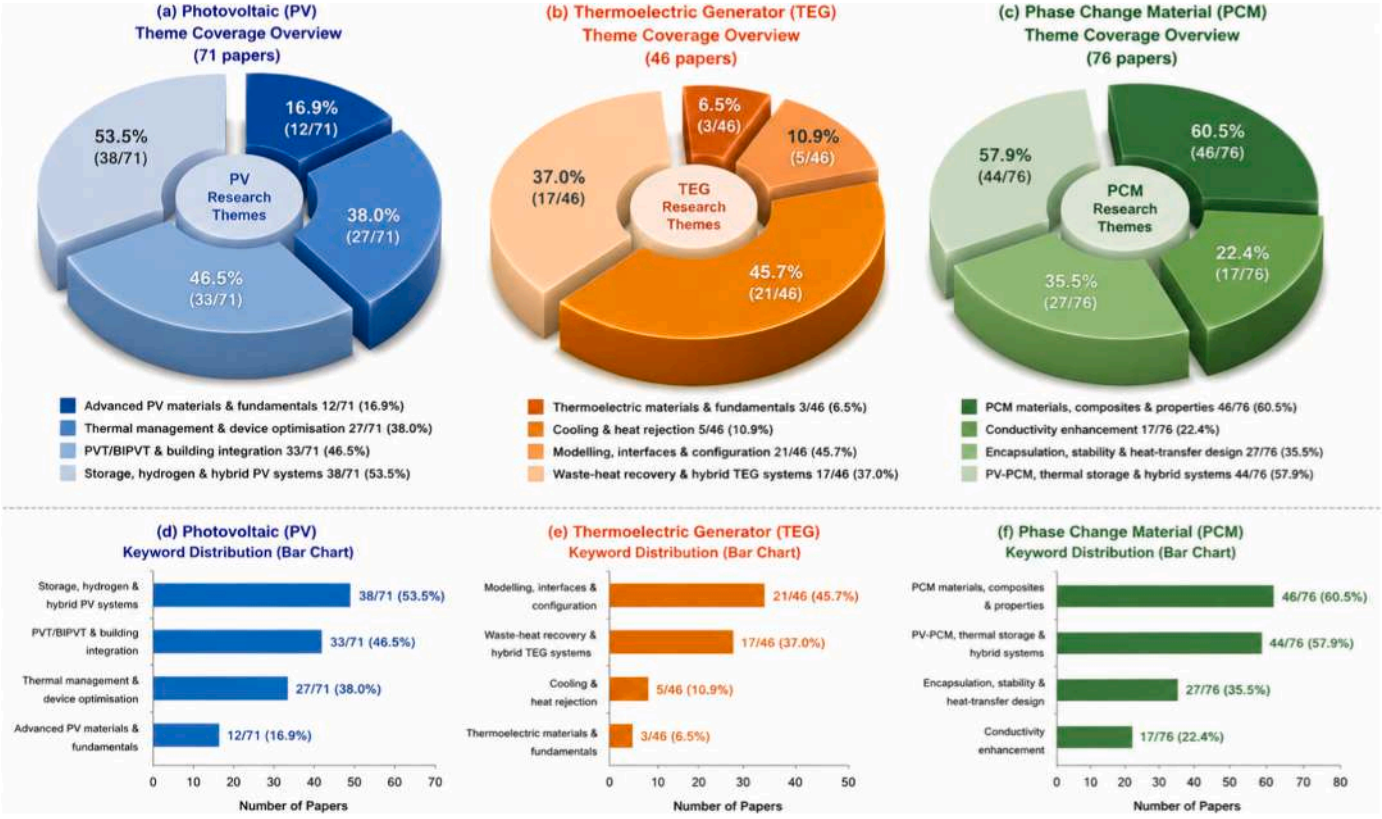


Fig. 3. Thematic distribution of PV, TEG, and PCM studies.

consolidated thematic category were recalculated using the union of unique reference identifiers rather than by simply summing the frequencies of the original categories. Consequently, duplicate counting of the same publication within the same thematic category was effectively avoided while preserving the multidimensional characteristics of interdisciplinary studies throughout the quantitative thematic analysis.

Based on this coding framework, the thematic distribution of the 121 core publications was quantified, as shown in Fig. 3. The denominator used to calculate the percentage within each technological domain corresponds to the number of unique publications assigned to that domain, namely 71 publications for PV, 46 for TEG, and 76 for PCM. Because non-exclusive multi-label coding was employed, one publication may contribute simultaneously to multiple domains and thematic categories. Accordingly, the percentages reported within each domain are independent and therefore do not sum to 100%.

The resulting thematic distribution provides an overview of the

Table 3
Standardized performance indicators and comparison criteria.

Indicator	Definition	Expression	Basis	Comparison rule
PV electrical efficiency, ηPV	PV power/incident solar power	$\eta_{PV} = P_{PV}/(GA_{PV})$	Active PV area	Match PV type, irradiance, area and temperature.
TEG efficiency, η_{TEG}	TEG power/hot-side heat input	$\eta_{TEG} = P_{TEG}/Q_h$	TEG hot-side heat input	Match ΔT, Th, Tc, load and Qh definition.
System electrical efficiency, η_{el}	Net PV + TEG power/solar input	$\eta_{el} = (P_{PV} + P_{TEG} - P_{aux})/(GA)$	Common system area	Use identical area basis and auxiliary-power treatment.
Thermal efficiency, η_{th}	Useful heat/solar input	$\eta_{th} = Q_u/(GA)$	Useful thermal output	Match heat-use definition, flow and temperature conditions.
Total energy efficiency, η_{total}	Electrical + useful thermal energy/solar input	$\eta_{total} = (P_{el} + Q_u)/(HA)$	Combined energy output	Use consistent boundaries; not equivalent to exergy efficiency.
Exergy efficiency, η_{ex}	Useful output exergy/solar exergy	$\eta_{ex} = Ex_{out}/Ex_{solar}$	Electrical + thermal exergy	Use the same reference environment and solar-exergy model.
Power density, PD	Power normalized by area	$PD = P/A$	Defined PV/TEG/collector area	Compare only on the same area basis.
Energy yield, EY	Energy generated over a stated period	$EY = \int P\,dt$ or E/A	Time period and area basis	Match reporting period and resource conditions.
TEG temperature difference, ΔT	Hot-side minus cold-side temperature	$\Delta T = T_h - T_c$	TEG interface temperatures	Use consistent sensor locations and operating state.
PV temperature reduction, ΔT_{PV}	PV temperature drop vs reference	$\Delta T_{PV} = T_{ref} - T_{test}$	Same PV/reference system	State reference, irradiance, ambient condition and time.
PCM storage capacity, Q_{PCM}	Sensible + latent heat stored	$Q_{PCM} \approx m[c_{p,s}\Delta T_1 + L + c_{p,l}\Delta T_2]$	PCM mass and useable range	Match PCM mass, latent heat and melting range.
Extended-generation duration	Time with output above a useful threshold	$P(t) \geq P_{threshold}$	Explicit output threshold	Do not equate measurable output with useful generation.

Note: Metrics are compared quantitatively only when definitions, system boundaries, normalization bases and operating conditions are equivalent. Otherwise, original values are retained for reference only.

current research emphasis within each technological domain and establishes the quantitative basis for the subsequent comparative discussion. Rather than representing publication trends over time, these statistics describe the relative distribution of research topics within the core systematic-analysis dataset and are therefore used to identify the dominant research directions, technological priorities, and knowledge gaps discussed throughout this review.

Given the diversity of experimental conditions, numerical models, and system configurations reported in the literature, direct comparison of published performance values may lead to misleading conclusions when different studies adopt inconsistent definitions, system boundaries, or operating conditions. To improve the consistency and reliability of cross-study comparisons, a standardized performance-comparison framework was established for the present review.

The principal performance indicators adopted throughout this review are summarized in Table 3, including PV electrical efficiency, TEG conversion efficiency, overall system electrical efficiency, thermal efficiency, total energy efficiency, exergy efficiency, power density, energy yield, TEG temperature difference, PV temperature reduction, PCM storage capacity, and extended-generation duration. These indicators were selected because they represent the most frequently reported metrics used to evaluate the thermal, electrical, and energy-storage performance of PV-TEG-PCM systems.

Rather than recalculating or normalizing the performance values reported in the original studies, the standardized framework defines consistent principles for interpreting and comparing these indicators across different investigations. Accordingly, the performance values presented throughout this review remain identical to those reported in the original publications, while their comparability is evaluated according to unified comparison criteria.

Particular attention was paid to distinguishing between performance indicators that represent different physical meanings and system boundaries. For example, PV electrical efficiency, TEG conversion efficiency, overall system electrical efficiency, thermal efficiency, total energy efficiency, and exergy efficiency describe different aspects of system performance and therefore cannot be compared directly without considering their respective definitions. Similarly, power density and energy yield were compared only when identical normalization areas, reporting periods, and operating conditions were explicitly specified.

For thermoelectric performance, comparisons were interpreted together with the corresponding hot-side temperature, cold-side temperature, temperature difference, electrical load condition, and heat-input definition because these parameters strongly influence TEG output characteristics. Likewise, PCM-related performance was evaluated with reference to PCM mass, latent heat, melting temperature range, storage capacity, and operating temperature range to ensure that comparisons reflect equivalent thermal-storage conditions rather than isolated numerical values.

When the original studies adopted different system boundaries, reference conditions, normalization methods, or metric definitions, the reported performance values were retained exactly as presented in the original publications but were not treated as directly equivalent. Instead, the methodological differences were explicitly identified in the corresponding comparison tables, and the associated studies were interpreted qualitatively rather than ranked solely according to numerical performance. This comparison strategy improves the transparency of the literature synthesis while reducing potential bias arising from inconsistent reporting practices among different studies.

3. Thermal management challenges in photovoltaic systems

3.1. Temperature-dependent degradation analysis of photovoltaic module conversion efficiency

The performance of PV systems is strongly affected by operating temperature. For crystalline-silicon PV modules, the maximum power output generally decreases as the module temperature rises above the standard test condition (STC) reference temperature of 25 °C. A commonly reported maximum-power temperature coefficient is approximately −0.4%/°C, although values for commercial crystalline-silicon modules typically range from about −0.2%/°C to −0.4%/°C depending on module technology, irradiance level, temperature range, and operating conditions [31]. Based on this coefficient range, an increase in module temperature from 25 °C to approximately 75 °C may correspond to an estimated power reduction of about 10-20% relative to STC. This range should be interpreted as a coefficient-based estimate rather than as a universal experimentally measured loss. As the module temperature increases progressively above 25 °C, electrical power losses become increasingly significant, particularly under high-temperature operating conditions. This negative temperature effect is therefore an important thermal-management challenge for PV systems because it directly influences energy yield and economic performance.

Temperature-dependent power degradation originates from multiple coupled physical mechanisms rather than from a single process. Based on previous studies [32–35], the mechanisms responsible for temperature-induced PV performance degradation can be summarized into three main aspects.

Firstly, elevated temperature narrows the semiconductor bandgap and exponentially increases the intrinsic carrier concentration, thereby increasing the diode saturation current. The resulting reduction in open-circuit voltage outweighs the slight increase in photocurrent, ultimately decreasing the maximum power output and conversion efficiency of the photovoltaic cell. The elevated operating temperature primarily originates from several heat-generation mechanisms within the PV cell. Among them, thermalization of above-bandgap photons is an intrinsic and unavoidable source of heat generation. However, this heat-generation mechanism should be distinguished from the subsequent electrical performance degradation caused by the elevated operating temperature [36]. Temperature-dependent behavior may also vary among PV technologies. For example, controlled light-soaking experiments on amorphous-silicon cells have shown that their performance depends on both illumination and characteristic temperature, with elevated temperature affecting the initial degradation and stabilization behavior [37]. Such technology-specific behavior should not be directly generalized to crystalline-silicon modules. By contrast, in crystalline-silicon cells, elevated temperature may enhance carrier recombination, shorten carrier lifetime, intensify lattice vibrations and carrier scattering, and consequently reduce carrier mobility and collection efficiency [38].

Secondly, temperature affects the principal electrical parameters of crystalline-silicon cells. The open-circuit voltage, V_{oc}, decreases significantly with increasing temperature and commonly exhibits a temperature coefficient of approximately −0.3%/°C to −0.4%/°C [39]. By contrast, the short-circuit current, I_{sc}, increases only slightly with temperature, typically by about 0.04%/°C to 0.05%/°C [39]. This modest increase in I_{sc} is insufficient to compensate for the substantially larger decrease in V_{oc}, leading to an overall reduction in conversion efficiency and maximum power output [31].

Finally, elevated temperature may also alter resistance-related losses and the fill factor of the solar cell. Temperature-dependent changes in the electrical contacts, interconnections, series resistance, and shunt resistance can modify the shape of the current-voltage characteristic and contribute to a reduction in the fill factor [38,39]. However, the magnitude and direction of these resistance changes depend on cell structure, contact materials, ageing condition, and measurement range. As the temperature increases, the maximum power point generally shifts toward a lower operating voltage and lower maximum power. Collectively, the reduction in V_{oc}, the deterioration in fill factor, and the resulting decline in maximum power dominate over the small increase in I_{sc}, thereby reducing the overall performance of crystalline-silicon PV modules [31].

3.2. Thermal stress analysis of photovoltaic modules

Temperature variations not only affect the electrical performance of photovoltaic (PV) modules but also induce thermal stress within the module structure. This stress mainly originates from the mismatch in the coefficients of thermal expansion (CTE) among silicon cells, solder joints, encapsulation layers, glass covers, and back sheet layers. Repeated thermal cycling accelerates degradation mechanisms such as solder fatigue, crack propagation, delamination, and interfacial debonding, ultimately shortening the service life and reducing the reliability of PV modules. Therefore, thermal management should be considered not only as a strategy for improving electrical efficiency but also as a key approach for mitigating thermomechanical degradation and extending module lifetime [40].

Mission-profile-based reliability assessment has become a widely adopted approach for evaluating the influence of thermal loading on PV power electronics and module reliability. As illustrated in Fig. 4, the method starts from actual operating conditions (e.g., solar irradiance and ambient temperature), which are converted into the electrical output of the PV system and subsequently into the thermal response of power devices through electro-thermal models. The resulting junction-temperature (T_j) history is then analyzed using thermal-cycle counting methods, such as the rainflow-counting algorithm, and cumulative fatigue damage is commonly estimated using lifetime prediction models combined with Miner's linear damage rule [34].

The accuracy of this assessment strongly depends on the temporal resolution of the mission profile. High-resolution environmental data are capable of capturing rapid irradiance and temperature fluctuations caused by transient weather conditions, thereby revealing more realistic junction-temperature variations and thermal-cycle distributions. In contrast, coarse-resolution data tend to smooth short-term fluctuations and may underestimate thermal cycling and cumulative fatigue damage, leading to an overly optimistic prediction of component lifetime [34].

The same principle applies to PV-TEG-PCM systems: the PCM layer acts as a thermal damper that reduces both the amplitude and frequency of temperature fluctuations experienced by the PV module. This thermal smoothing effect can reduce the severity of thermal cycling and thus extend the operational lifetime of the PV system, a benefit that mission-profile-based assessment can quantify. For PV-TEG-PCM hybrid systems, reducing module temperature and suppressing temperature fluctuations are expected to alleviate thermal stresses and decrease the severity of thermal cycling. Consequently, mission-profile-based reliability assessment provides a valuable framework for evaluating not only the electrical performance but also the long-term reliability benefits of PV-TEG-PCM systems. Nevertheless, the integration of PCM thermal storage and thermoelectric coupling into reliability models remains limited and deserves further investigation [33,34].

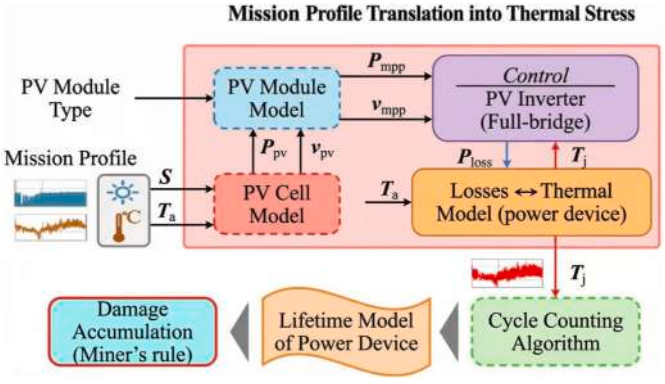


Fig. 4. Thermal stress and reliability analysis of PV inverter power devices based on mission profiles.

3.3. Review of existing cooling and heat-dissipation technologies

Thermal management technologies for photovoltaic (PV) systems can generally be classified into active and passive cooling approaches. Active cooling removes excess heat through externally driven air or liquid circulation, whereas passive cooling relies on natural convection, radiation, heat sinks, or phase change materials (PCMs) without additional energy consumption. Although passive methods are simple and energy-efficient, their cooling capacity is often limited under high-temperature conditions. Consequently, active cooling has become the primary approach for achieving effective PV thermal regulation.

Early active cooling mainly employed forced-air convection. Teo et al. [35] demonstrated that finned air ducts combined with forced ventilation reduced PV operating temperature from 68 °C to 38 °C and improved electrical efficiency from 8.6% to 12.5%, confirming the feasibility of air cooling. Owing to the higher specific heat capacity of water, subsequent studies shifted toward liquid cooling. Bahaidarah et al. [36] showed that water-cooled PV/T systems reduced module temperature by approximately 20% while increasing the total energy output to nearly four times that of conventional PV modules, demonstrating the superior heat-removal capability of liquid cooling.

Following the successful implementation of liquid cooling, research focused on optimizing heat-transfer pathways and temperature uniformity. Yin et al. [37] incorporated functionally graded materials and optimized serpentine flow channels to reduce surface temperature gradients, maintaining PV electrical efficiency at more than 95% of its room-temperature value under high-temperature conditions. Beyond conventional rear-side cooling, Robles-Ocampo et al. [38] proposed a bifacial PV/T configuration combining optical reflection and water cooling, enabling simultaneous thermal regulation and bifacial power generation, with approximately 40% higher electrical output than conventional PV/T systems.

The development of PV thermal management has subsequently evolved from module cooling to integrated energy utilization. Pantic et al. [39] demonstrated that building-integrated photovoltaic/thermal (BIPV/T) systems can simultaneously provide electricity, space heating, ventilation, and domestic hot water. Although adding a front glass cover increased PV operating temperature and reduced electrical efficiency, it produced higher-grade thermal energy suitable for domestic hot water applications. This transition from improving heat dissipation to cascading waste-heat utilization illustrates the evolution of PV/T technology toward integrated building-energy systems.

Overall, existing cooling technologies have progressed from simple air cooling to liquid cooling, structural optimization, and system-level energy integration. While active cooling effectively suppresses PV temperature and improves overall energy utilization, it generally requires additional pumping or fan power and increases system complexity. These limitations have motivated growing interest in passive thermal-management strategies, particularly PCM-assisted cooling, which is discussed in the following section.

3.4. Key technical bottlenecks in thermal management

The primary technical bottleneck in the thermal management of photovoltaic-thermal (PVT) systems stems from the intrinsic trade-off between thermal management efficiency and long-term stability. At the materials level, although advanced thermal working fluids such as nanofluids and phase change materials (PCMs) have demonstrated potential for enhancing heat transfer, their long-term operational reliability remains a significant challenge. For instance, nanofluids are susceptible to nanoparticle agglomeration and aging under high concentrations and prolonged operation, leading to a degradation in thermal conductivity. Similarly, PCMs may suffer from leakage and deterioration of thermophysical properties after repeated phase change cycles. These inherent material-level deficiencies directly exacerbate the complexity of multi-physics coupling control at the system level. This is

particularly evident in cross-disciplinary applications like hydrogen production or building-integrated photovoltaics (BIPV), where the PVT system must precisely coordinate dynamic energy interactions among photovoltaic, photothermal, energy storage, and other subsystems. However, existing research still lacks sophisticated dynamic models and real-time optimal control strategies that can adapt to a wide range of variable on-site operating conditions.

Ultimately, these technical challenges manifest as insufficient adaptability of the system to diverse environmental conditions in practical applications. Under extreme climates, such as high temperature, high humidity, or severe cold, PVT modules are prone to issues like uneven thermal stress distribution, material corrosion, or surface frosting, which severely limits the technology's geographical adaptability and widespread adoption. These interconnected technical bottlenecks culminate in a significant economic barrier impeding the large-scale commercialization of PVT technology. To overcome material performance degradation, achieve efficient control of complex systems, and enhance environmental resilience, it is often necessary to employ high-cost technical solutions and components, such as hybrid nanofluids, composite phase change materials (cPCMs), microchannels, and enhanced fins. This results in a considerably higher initial capital cost (ICC) and subsequent operational and maintenance expenses for integrated PVT systems compared to conventional separate photovoltaic or solar thermal collector systems. Concurrently, the suboptimal real-world operational performance, caused by material aging and inadequate environmental adaptability, leads to a lower-than-expected return on investment and a significantly extended payback period (PBP). Therefore, the high initial investment and prolonged payback period are not isolated economic issues but are direct economic manifestations of the core technical challenges in PVT thermal management. They constitute the critical obstacles that must be surmounted for the technology to transition from laboratory-scale research to widespread market application.

4. Phase-change materials (PCM) and PV/T-PCM systems

4.1. Thermal energy storage characteristics of phase-change materials (PCM)

Phase change materials (PCMs) store and release thermal energy through reversible solid-liquid phase transitions, providing high latent-heat storage over a narrow temperature range and making them one of the most effective passive thermal-management technologies for photovoltaic (PV) systems [40]. Compared with sensible heat storage, PCMs effectively suppress PV temperature fluctuations and improve thermal stability, making them particularly suitable for PV/T-PCM and PV-TEG-PCM systems. Their performance depends not only on latent heat but also on melting temperature, thermal conductivity, cycling stability, leakage resistance, material compatibility, and fire safety [41–43]. Therefore, practical PCM selection requires balancing thermal performance, long-term reliability, manufacturability, and economic feasibility rather than optimizing a single property.

Despite these advantages, several challenges continue to limit the large-scale application of PCMs. Leakage during repeated melting and solidification, intrinsically low thermal conductivity, supercooling, phase segregation, flammability, thermal degradation, and encapsulation cost remain the principal obstacles in PV thermal-management systems [44–51]. Although encapsulation can effectively suppress leakage and improve cycling stability, it inevitably increases thermal resistance and manufacturing complexity. Similarly, enhancing thermal conductivity often compromises latent-heat storage capacity, indicating that PCM optimization is fundamentally a multi-objective engineering problem.

As shown in Fig. 5, PCMs are generally classified into organic, inorganic, and eutectic materials. Organic PCMs exhibit excellent chemical stability and cycling durability but relatively low thermal conductivity, whereas inorganic PCMs provide higher thermal conductivity and energy-storage density but are more susceptible to supercooling and phase segregation. Eutectic PCMs combine multiple components to achieve well-defined melting temperatures and stable phase-transition behaviour, making them attractive for applications requiring precise thermal regulation [52–58]. Consequently, PCM selection should be determined by the operating temperature range and

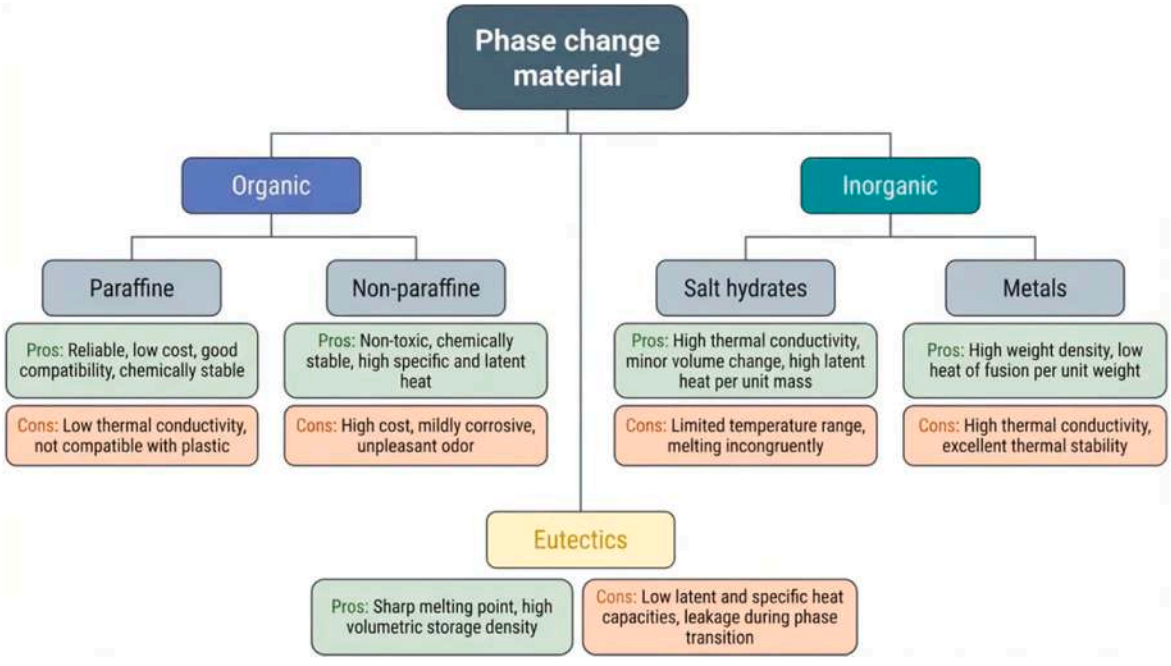


Fig. 5. Classification of phase change materials and their associated advantages and disadvantages.

reliability requirements of the target application.

To overcome the intrinsic limitations of conventional PCMs, composite design, nanoparticle incorporation, and encapsulation technologies have been widely investigated [59–81]. High-conductivity fillers, such as expanded graphite, carbon nanotubes, metal foams, and nanoparticles, significantly improve heat-transfer capability and structural stability, although excessive filler loading generally reduces latent-heat storage capacity and increases material cost [60–72]. Meanwhile, microencapsulation effectively prevents leakage and enhances long-term cycling stability but introduces additional thermal resistance, requiring careful optimization of shell materials and thickness [73–81].

Finally, PCM selection should match the operating temperature of the target application. Organic and hydrated-salt PCMs are generally suitable for low-temperature PV thermal management, whereas eutectic and molten-salt PCMs are more appropriate for medium- and high-temperature thermal-storage systems [82–86]. Therefore, practical material selection should be regarded as a multi-objective optimization process that simultaneously considers melting temperature, latent heat, thermal conductivity, cycling stability, leakage resistance, structural compatibility, and economic feasibility, thereby providing the material basis for subsequent PV/T-PCM and PV-TEG-PCM system design.

4.2. Advantages of PCM integration in photovoltaic systems

The integration of phase change materials (PCMs) with photovoltaic (PV) systems enhances solar-energy utilization by simultaneously regulating PV temperature and recovering waste heat. Through latent-heat storage, PCM suppresses PV temperature fluctuations, stabilizes electrical efficiency, and enables the subsequent utilization of stored thermal energy in downstream applications [87]. Consequently, recent studies have increasingly combined PCM-assisted PV/T systems with heat pumps to improve overall energy efficiency rather than dissipating the recovered heat.

Representative PV/T-heat pump configurations are summarized in Fig. 6. Direct-expansion PV/T heat pumps (DX-PVT-HP, Fig. 6a) allow refrigerants to evaporate directly within the PV collector, whereas indirect-expansion systems (IDX-PVT-HP, Fig. 6b) employ an intermediate heat-transfer fluid, offering greater operational flexibility and lower refrigerant-leakage risk [88]. Building on these configurations, PV/T systems have also been integrated with air-source heat pumps (ASHP, Fig. 6c) and ground-source heat pumps (GSHP, Fig. 6d). Air-source systems achieved thermal and electrical efficiencies of 18.9% and 10.9%, respectively, with an overall efficiency improvement of 52%, while ground-source configurations reduced borehole length by 18% and land occupation by 89% [88,89]. These results demonstrate that effective thermal matching between PCM melting temperature and PV operating temperature is essential for maintaining long-term thermal regulation.

Building integration further extends the benefits of PCM-assisted PV/T systems. Long-term experimental studies reported total system efficiencies of up to 39.4% and building energy savings of 64.2% through façade-integrated PV/T-PCM systems [90–93]. As illustrated in Fig. 7, roof-integrated systems (Fig. 7a–b) optimize cooling hardware installation, while façade-integrated systems (Fig. 7c–d) combine architectural integration with improved electrical performance through adjustable PV louvers and building-envelope design [94,95].

Beyond buildings, PCM-assisted PV/T systems have been successfully applied to agricultural drying, thermoelectric energy harvesting, and solar-driven hydrogen production [96–103]. In solar dryers, PCM extends drying during low-irradiance periods and significantly improves thermal efficiency [96–101]. In PV/T-TEG systems, PCM maintains the temperature difference required for thermoelectric generation through

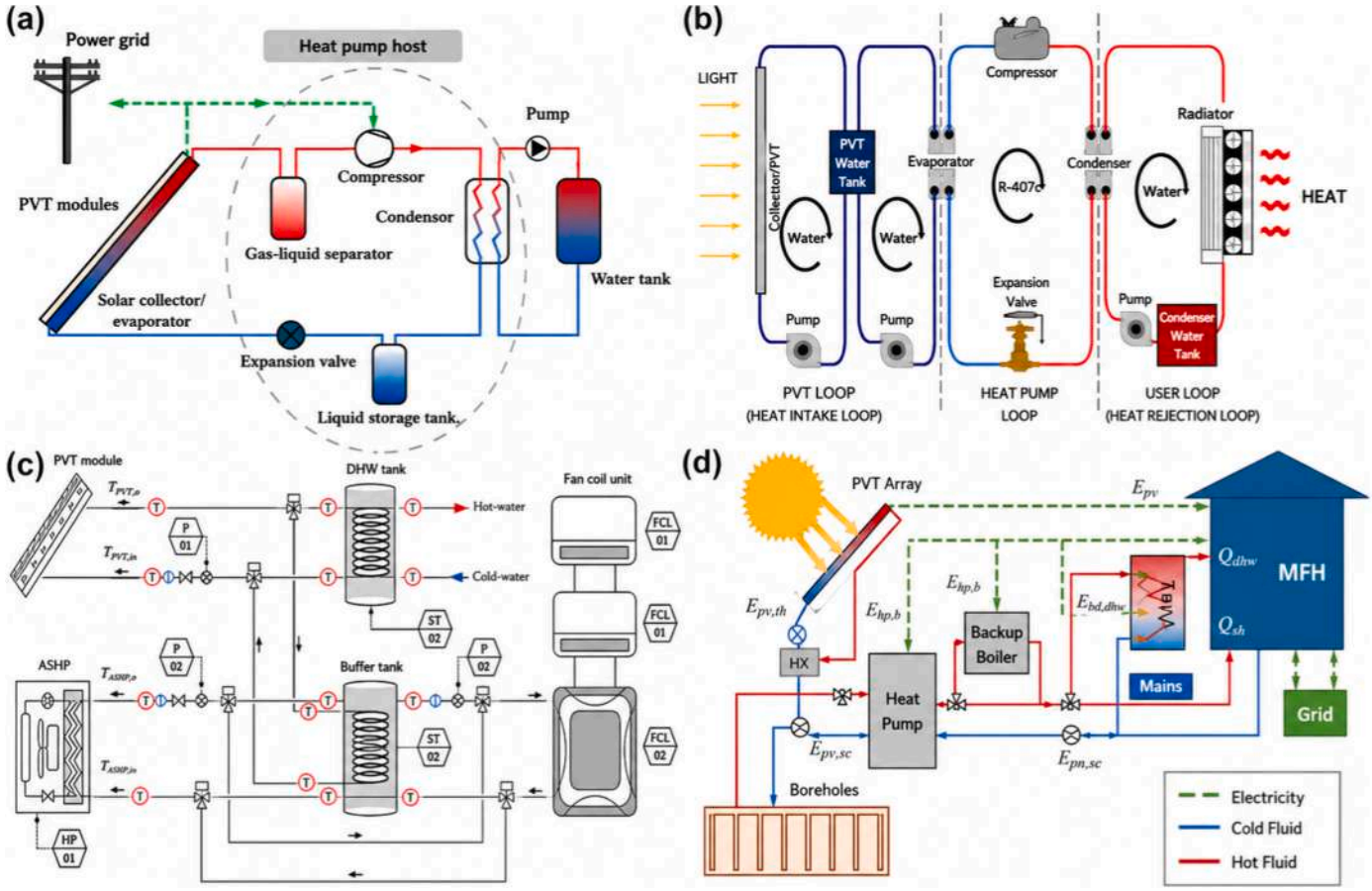


Fig. 6. Schematic diagrams of (a) DX-PVT-HP system [86], (b) IDX-PVT-HP system, (c) PVT integrated ASHP system [88], (d) PVT integrated GSHP system [89].

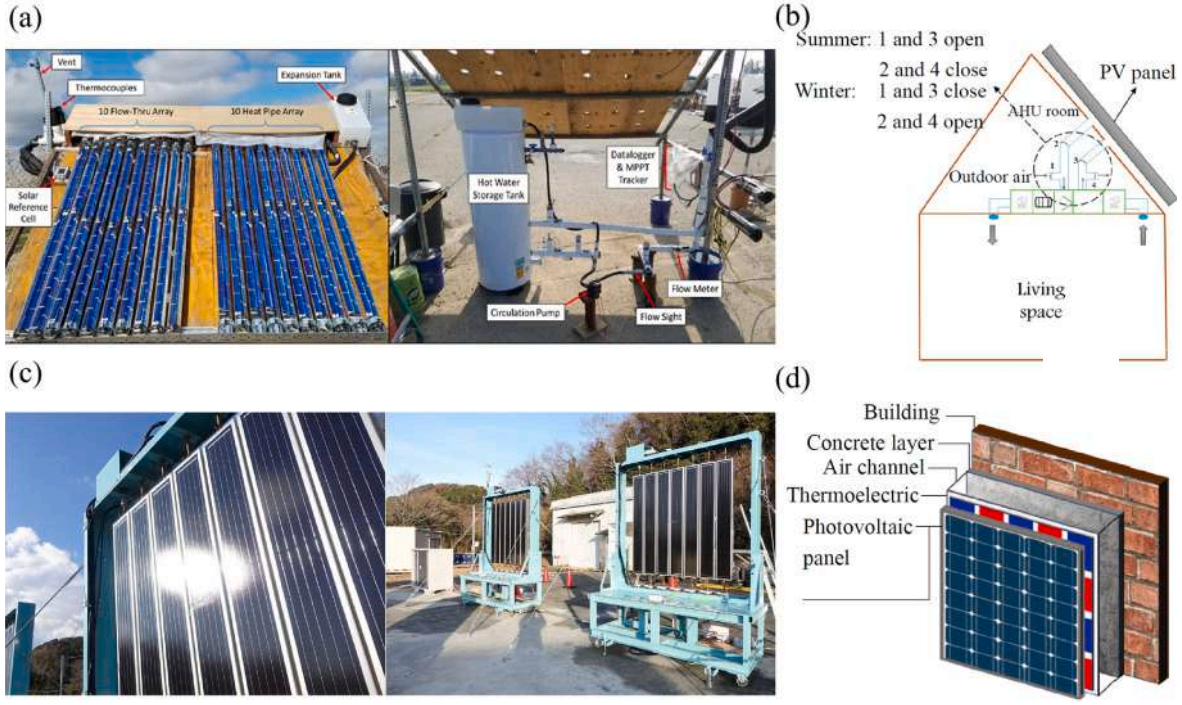


Fig. 7. (a) Roof integrated BIPVT test installation view, (b) Schematic diagram of roof integrated BIPVT [88], (c) Façade integrated BIPVT test installation view [94], (d) Schematic diagram of facade integrated BIPVT [95].

passive heat storage [99,101]. In hydrogen-production systems, PCM buffers thermal fluctuations and improves the operational stability of electrolyzer under variable solar conditions [102,103]. These applications demonstrate that the primary value of PCM extends beyond PV cooling to efficient thermal-energy storage and cascaded energy utilization.

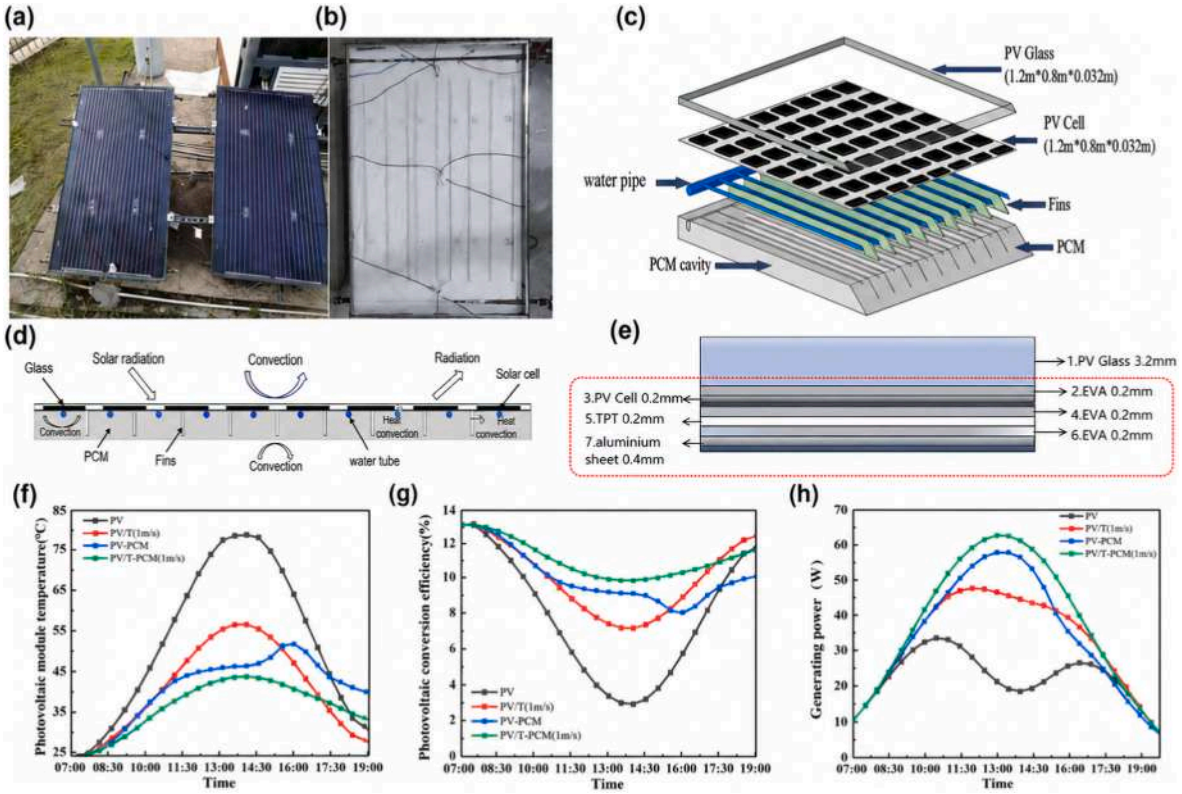


Fig. 8. (a) PV modules in system. (b) Internal structures of PV/T-PCM system. (c) Geometric model of PV/T-PCM system. (d) Schematic of heat transfer processes between system and external environment. (e) Schematic of simplified model of PV module. (f) Temperatures of PV modules in various PV systems. (g) Photovoltaic conversion efficiencies of various PV systems. (h) Variations of output power with time in various PV systems [32].

Overall, these studies indicate that PCM integration has progressed from passive PV temperature regulation toward broader applications in thermal-energy storage and renewable-energy integration, including building, agricultural, thermoelectric, and hydrogen-production systems. This broader body of work provides a useful basis for the further development of integrated PV/T-PCM systems by linking PV thermal regulation, thermal-energy storage, and heat-transfer enhancement.

4.3. System configuration design of PV/T-PCM

The configuration design of photovoltaic/thermal-phase change material (PV/T-PCM) systems aims to simultaneously improve photovoltaic electrical output and thermal energy utilization through the coordinated integration of phase change materials (PCMs), heat-transfer structures, and cooling channels. Compared with conventional PV modules, PV/T-PCM systems combine heat extraction, latent heat storage, and thermal energy recovery into a unified thermal-management framework, thereby suppressing module temperature rise while extending the utilization of recovered thermal energy. Consequently, system configuration should simultaneously optimize the arrangement of PCM modules, flow channels, and heat-transfer components, together with PCM thickness, installation position, thermal-contact resistance, and compatibility with auxiliary devices such as heat exchangers or thermoelectric generators. Excessive PCM thickness increases thermal resistance and delays heat dissipation, whereas insufficient PCM volume shortens the effective thermal-buffering period. Therefore, an effective configuration should balance heat-transfer efficiency with thermal-storage capacity to maintain stable PV operating temperatures and maximize comprehensive energy utilization.

Fig. 8 illustrates the configuration concept and operating mechanism of a representative PV/T-PCM system. Fig. 8a–c presents the system implementation, internal PCM container, and multilayer geometric model, showing the integration of PV modules, PCM, metal fins, and heat-exchange pipes into a coupled thermal-management structure. Based on this structural design, Fig. 8d describes the principal heat-transfer processes between the PV/T-PCM system and the surrounding environment, whereas Fig. 8e simplifies the multilayer PV module into an equivalent thermal-mass model for numerical analysis. Building upon these structural and thermal models, Fig. 8f–h compare the operating temperatures, photovoltaic conversion efficiencies, and output power of different PV systems. The results demonstrate that PCM integration effectively suppresses module temperature rise during periods of increasing solar irradiance, leading to improved photovoltaic conversion efficiency and electrical output, thereby confirming the effectiveness of latent heat storage for PV thermal management [104].

The thermal and electrical benefits illustrated in Fig. 8 have been consistently validated by subsequent experimental studies. Akshayveer et al. [32] reported that combining PCM with an air-cooled PV/T collector reduced module temperature by 35% while improving electrical efficiency by 14.12%, demonstrating the synergistic effect of latent heat storage and convective cooling. Hossain et al. [105] further optimized the heat-transfer pathway using a double-serpentine flow channel, achieving a maximum thermal efficiency of 87.72%, which highlights the importance of flow-channel geometry and PCM position. Carmona et al. [106] confirmed that retrofitting conventional PV modules with PCM provides a practical strategy for improving thermal regulation without fundamentally modifying existing PV structures. Yang et al. [107] showed that integrating PCM around the heat-transfer tubes reduced conductive heat losses and increased primary energy efficiency by 14% through delayed heat release. Further optimization of the PCM-water configuration by Maatallah et al. [108] and Li et al. [109] increased total system efficiency by 40.59%, enhanced electrical output by approximately 17.33%, and improved daily energy yield by 74.3%. Collectively, these studies demonstrate that the performance improvement of PV/T-PCM systems originates from the coordinated optimization of structural configuration, heat-transfer pathways, PCM position and working-fluid design rather than from any individual design modification alone.

Beyond structural optimization, the effectiveness of PV/T-PCM systems is fundamentally determined by the selection and integration of PCMs. Among the key thermophysical properties, melting temperature and latent heat primarily govern the thermal-storage behaviour. Existing studies consistently indicate that the melting temperature should be selected to match the normal operating temperature of PV modules so that latent heat can be effectively utilized during periods of high solar irradiance. For example, Carmona et al. [110] employed RT-35 (29-36 °C), whose phase transition occurs within the typical operating range of PV modules, thereby maintaining stable module temperatures while improving electrical and hybrid efficiencies. Likewise, PCMs with relatively high latent heat provide prolonged heat-storage and heat-release capability, enabling sustained cooling under fluctuating solar irradiation and improving the recovery of useful thermal energy. Therefore, the melting temperature and latent heat should be optimized simultaneously according to climatic conditions and system operating characteristics rather than being considered independently.

Besides intrinsic thermophysical properties, the practical performance of a PV/T-PCM system strongly depends on heat-transfer enhancement and module integration. Because most organic PCMs possess low thermal conductivity, efficient heat transfer requires appropriate structural enhancement, such as metallic fins, aluminium containers, or optimized heat-transfer channels, to reduce thermal resistance and improve heat distribution within the PCM [87,110]. However, increasing heat-transfer capability generally introduces additional material consumption, structural complexity, and manufacturing cost. Consequently, the design of PV/T-PCM systems should balance thermal conductivity enhancement with compactness, weight, and engineering feasibility instead of pursuing maximum heat-transfer performance alone.

Long-term operational reliability represents another critical consideration for practical applications. Encapsulation plays a central role in preventing PCM leakage while maintaining sufficient thermal contact with the PV module. Representative encapsulation approaches include flexible polymer films, aluminium foil packages, aluminium-alloy containers, and microencapsulated PCM particles, each providing different trade-offs between leakage prevention, heat-transfer efficiency, mechanical durability, and manufacturing complexity [105,110,111]. In addition, cycling stability, supercooling, and phase segregation directly influence the long-term effectiveness of thermal energy storage. Organic PCMs reported in the reviewed studies generally exhibit favorable thermal stability with relatively limited phase segregation, whereas inorganic PCMs remain susceptible to supercooling and phase separation, leading to delayed heat release and gradual deterioration of thermal-storage performance during repeated melting–solidification cycles [112]. These findings indicate that long-term durability should be evaluated together with encapsulation performance and leakage resistance, rather than relying solely on short-term thermal performance.

4.4. Simulation of heat conduction and phase-change processes

Numerical simulation of heat transfer and phase change processes is crucial in engineering thermal management systems, especially in applications involving phase change materials (PCM). Accurate prediction of the temperature field and phase change behaviour is a prerequisite for optimizing system performance. Computational fluid dynamics (CFD) is widely used, but its default models may introduce significant errors when handling nonlinear phase change. Therefore, introducing user-defined function (UDF) models, which allow flexible definition of the energy-temperature relationship, can significantly improve prediction accuracy. For example, when simulating the thermal behaviour of a PCM container, the UDF model generates a gradual enthalpy-temperature curve by approximating differential scanning calorimetry (DSC) data, whereas the default CFD model assumes a constant energy

change. The latter can lead to prediction inaccuracies, particularly during the initiation and conclusion stages of the phase change.

Model validation was conducted independently for the melting and solidification processes to evaluate performance under different thermal conditions. The simulations used a computational model verified for grid independence, containing 12,320 hexahedral elements, with a convergence time step of 10 s set to ensure computational stability. The initial temperature field was set according to experimental conditions: for the melting process, the initial temperature was 8 °C, with a gradually increasing air temperature and a convective heat transfer coefficient of 5 W/(m^2·K) [113]; for the solidification process, the initial temperature was 59 °C, with a gradually decreasing air temperature and a convective heat transfer coefficient of 8 W/(m^2·K). The material's specific heat capacity was set to 2550 J/(kg·K). Model deviation was quantified by comparing experimental data from monitoring points with simulation results and calculating the root mean square error (RMSE).

In the melting simulation, the UDF model demonstrated significant advantages. Its predicted temperature trend closely matched the experimental data, with an error of only 1.3%, compared to a 12.3% error for the default model. This difference stems from the UDF model's gradual treatment of the latent heat of phase change, avoiding the prediction bias introduced by the default model during the initial melting stage, which would accumulate and affect the accuracy of subsequent thermal dynamics. Furthermore, the UDF model accurately captured the onset and end temperatures of melting, confirming the reliability of the method based on approximating the energy curve from DSC data, although minor deviations at the end of the simulation might originate from specific heat capacity variations not accounted for in the model.

In the solidification simulation, the UDF model also performed superiorly. Although the gradient of its energy-temperature curve at the solidification point (298 K) was steeper compared to the DSC data, leading to minor differences in the initial prediction stage, the overall temperature trend still accurately reflected the input curve. The RMSE for the UDF model was 0.5%, compared to 6.3% for the default model, with an average deviation of about 2 °C. This highlights the UDF model's advantage in capturing the physics of the phase change, even when the input curve itself has some inherent inaccuracies.

In summary, the UDF model demonstrates higher accuracy and flexibility in simulating heat conduction and phase change processes, particularly for handling nonlinear phase change, where its prediction error is significantly lower than that of the default model. This provides a reliable numerical tool for engineering applications, and it is recommended to prioritize custom models in related research. Future work could focus on optimizing the method for approximating the energy-temperature curve to further enhance prediction accuracy.

4.5. Performance testing and comparative analysis

Performance testing provides an essential basis for evaluating the practical feasibility and optimization potential of photovoltaic/thermal (PVT) systems. However, direct comparison among published studies remains challenging because electrical efficiency, thermal efficiency, operating conditions, and testing protocols are often defined differently. Variations in solar irradiance, ambient temperature, working-fluid type, flow rate, and auxiliary power consumption further complicate cross-study comparisons. Therefore, standardized testing methodologies and unified performance indicators are indispensable for objectively evaluating different PV/T-PCM technologies and identifying their practical advantages and limitations [111,114].

Representative studies summarized in Table 4 indicate that the performance of PV/T-PCM systems depends not only on PCM material properties but also on module configuration, cooling strategy, and environmental conditions. Experimental studies generally demonstrate lower operating temperatures and higher electrical efficiency after PCM integration, while numerical studies provide valuable insights into heat transfer behavior and structural optimization. Nevertheless, the reported improvements vary considerably because of differences in PCM melting temperature, latent heat, thermal conductivity enhancement strategies, encapsulation methods, and system integration approaches [111,114,115].

From the perspective of practical engineering applications, electrical efficiency should not be considered independently from thermal performance. Water-cooled PV/T-PCM systems generally achieve higher thermal efficiencies because of improved heat extraction, whereas air-cooled systems usually offer simpler structures and lower installation costs. Likewise, passive PCM cooling eliminates additional pumping power but may suffer from limited heat dissipation after complete melting. Consequently, different thermal management strategies involve inevitable trade-offs among electrical output, thermal recovery, system complexity, and economic performance [92,115].

Moreover, the reported electrical and thermal efficiencies should be interpreted with careful attention to their definitions. Some studies define thermal efficiency based on the heat extracted by the working fluid, while others report useful heat output after deducting storage losses. Similarly, total efficiency may be reported either as a direct summation or as a weighted sum. This variability in definition complicates cross-study comparison and underscores the need for standardized reporting protocols in future PV/T-PCM research.

Beyond conventional electrical and thermal efficiencies, comprehensive system evaluation should incorporate exergy efficiency, life-cycle assessment (LCA), economic indicators, and long-term operational reliability. Exergy analysis reflects the quality of recovered thermal energy rather than merely its quantity, while LCA and techno-economic analysis evaluate the environmental impacts and economic feasibility throughout the system lifetime. These complementary metrics provide a more comprehensive assessment framework for comparing different PV/T-PCM configurations under realistic operating scenarios [123].

Overall, performance testing should extend beyond reporting efficiency improvements to include material durability, thermal cycling stability, leakage resistance, and long-term operational reliability. Future studies are encouraged to establish standardized testing protocols while integrating electrical performance, thermal management, exergy efficiency, economic analysis, and reliability evaluation into a unified assessment framework. Such an approach will facilitate objective comparison among different PV/T-PCM and PV-TEG-PCM systems and provide more practical guidance for large-scale engineering deployment.

From the perspective of long-term reliability, the performance of microencapsulated PCMs is influenced by multiple factors, including core-shell material compatibility, fabrication methods, and operating conditions [44]. However, long-term outdoor validation and lifetime assessment of PCM-based thermal management systems remain limited [44]. Moreover, standardized testing protocols and durability evaluation criteria have yet to be established. The degradation mechanisms of PCMs under repeated thermal cycling-particularly crack formation, void generation, and interfacial debonding-also require more systematic investigation to support reliable long-term applications [44,50].

5. Challenges for post-sunset power generation in photovoltaics

5.1. High dependence of PV power output on solar irradiance

The operational performance of photovoltaic systems is strongly influenced by environmental conditions, particularly solar irradiance and module temperature [113], while the corresponding electrical response can be described using the photovoltaic equivalent-circuit model [124]. Fig. 9 provides a multiscale overview of this dependence, ranging from module-level energy conversion to system deployment and electrical characteristics. Fig. 9a summarizes the energy-balance pathways of a PV module, showing that only part of the

Table 4
Comparison of representative PV-TEG-PCM systems.

Study	PV	TEG	PCM	PCM size	Configuration	Conditions	Validation	Electrical gain	TEG output	Thermal result	Duration	Limitation
[116]	5 W mono-Si	22 Bi_2Te_3 TEC1-12706	Microencapsulated n-octadecane	5 mm	TEGs on PV rear; PCM on TEG cold side	Indoor chamber; 500 W lamp; 20 °C initial	3-D FLUENT validated by prototype	≈9.5% higher cumulative PV output at 1000 W m^{-2}over 1.5 h	Peak ≈0.55 W m^{-2}	Peak ΔT ≈1.6 K	≈5 h	Very low ΔT/ output; melted PCM may add resistance
[99]	-	-	Encapsulated pellets + metal powder	-	FPC-PV-PCM-TEG with pulsing heat pipes/cartridges	Outdoor horizontal test; Ankara summer day	Quarter-scale prototype vs model	-	-	Max total exergy ≈60 W m^{-2} at 800 W m^{-2}	≈24 h	Extended architecture not directly comparable
[117]	40 × 40 mm Si	Bi_2Te_3; 128 pairs	RT28HC paraffin	5-25 mm	PV-TEG-PCM vs PV-PCM-TEG	3-D transient FE; 298.15 K initial	Submodels validated; no full prototype	Average PV efficiency increased with PCM height	0.0156 → 0.0292 W peak	Max PV-temperature reduction ≈19.22 K	300 s	Only 5 min; no integrated experiment
[118]	Mono-Si	3 mm equivalent layer	RT35, RT42, n-tetracosane	20 mm	PV-graphene-TEG-PCM	3-D FLUENT; summer solstice, 06:00-19:00	Temperature checked against Darkwa et al.	Peak PV-efficiency gain +0.87 to +1.04 pp	-	Peak reduction 12.04-14.33 °C	13 h	Numerical; indirect validation; homogenized TEG
[119]	Mono-Si YLE40	20 BiTe TEG1-PB-12611-6.0	RT55 paraffin	4 kg	PV/T-PCM-TEG with serpentine copper tube	Outdoor, Isparta; ≈10:00–19:40	Cases compared with same-day standard PV	Total power +10.29% to +14.22%	Peak 1.43-1.82 W	PV peak ≈71–75 °C vs 85–87 °C	≈9 h 40 min	Cases tested on different days
[120]	Four commercial PV cells	-	-	-	PV-TEG-PCM with night-sky radiative cooling	Full day-night cycle	Design, optimization and prototype test	Daytime PV and nighttime generation demonstrated	-	PCM limited overheating and stored heat	24 h	Key PV/PCM/TEG parameters NR
[121]	Mono-Si GHGN-D50WK	18 Bi_2Te_3 SP1848-27145	50% paraffin + 50% petroleum jelly	15 mm	PV, PV/T and dual-loop PCM-TEG variants	Outdoor Wuhan; 11 Mar 2024, 10:00-16:30	Five systems tested simultaneously	Best mean η_{el} = 12.97%; +45.08% vs PV	Peak ≈0.67 W; max ΔT ≈9.6 °C	Max PV reduction 10.8 °C vs PV	6.5 h	Compressor power excluded; single-day test
[122]	Crystalline-Si cell	Commercial Bi_2Te_3	ZJ-PCM-A-30/-38	10 mm baseline	Heat-accumulating PV-TEG-PCM	Indoor xenon lamp; 1000 W m^{-2}	Comparative experiments	PV efficiency 12.5%→14.2% (+1.7 pp)	≈0.112 mW at 2.8 Ω	PV temperature −2.4 °C at 1.5 h	5 h	Very small TEG output; short constant-irradiance tests

Note: pp = percentage points. Values are not directly comparable because operating conditions and system boundaries differ.

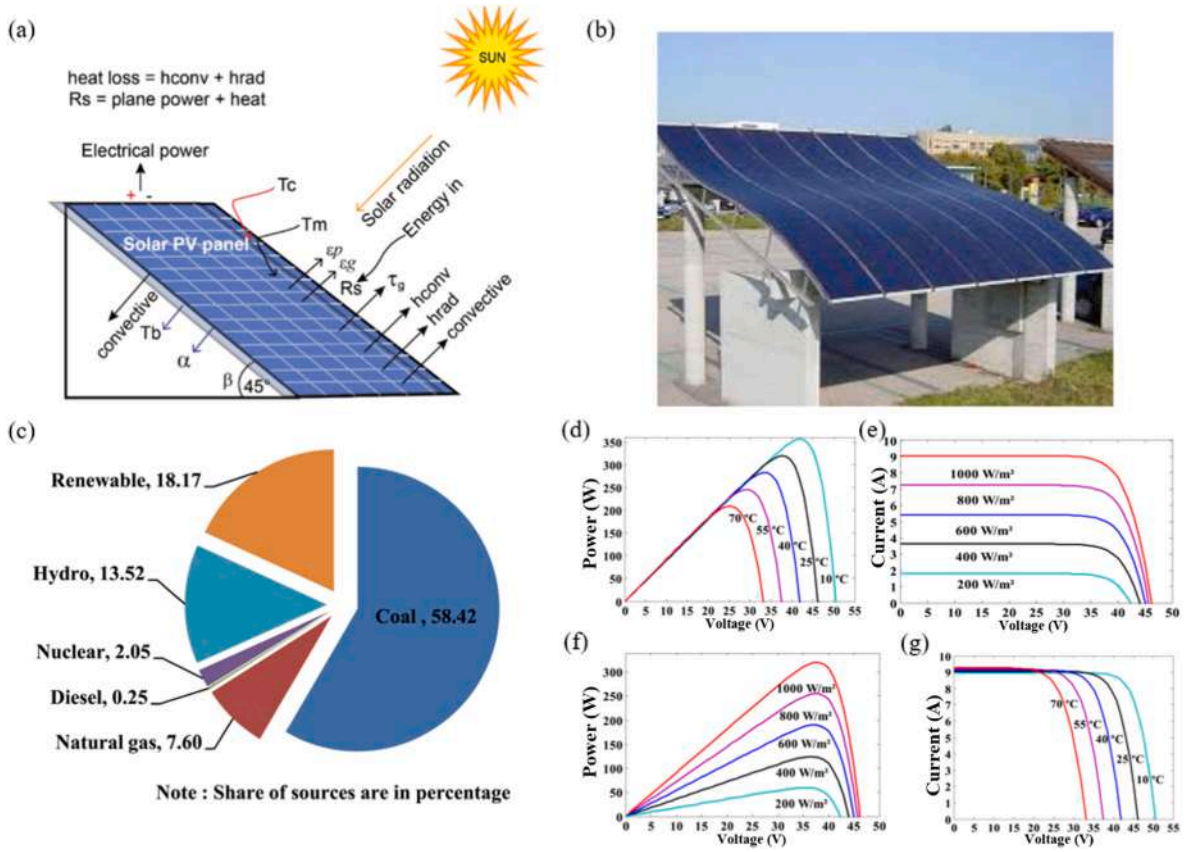


Fig. 9. (a) Adapted from Ref. [125], which was based on earlier module energy-balance models. (b) Flexible thin-film solar panels [126]. (c) Share of sources are in percentage. (d) P-V characteristics, varying temperature at constant solar radiation. (e) I-V characteristics, varying irradiance at constant temperature. (f) P-V characteristics, varying solar radiation at constant temperature. (g) I-V characteristics, varying temperature at constant solar radiation [127].

incident solar energy is converted into electricity, while the remainder is dissipated through thermal and optical losses. Fig. 9b and c further illustrate the diversity and large-scale deployment of photovoltaic technologies, including flexible thin-film modules and their contribution to national power-generation portfolios. However, regardless of module configuration or deployment scale, installed capacity does not guarantee stable power output, because instantaneous PV generation remains highly sensitive to meteorological conditions. The temperature-dependent P-V characteristics shown in Fig. 9d indicate that lower module temperatures generally increase the maximum output power. In comparison, solar irradiance has a more direct effect on photocurrent and therefore on instantaneous power generation. At a constant temperature of 25 °C, increasing irradiance from 200 to 1000 W/m^2 produces an approximately proportional increase in short-circuit current, while the open-circuit voltage changes only modestly (Fig. 9e) [124]. The corresponding P-V curves exhibit a substantial increase in maximum power as irradiance rises (Fig. 9f) [124]. The I-V characteristics shown in Fig. 9g further indicate that lower module temperatures also increase the operating voltage. These results demonstrate that, although temperature modifies PV voltage and efficiency, solar irradiance remains the primary environmental driver of instantaneous photovoltaic power output.

The irradiance-dependent electrical trends shown in Fig. 9e and f can be further explained by semiconductor physics and the photovoltaic equivalent-circuit model. In particular, the short-circuit current exhibits a linear dependence on irradiance, reflecting the proportionality between photogenerated current and incident photon flux [128]. Conversely, the open-circuit voltage increases more gradually (approximately logarithmically) with increasing irradiance [128,129]. The resulting displacement of the maximum power point, predominantly driven by current enhancement, underscores the module's pronounced sensitivity to instantaneous solar irradiance [124,128].

A more nuanced non-linear relationship exists between irradiance and conversion efficiency. Efficiency generally increases as irradiance rises from low-light conditions, although its variation at moderate and high irradiance depends on cell technology, operating temperature, and device characteristics [128]. This nonlinear behavior arises primarily from the competing effects of series and shunt resistances. At low irradiance, shunt-related leakage can account for a larger proportion of the photogenerated current, thereby exerting a more pronounced influence on conversion efficiency [128]. Conversely, series resistance losses intensify quadratically at high irradiance due to elevated current. This dynamic balance explains the non-linear efficiency response.

In summary, irradiance-dependent power output represents a cornerstone of photovoltaic science. Its manifestations include an approximately linear current response, a logarithmic voltage response, and a nonlinear efficiency variation influenced by series and shunt resistances. A profound understanding of these principles is indispensable for robust energy yield modeling, technology selection for specific geographical contexts, and advanced maximum power point tracking (MPPT) algorithm development to optimize field performance. Most importantly, when solar irradiance falls to zero after sunset, direct photovoltaic generation ceases, making energy storage, grid support, or complementary generation indispensable for continuous nighttime power supply.

5.2. *Impact of night-time power interruption on energy system stability*

Against the backdrop of the global clean-energy transition, maintaining reliable electricity supply during nighttime has become an

increasingly important challenge. As photovoltaic generation ceases after sunset while climate change and extreme weather continue to increase nighttime cooling demand, the mismatch between renewable-energy supply and electricity demand is expected to intensify, making nighttime resilience a critical consideration for future low-carbon energy systems.

Nighttime electricity demand is highly temperature-sensitive and becomes increasingly vulnerable during compound hot-day/tropical-night events. Rising nighttime temperatures substantially increase cooling loads while simultaneously reducing the operational flexibility of renewable-dominated power systems. Combined with the intrinsic intermittency of solar and wind resources, these factors increase the risks of voltage instability, frequency deviations, localized overloading, and even large-scale power interruptions under extreme weather conditions [130].

Fig. 10 illustrates these challenges from both the energy-supply and climatic perspectives. Fig. 10a presents the electrical characteristics of a representative photovoltaic module, while Fig. 10b shows a green data-center architecture integrating photovoltaics, batteries, hydrogen production, fuel cells, heat pumps, and district-heating networks to enhance operational resilience [127]. Fig. 10c illustrates the voltage characteristics of the battery-storage subsystem, highlighting its role in supporting nighttime electricity supply. On the demand side, Fig. 10d classifies different types of heat extremes in Beijing, and Fig. 10e compares the corresponding diurnal variations in air temperature and relative humidity, showing that compound hot-day/tropical-night events maintain elevated nighttime temperatures and prolong cooling demand. Figs. 10f and g further present the associated 500-hPa geopotential-height anomaly and upper-level zonal-wind pattern, indicating persistent atmospheric circulation conditions that intensify nighttime heat stress and electricity demand [130].

Mitigating nighttime power interruptions requires coordinated deployment of energy storage, demand-side management, diversified renewable-energy resources, and smart-grid technologies. Battery and pumped-hydro storage can shift surplus daytime electricity to nighttime operation, while integrated multi-energy systems combining photovoltaics, batteries, hydrogen, fuel cells, and heat pumps provide greater operational flexibility under renewable-energy shortages [127]. Nevertheless, conventional storage technologies increase both system cost and operational complexity. These limitations have stimulated growing interest in complementary technologies capable of extending renewable-energy utilization beyond daylight hours. Among them, PV-TEG-PCM systems can utilize PCM to store part of the daytime thermal energy and sustain a temperature difference across thermoelectric generators after sunset, thereby providing supplementary electricity generation and enhancing the resilience of future low-carbon energy systems.

5.3. *Limitations of energy storage technologies for night-time supply*

To illustrate these system-level considerations, Fig. 11 brings together three representative studies conducted at different application scales: an integrated energy supply–storage system, an off-grid rural household energy system, and a photovoltaic–energy-storage–grid system for electric-bus charging. Rather than representing a single continuous case, these examples collectively illustrate how system architecture, energy-carrier matching, renewable-output variability, and operational coordination influence the use of stored solar energy beyond daylight hours.

On the integrated-energy-system scale, Figs. 11a and b present the case-study configuration and the generic structure of an energy supply-storage system considered in year-round operational planning [131]. In the case study, photovoltaic generation is coupled with a water electrolyzer, a metal-hydride hydrogen storage tank, and a hydrogen-fueled combined heat and power unit, enabling the coordinated management of electricity, heat, and hydrogen over annual time scales [131]. These panels therefore highlight the operational complexity created by the interaction of multiple generation, conversion, and storage devices rather than identifying a single storage-related bottleneck.

At the household scale, Figs. 11c and d show the operating principle and physical implementation of an off-grid photovoltaic-based thermal-energy-storage system designed for rural households in remote cold regions [132]. In this configuration, nighttime space-heating demand is primarily met through thermal storage, whereas household electrical loads are supported by a compact battery. This energy-grade matching strategy avoids storing all photovoltaic output in electrical form and illustrates how direct PV-to-heat conversion and appropriately sized electrical storage can improve local solar-energy utilization while

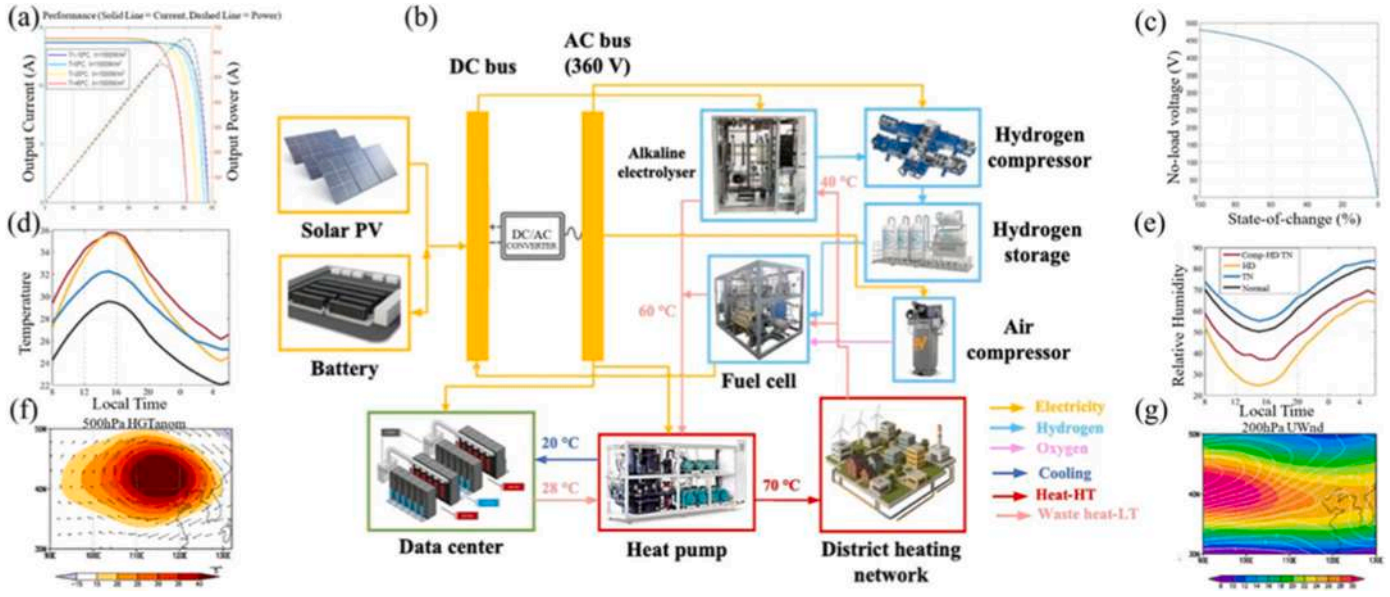


Fig. 10. (a). Characteristics of Telesun TP7F78M-H-590 datasheets in a solar panel. (b). Schematic diagram of the designed green data center. (c). Voltage profile of the battery [127]. (d). Characteristics of heat extremes in Beijing. (e). Diurnal cycle of surface air temperature and surface relative humidity during the pure-HD (yellow), pure-TN (blue), comp-HD/TN (red) and normal (black) days. (f). Geopotential height anomaly at 500 hPa during comp-HD/TN (unit: gpm). (g) Climatological zonal wind (shading; unit: m/s) and the zonal wind during comp-HD/TN (contour interval: 2 m/s) [130].

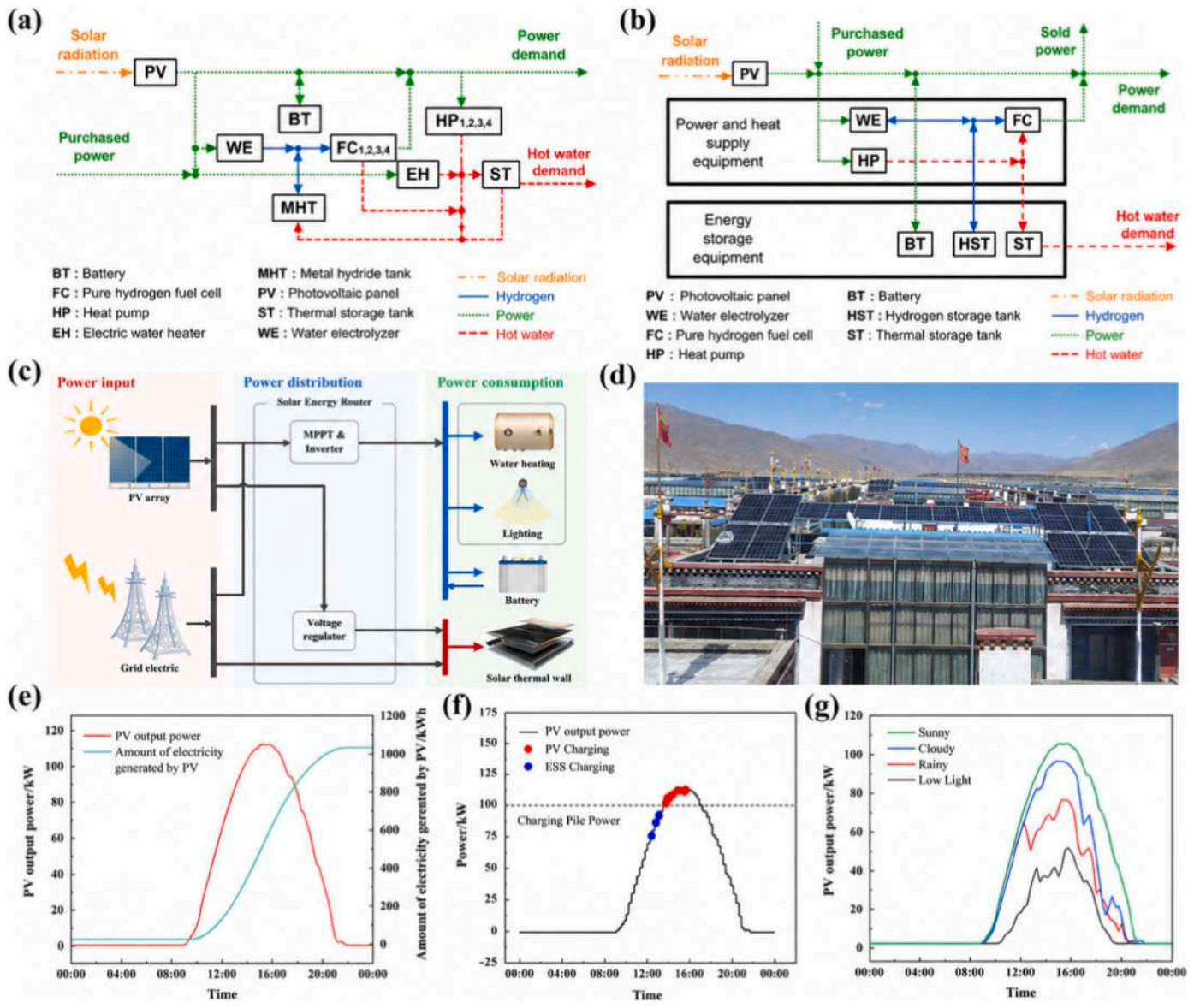


Fig. 11. (a). Configuration of energy supply-storage system in case study. (b). Generic structure of energy + supply-storage system [131]. (c). System operation principle. (d). Case studies [132]. (e). PV output power curve. (f). Time distribution of electricity supply mode and PV output power curve. (g). PV output-power curves under different weather conditions [133].

limiting battery requirements [132].

At the transport-system scale, Fig. 11e–g presents the time-varying photovoltaic output, the temporal alposition of different electricity-supply modes, and photovoltaic-output profiles under different weather conditions in an electric-bus charging application [133]. The variations shown in these panels affect the relative contributions of photovoltaic generation, the energy-storage system, and the power grid. They therefore demonstrate the importance of coordinating charging schedules, storage capacity, and grid support with the availability of photovoltaic power [133], rather than directly indicating electrolyzer inefficiency or unavoidable storage oversizing.

Taken together, the three studies indicate that the effectiveness of nighttime solar-energy utilization is not determined by storage capacity alone. It also depends on the selected energy carriers, conversion pathways, weather-dependent generation, storage sizing, load characteristics, and operational scheduling. Energy storage remains essential for shifting solar energy beyond daylight hours, but its technical and economic value is strongly dependent on the system configuration and application. These case studies do not, by themselves, establish that storage-based nighttime supply is universally impractical; instead, they emphasize the need for application-specific system integration and coordinated operation.

Within this context, PCM-assisted thermoelectric generation should be positioned as a supplementary energy-recovery pathway rather than as a replacement for electrochemical or grid-scale energy storage. Its potential contribution lies in recovering thermal energy that would otherwise be dissipated and extending electricity generation after sunset through relatively simple solid-state conversion devices. The practical significance of this contribution depends on the achievable nighttime power output, generation duration, stored thermal energy, and sustained temperature difference across the thermoelectric generator, as examined in the following section.

5.4. PCM-assisted post-sunset energy harvesting in PV-TEG systems

Unlike conventional PV systems, which cease electricity generation immediately after sunset, PV-TEG-PCM systems seek to extend energy utilization by storing surplus daytime thermal energy within the PCM

and subsequently converting part of this stored heat into electricity through thermoelectric generation. This approach complements rather than replaces conventional electrical energy-storage technologies, as the available nighttime electrical output is ultimately limited by the finite thermal energy stored during daytime operation. Consequently, the practical capability of post-sunset electricity generation is governed by the coupled effects of PCM thermal storage, heat-transfer processes, ambient conditions, and TEG conversion efficiency rather than by the simple integration of PCM and TEG components.

Representative studies demonstrate that PCM can effectively shift thermal energy from daytime to nighttime operation. A capric-acid PCM with a melting temperature of 30.1 °C and a latent heat of 158 kJ kg^{-1} achieved a heat-storage capacity of 12,050.63 kJ, representing a 20.6% improvement over a conventional PV/T system while sustaining latent-heat release for approximately 400 min [99]. Similarly, PCM-assisted systems achieved daily thermal storage of 33.86-34.97 MJ day^{-1}, corresponding to 52.27-53.25% of the converted thermal energy, with heat-utilization ratios of 89.05-91.91% and heat-release efficiencies of 82.86-88.41%, indicating that more than half of the harvested daytime thermal energy can be temporally shifted through latent-heat storage [132]. These results confirm that PCM substantially enhances temporal thermal-energy utilization, thereby providing the thermal driving force required for post-sunset thermoelectric generation.

The effectiveness of nighttime thermoelectric generation further depends on PCM design and thermal-boundary conditions. PCM melting temperatures ranging from 28 to 57 °C have been reported to provide sustained temperature differences between the PCM and the ambient environment under different climatic conditions [96,132]. Under optimized conditions, a PCM with a phase-transition temperature of approximately 38 °C and a latent heat of 215 J g^{-1} enabled a TEG efficiency of 0.42%, sustained nighttime voltages of approximately 0.20 V, and increased the overall system efficiency by 1.2-2.61% [96,120]. Increasing the PCM thickness from 10 mm to 30 mm further increased the nighttime voltage from approximately 0.16 V to 0.20 V, whereas excessively high phase-transition temperatures reduced latent-heat utilization because complete melting could not be achieved during daytime operation [120]. These observations indicate that nighttime electrical performance is determined by the combined optimization of PCM thermophysical properties, PCM geometry, and thermal management rather than by thermal-storage capacity alone.

Beyond hot-side thermal storage, the achievable TEG output also depends critically on cold-side temperature regulation. Ambient conditions and cold-side heat dissipation further determine the effective temperature difference across the TEG. The thermal driving force is produced not by the absolute temperature of the PCM alone, but by the difference between the PCM-controlled hot side and the ambient- or radiatively cooled cold side. Favorable low-temperature environments can significantly increase this difference. In one reported climatic condition, the ambient temperature decreased to −21.3 °C, thereby enhancing the potential temperature gradient between the PCM and the surroundings [132]. Radiative cooling can similarly reduce the cold-side temperature during clear nights. A radiative-cooling-enhanced RC-TEG-PCM-PV configuration produced a nighttime voltage of approximately 240 mV during summer conditions, associated with a maximum temperature difference of approximately 5.4 K. These results show that cold-side thermal management, including heat-sink design, natural or forced convection, and radiative heat rejection, is as important as PCM storage capacity in sustaining nighttime TEG operation.

Recent studies further highlight both the potential and the current limitations of PV-TEG-PCM systems. Hou et al. [134] developed a tri-mode photothermal-phase-change-radiative-cooling system capable of alternating between daytime photovoltaic operation and nighttime thermoelectric generation, thereby demonstrating the feasibility of extending energy harvesting beyond daylight hours. Zhang and Xuan [135] showed that the performance of PV-TEG hybrid systems is fundamentally governed by the available temperature difference and effective thermal management under fluctuating solar radiation. Benghanem et al. [136] further demonstrated that incorporating a commercial Bi_2Te_3 thermoelectric module does not necessarily improve the net electrical output of a photovoltaic system when the available temperature difference is insufficient. Collectively, these studies indicate that practical PV-TEG-PCM performance relies on the coordinated optimization of thermal storage, heat transfer, cold-side heat rejection, and thermoelectric conversion, rather than on simply integrating additional functional components.

Overall, the reviewed literature consistently demonstrates that PCM effectively extends the temporal utilization of solar-derived thermal energy and enables supplementary low-power post-sunset thermoelectric energy harvesting.

5.5. Novel energy compensation pathways driven by night-time demand

Nighttime electricity demand exhibits distinct characteristics from daytime demand because heating, cooling, and other continuous loads may remain significant after sunset. Meanwhile, fluctuations in renewable generation increase the uncertainty of residual load balancing, requiring coordinated operation of controllable generation, energy storage, grid interaction, and demand-side resources. To address these challenges, advanced energy-management strategies have been developed, including scheduling optimization, multi-stage dispatch, decomposition-based planning, and predictive control, as summarized in Fig. 12.

Fig. 12a–b compare centralized and decentralized scheduling strategies for a representative day in February [137]. The results demonstrate that different scheduling architectures influence the response of flexible devices, such as heat pumps and combined heat and power units, to residual-load variations and consequently affect electricity exchange between buildings and the grid [137].

However, uncertainties in renewable generation and load forecasts may reduce the effectiveness of predetermined schedules. Therefore, two-stage day-ahead and intraday dispatch strategies have been developed to incorporate updated forecasting information. Fig. 12c–d illustrate an integrated electricity–heating system and the corresponding day-ahead/intraday forecasts of wind power, photovoltaic generation, electrical load, and thermal load [138]. By correcting initial schedules with updated information, these approaches improve coordination among renewable generation, storage, conversion devices, and demand-response resources [138].

For large-scale energy systems involving numerous devices and intertemporal constraints, optimization complexity becomes a major challenge. Decomposition-based approaches, such as Dantzig-Wolfe reformulation and column generation, address this issue by dividing large optimization problems into smaller subproblems while preserving critical constraints associated with energy storage and unit operation [140–142]. These methods are particularly valuable for long-term planning problems with strong seasonal variations and complex energy interactions.

For short-term operation under uncertainty, model predictive control (MPC) provides an adaptive solution through continuous forecasting and rolling optimization. Figs. 12e and f present a stochastic MPC framework for a concentrated solar power–photovoltaic hybrid system with thermal energy storage, showing the coordination of receiver operation, thermal storage charging/discharging, photovoltaic-assisted heating, and grid interaction under changing conditions [139].

Overall, nighttime energy management requires complementary strategies at different temporal and spatial scales. Scheduling methods improve coordination among flexible loads [137], multi-stage dispatch enhances adaptability to forecast updates [138], decomposition techniques improve computational efficiency for large-scale optimization [140–142], and predictive control enables real-time adjustment under uncertainty [139]. The effectiveness of these approaches depends on system scale, forecast accuracy, storage capacity, tariff structures, and

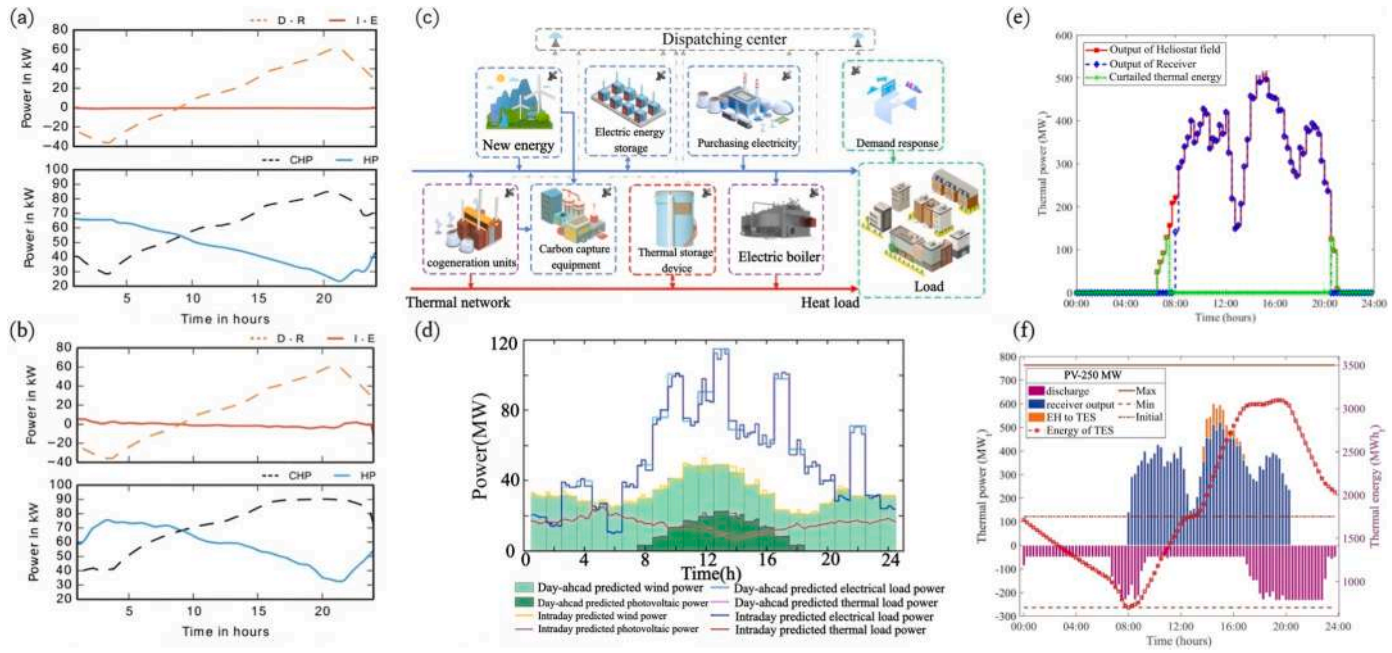


Fig. 12. (a), (b). Schedule profiles for a random day in February using the centralized and decentralized approach [137]. (c). Operation structure framework of integrated electricity and heating systems. (d). Day-ahead and intraday forecasted values for wind power, photovoltaic power, and electrical heat load [138]. (e). Thermal energy flows in receiver startup process. (f). Thermal storage, charging/discharging thermal power of TES system with PV 250 MW [139].

computational requirements. Nevertheless, most existing strategies focus primarily on electrical or thermal energy scheduling, while extending renewable energy availability beyond daylight hours remains challenging. This limitation motivates the exploration of PV-TEG-PCM systems, where PCM thermal storage and thermoelectric conversion provide an additional pathway for utilizing residual solar energy after sunset.

6. Thermoelectric generation (TEG)

6.1. Principle of thermoelectric power generation and its potential in PV systems

Thermoelectric generators directly convert a temperature difference into electricity through the Seebeck effect. Their solid-state structure, absence of moving components, and relatively low maintenance requirements make them attractive for recovering low-grade heat. In photovoltaic-thermoelectric systems, a TEG is generally attached to the rear surface of the PV module or incorporated into a PV/T thermal-management structure. The heat transferred from the PV module establishes the TEG hot-side temperature, whereas a heat sink, cooling fluid, ambient air, or another heat-rejection device controls the cold-side temperature. Therefore, the electrical contribution of the TEG depends not simply on the amount of heat absorbed from the PV module, but on the temperature difference that can be maintained across the thermoelectric module.

To better clarify the design basis of PV-TEG and PV-TEG-PCM systems, the governing thermoelectric relations should be explicitly considered. For a TEG module operating between a hot side temperature T_h and a cold side temperature T_c, the temperature difference is expressed as

$$\Delta T = T_h - T_c \tag{6-1}$$

and the open-circuit Seebeck voltage is

$$V_{oc} = \alpha \Delta T \tag{6-2}$$

where α is the effective Seebeck coefficient of the thermoelectric module. When an external load resistance R_L is connected, the output current can be written as

$$I = \frac{\alpha \Delta T}{R_i + R_L} \tag{6-3}$$

where R_i is the internal electrical resistance of the TEG. The electrical output power is therefore

$$P_{TEG} = I^2 R_L = \frac{(\alpha \Delta T)^2 R_L}{(R_i + R_L)^2} \tag{6-4}$$

and the maximum output power is obtained when $R_L = R_i$:

$$P_{max} = \frac{(\alpha \Delta T)^2}{4R_i} \tag{6-5}$$

These relations indicate that TEG output is highly sensitive to the maintained temperature difference across the module. Therefore, in PV-TEG and PV-TEG-PCM systems, simply increasing heat absorption from the PV back surface is insufficient; the system must also maintain a sufficiently low cold-side temperature and minimize thermal losses.

The thermoelectric material performance is commonly evaluated using the dimensionless figure of merit:

$$ZT = \frac{\alpha^2 \sigma T}{k} \tag{6-6}$$

where σ is electrical conductivity, k is thermal conductivity, and T is the average absolute temperature of the TEG. A higher ZT generally corresponds to improved thermoelectric conversion capability, but the practical efficiency of a TEG module remains strongly constrained by the achievable ΔT, contact resistance, and heat-rejection condition.

The conversion efficiency of a TEG can be expressed as

$$\eta_{TEG} = \frac{P_{TEG}}{Q_h} \tag{6-7}$$

where Q_h is the heat absorbed at the hot side. For a simplified TEG model, the hot-side and cold-side heat flows can be described as

$$Q_h = \alpha I T_h - \frac{1}{2} I^2 R_i + K\Delta T \tag{6-8}$$

$$Q_c = \alpha I T_c + \frac{1}{2} I^2 R_i + K\Delta T \tag{6-9}$$

where K is the thermal conductance of the TEG. These equations show that heat transport in a TEG includes Peltier heat, Joule heat, and conductive heat leakage. Therefore, the net electrical output depends not only on the thermal energy supplied by the PV module or PCM, but also on how effectively the cold side rejects heat to the surrounding environment.

The preceding relations indicate that the electrical and thermal performance of a PV-TEG-PCM system is governed by the coupled effects of heat absorption, cold-side heat rejection, and the resulting temperature difference across the TEG. These relationships are experimentally illustrated in Fig. 13, which compares five PV/T-TEG-PCM configurations [121]. Fig. 13a–b presents the experimental platform and the corresponding structural arrangements, establishing the different heat-transfer and cooling pathways. Their thermal responses are subsequently compared in Fig. 13c–f: Fig. 13c shows the PCM temperature evolution, Fig. 13d presents the TEG hot-and cold-side temperatures and the resulting (ΔT), and Fig. 13e–f report the working-fluid outlet temperature and PV-cell temperature, respectively. These thermal characteristics are reflected in the system-level performance shown in Fig. 13g–h, which present the thermal output and efficiency and the electrical generation and efficiency, respectively. Overall, the actively cooled PV/T-PCM-TEG configuration achieved the most favorable combined performance because enhanced heat rejection reduced the PV temperature while maintaining a larger and more stable (ΔT) across the TEG [121].

In practical systems, thermal contact resistance at the PV-TEG and TEG-heat-sink interfaces reduces the effective temperature difference across the thermoelectric legs. Thermal-interface materials, adequate clamping pressure, uniform contact, and effective cold-side cooling can therefore improve both PV and TEG performance [143]. Interface quality and heat rejection have consequently been identified as key determinants of PV-TEG system performance [144].

PV-TEG integration nevertheless involves a fundamental trade-off. Improved heat removal can reduce PV temperature, whereas the TEG and its interfaces may introduce additional thermal resistance. Moreover, active cooling can increase (ΔT) but also consume parasitic power. System optimization must therefore consider PV temperature, TEG output, interface resistance, cooling capacity, parasitic energy consumption, and electrical load matching simultaneously. Although advances in thermoelectric materials, module geometry, thermal interfaces, heat sinks, and control strategies have improved PV-TEG performance, practical deployment remains limited by the moderate temperature differences available in PV systems, relatively low conversion efficiency, high material and integration costs, and added system complexity. Thermoelectric generators should therefore be regarded mainly as supplementary devices for PV thermal management and waste-heat recovery rather than as independent sources of substantial or continuous electrical power.

6.2. *Algorithm approach for reconstruction and optimization of TEG system*

Non-uniform temperature distributions cause electrical mismatch among thermoelectric generator (TEG) modules, thereby reducing array

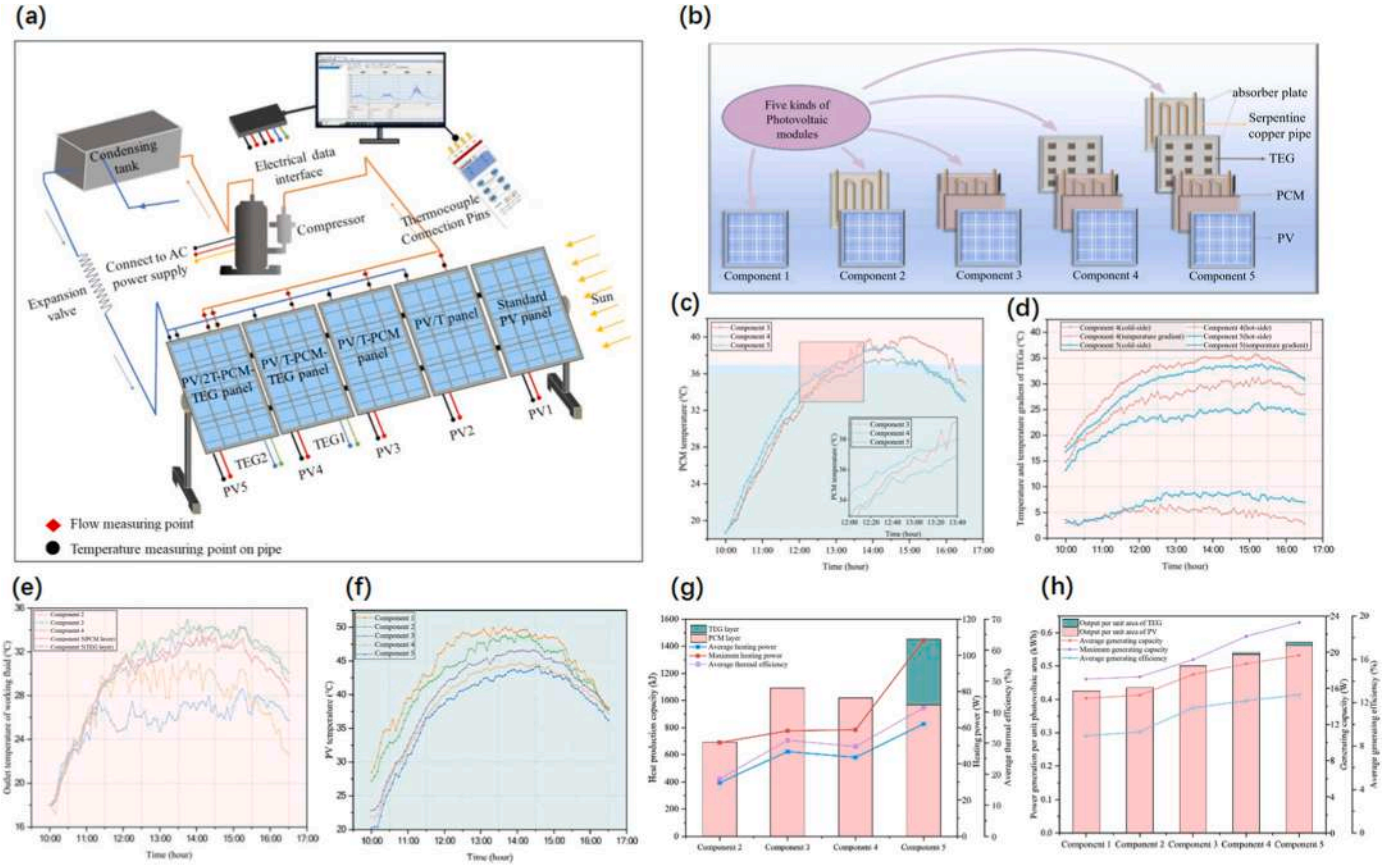


Fig. 13. (a) Experimental layout of five PV/T-TEG configurations; (b) schematics of the five configurations; (c) PCM temperature; (d) TEG hot- and cold-side temperatures and temperature difference; (e) working-fluid outlet temperature; (f) PV-cell temperature; (g) thermal output and efficiency; and (h) electrical generation and efficiency [121].

power and producing multiple operating points. Dynamic electrical reconfiguration provides an effective means of mitigating this mismatch by modifying the series-parallel connections or switching states of individual modules according to the instantaneous thermal field. Because the resulting optimization problem is nonlinear, discrete, and highly dependent on operating conditions, metaheuristic algorithms have been widely adopted to identify high-power array configurations.

Representative approaches include the improved immune genetic algorithm (IIGA), jellyfish search algorithm, energy valley optimizer, and RIME optimizer [145–148]. These algorithms search for module arrangements or switching combinations that maximize array output power while avoiding local optima. Their performance is generally evaluated in terms of maximum power, convergence speed, solution stability, and computational cost. Among them, IIGA improves the balance between global exploration and local exploitation through population-diversity preservation and elite-selection mechanisms. Tests on 9 × 9 and 15 × 15 TEG arrays under non-uniform temperature distributions showed that it could identify high-power configurations with improved convergence and stability [145]. Other studies evaluated alternative optimizers under multiple symmetric and asymmetric temperature patterns, using numerical simulation or hardware-in-the-loop validation [146–148]. Collectively, these studies demonstrate that algorithmic reconfiguration can reduce electrical mismatch, although its effectiveness depends on array scale, thermal-pattern complexity, switching constraints, and real-time computational requirements.

Beyond TEG-array reconfiguration, algorithm-assisted optimization has also been applied at the integrated PV-TEG system level, as summarized in Fig. 14. Fig. 14a presents an experimental PV-TEG platform used to evaluate the effects of cooling and photovoltaic enhancement strategies [149]. Fig. 14b shows a grid-connected PV configuration incorporating an artificial-bee-colony perturb-and-observe MPPT algorithm, a DC-DC boost converter, and a grid-side inverter [146]. The algorithm adjusts the converter duty ratio under partial shading to improve impedance matching and power extraction. The resulting system-level responses are shown in Fig. 14c–g. Fig. 14c compares PV-TEG power output and relative enhancement under different cooling and optical-concentration strategies, showing that the combined benefit varies with irradiance [149]. Fig. 14d illustrates the structure of a PCM-integrated PV/T panel containing labyrinth channels, a serpentine heat exchanger, and paraffin-based thermal storage [119], while Fig. 14e presents the corresponding power enhancement obtained using different photovoltaic strategies [119].

The studies summarized above involve two related but distinct optimization levels. TEG-array reconfiguration directly reduces module-level electrical mismatch caused by non-uniform temperature distributions, whereas MPPT, cooling, optical concentration, and PCM integration optimize energy conversion and thermal management at the system level. Combining these approaches may improve overall energy harvesting, but practical implementation requires coordinated control of switching losses, sensor accuracy, parasitic cooling consumption,

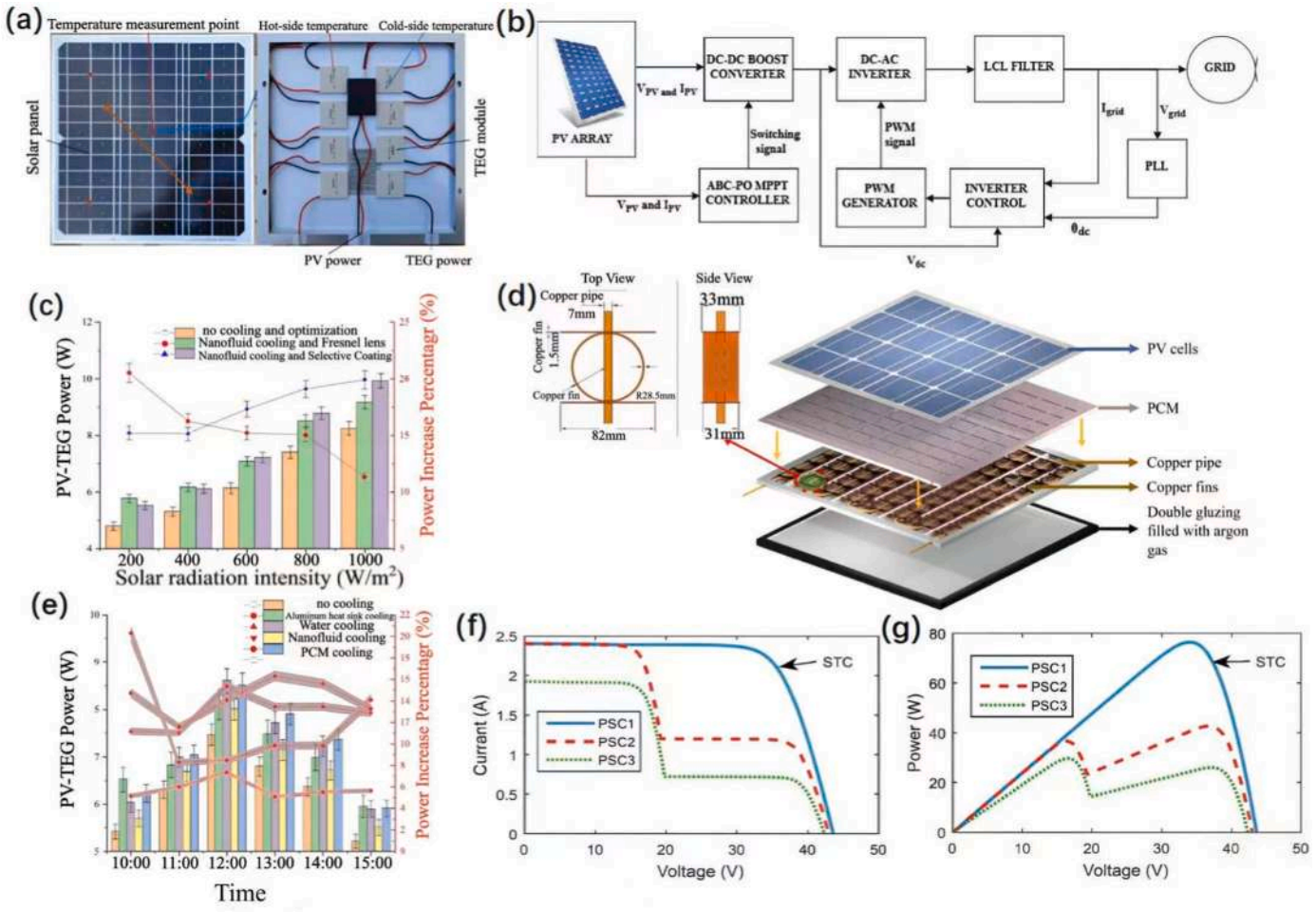


Fig. 14. (a) Experimental configuration of the PV-TEG system [149]. (b) General configuration of the grid-connected PV system employing ABC-PO MPPT [146]. (c) PV-TEG power output and power enhancement under different cooling and photovoltaic optimization strategies [149]. (d) Structural design of the PCM-filled PV/T panel [119]. (e) PV-TEG output under different photovoltaic enhancement strategies [119]. (f) I-V and (g) P-V characteristics under different partial-shading patterns [146].

algorithm execution time, and hardware complexity.

6.3. Thermal management and material innovation of thermoelectric generation

Thermoelectric devices convert thermal energy directly into electrical energy through the Seebeck effect, a process that depends critically on maintaining a sufficient temperature gradient across the thermoelectric legs. Efficient thermal management is therefore essential not only for device performance stability but also for maximizing the thermoelectric figure of merit (*ZT*). Material innovation has become a key driver in this domain, where novel thermoelectric structures refine thermal conductivity control and enhance thermoelectric conversion efficiency.

Recent material advances, including nanostructured, organic, and hybrid thermoelectric materials, have achieved higher *ZT* values through precise microstructural engineering [150–152]. Nanocomposites, for example, suppress lattice thermal conductivity while preserving electrical conductivity, enabling efficient heat capture and dissipation in thermal management applications. Additive manufacturing and thin-film deposition techniques have further reduced production costs, accelerating the transition of material innovations from laboratory development to commercial deployment. These material-level improvements directly support the integration of thermoelectric devices into hybrid renewable-energy systems, including solar PV/T-PCM-TEG configurations.

Gürbüz et al. [119] experimentally investigated a PV/T-PCM-TEG system incorporating RT55 paraffin, Al_2O_3 additives, and copper fins to demonstrate the combined benefits of material and structural optimization. As shown in Fig. 15a, the system uses city tap water circulated via a brushless pump (≈1 W) to manage temperature gradients across the thermoelectric modules. Three configurations were compared: Case 1 (base PCM-filled PV/T panel), Case 2 (10 wt% Al_2O_3-enhanced PCM), and Case 3 (Al_2O_3-enhanced PCM with copper fins). The technical configurations are illustrated in Fig. 15b. Fig. 15c quantifies the power output enhancements relative to standard PV panels, showing that Case 3 achieved the highest and most stable gains-initially 11.6% and 6.5% near day-end-owing to combined material and structural refinements. Diurnal power variations (Fig. 15d) exhibit a consistent decline from 10:00 to 12:00, followed by gradual reduction until 18:00, attributable

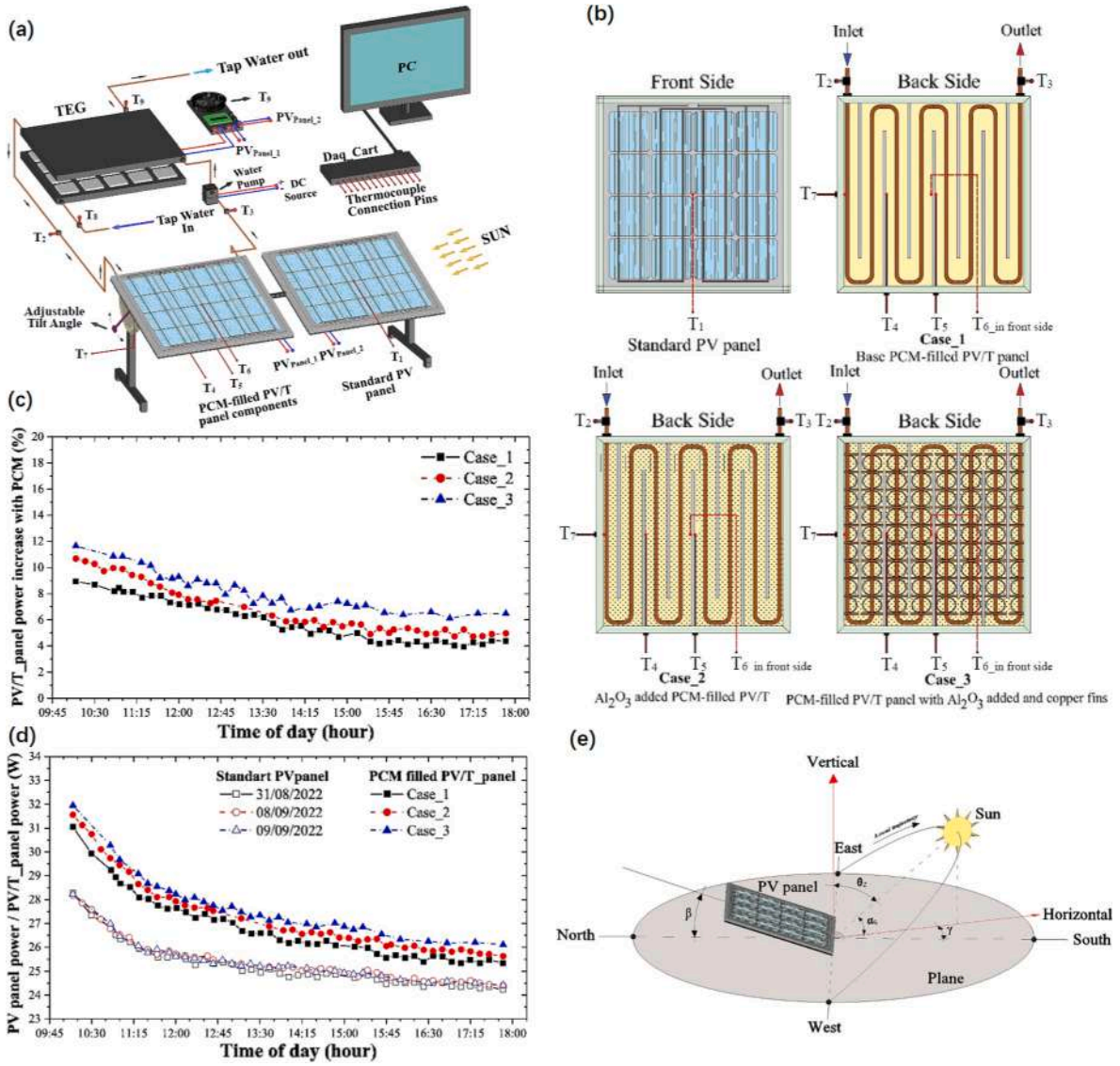


Fig. 15. (a) Experimental layout of the solar PV/T-PCM-TEG hybrid system; (b) technical drawings of the standard PV panel and three PCM-filled PV/T configurations; (c) relative power enhancement of the PV/T configurations; (d) diurnal power variation of the different configurations; and (e) angular adjustment scheme of the manual solar-tracking module [119].

to rising surface temperatures from solar radiation-a behavior well-documented in previous studies. To maximize energy capture, manual solar tracking was implemented using azimuth and tilt angles (ϒ and β in Fig. 15e), adjusted diurnally to optimize solar incidence. These results demonstrate that simultaneous enhancement of PCM thermal conductivity and internal heat spreading produces more stable power improvements than either strategy alone, although all reported gains should be interpreted as case-specific rather than universal performance benchmarks.

Despite these advances, several challenges remain. Cost control, manufacturing scalability, and environmental adaptability must be addressed to enable broader deployment.

6.4. TEG system integration performance evaluation

The photovoltaic-thermoelectric (PV-TEG) hybrid system represents an innovative approach in renewable energy technology, offering enhanced system output power and energy density. However, its economic feasibility remains a critical constraint on widespread adoption. While TEG integration increases power generation capacity, associated costs hinder scalability. Preliminary studies indicate that although PV-TEG systems outperform standalone PV systems in output power and energy density, their levelized cost of electricity (LCOE) is substantially higher. This performance-cost trade-off necessitates rigorous economic analysis for system evaluation and informs subsequent optimization strategies.

Theoretical analyses have characterized the performance and cost profiles of PV-TEG systems. An ideal PV system achieves a maximum power density of 191.2 W/m^2, whereas a practical PV-TEG hybrid system yields an average power density of 164.4 W/m^2. Within the hybrid system's total output of 481 W, the TEG module contributes merely 78.6 W (16.12% of total generation). Cost analysis reveals an LCOE of 6.80 INR/kWh for the hybrid system-significantly exceeding both ideal PV (3.96 INR/kWh) and practical standalone PV (5.08 INR/kWh) systems. This economic penalty arises from the TEG module's limited output, constrained by a maximum hot-to-cold side temperature difference (ΔT) of only 6 °C. Such a minimal ΔT inherently restricts thermoelectric conversion efficiency and system-level economic viability. Experimental validation confirms the direct correlation between TEG output power and ΔT, demonstrating that a 6 °C gradient caps power generation at suboptimal levels. Although elevated ambient temperatures may enhance PV output, TEG performance gains remain thermodynamically limited. Energy density comparisons further illustrate this constraint: the hybrid system (164.2 W/m^2) exceeds practical standalone PV (140.5 W/m^2) but falls short of ideal PV (191 W/m^2). While TEG integration improves spatial efficiency (e.g., 13.4% footprint reduction), overcoming its performance bottleneck requires thermal management optimization.

The economic performance of PV-TEG and PV-TEG-PCM systems varies substantially with system configuration, climatic conditions, component prices, cooling-energy consumption and assessment boundaries. Under the real environmental and market conditions of the Atacama Desert, Montero et al. calculated an LCOE of approximately 0.071 USD kWh^{-1} for a hybrid PV-TEG system. However, the hybrid system was not economically competitive with conventional PV at the prevailing TEG cost and conversion efficiency [153].

For a PV-PCM/TEG-T system, Li et al. reported an electricity-generation cost of 0.076 USD kWh^{-1} and identified the purchase of TEG modules as the dominant additional capital expenditure [154]. Although the energy payback time was shorter than those of the selected reference PV and PV-PCM-T systems, the unit electricity-generation cost remained higher because of the additional TEG investment.

Economic outcomes reported for other PV/T-TEG configurations are also system-specific. For example, a nanofluid-cooled PV/T-TEG system investigated under the climatic conditions of Calicut achieved an LCOE of 0.04945 USD kWh^{-1} and an energy payback time of 2.57 years [155]. Such results should not be directly transferred to other climates or system scales without harmonizing component lifetime, discount rate, cooling-energy consumption, maintenance cost and area definitions.

For PV-PCM systems, economic feasibility depends on the balance between temperature reduction, additional electricity generation, PCM cost, cycling durability and local climatic conditions [156]. Therefore, universal claims such as fixed LCOE reductions or payback-period improvements should be avoided unless identical economic boundaries are used.

It should be noted that the power density and LCOE values cited here are based on specific system configurations and local economic conditions (including local labor costs, material prices, and grid electricity tariffs). LCOE comparisons across different geographical regions and system scales should therefore be made with appropriate adjustments for local factors. Furthermore, the power density metric is sensitive to the area basis used (PV module area vs. total system footprint), which should be explicitly specified in any comparative assessment.

The cost sensitivity of PV-TEG-PCM systems is governed by several interdependent factors. The purchase of thermoelectric generators constitutes the dominant capital expenditure in these systems, making the number and configuration of TEG modules a critical economic parameter [153]. Technical and economic feasibility analyses have shown that hybrid PV-TEG systems have become profitable after a period of approximately 5.5 years, with profitability strongly influenced by system design parameters and operating conditions [153].

The economic viability of PCM integration is similarly sensitive to material costs and performance gains. Techno-economic assessments of PV modules integrated with PCM have been conducted across diverse climate conditions, revealing that economic feasibility depends on the balance between cooling performance improvement and additional material costs [156]. The high initial cost of PCM materials may not be fully offset by the resulting energy improvement unless the PCM is deployed at scale or achieves extended operational lifetime benefits [156]. Upscaling PCM production and realizing higher lifetime earning potential are therefore critical pathways to improving economic feasibility [156].

Furthermore, the economic performance of PV-TEG-PCM systems exhibits strong climatic dependence. In regions with high global horizontal irradiation, such as the Atacama Desert in Chile (GHI>2500 kWh/m^2/year), the economic feasibility of hybrid PV-TEG systems has been demonstrated under real environmental and market conditions [153]. However, in less favorable climates, the limited temperature gradient across the TEG may render the additional investment uneconomical, underscoring the need for advanced thermoelectric materials or improved cooling approaches to achieve economic viability.

A critical factor often overlooked in PV-TEG and PV-TEG-PCM design is the thermal contact resistance at the PV/TEG, TEG/PCM, and TEG/heat-sink interfaces. The effective temperature difference across the thermoelectric legs is generally smaller than the externally measured temperature difference because part of the thermal potential is lost across interfacial layers, adhesives, encapsulants, air gaps, and heat-spreading plates. This can be expressed qualitatively as

$$\Delta T_{TEG} < \Delta T_{system} \tag{6-10}$$

or, using a thermal-resistance network,

$$Q = \frac{T_{PV} - T_{amb}}{R_{PV-TEG} + R_{TEG} + R_{TEG-cold} + R_{sink}} \tag{6-11}$$

where $R_{PV\text{-}TEG}$, R_{TEG}, $R_{TEG\text{-}cold}$, and R_{sink} represent the thermal resistances of the PV-to-TEG interface, the TEG module, the cold-side interface, and the heat sink or cooling boundary, respectively. A high contact resistance reduces the heat flux entering the TEG and lowers the actual ΔT, thereby decreasing the Seebeck voltage and output power.

The cold-side boundary condition is equally important. If the TEG cold side is exposed only to natural convection, the cold-side

temperature may rise rapidly, causing ΔT to collapse. In contrast, forced air cooling, water cooling, heat pipes, radiative cooling, or PCM-assisted thermal buffering can maintain a lower T_c and improve TEG performance. However, active cooling may introduce parasitic power consumption, while passive cooling is strongly dependent on ambient temperature, wind speed, effective sky temperature (for radiative cooling), and heat-sink geometry. Therefore, the design of PV-TEG-PCM systems must co-optimize heat supply from PV/PCM and heat rejection from the TEG cold side.

For PV-TEG-PCM systems, PCM position also affects the cold-side boundary condition. If the PCM is placed behind the TEG, it may serve as a thermal buffer and extend heat release after sunset, but it can also reduce the instantaneous heat-rejection rate if its temperature rises during charging. If PCM is placed between the PV and TEG, it may stabilize PV temperature but weaken the hot-side temperature available to the TEG. Hence, PCM integration should be evaluated using a coupled thermal-resistance framework rather than assuming that additional PCM always increases TEG power.

To circumvent the intrinsic performance bottlenecks arising from constrained temperature gradients (ΔT) and their concomitant economic liabilities, recent research has introduced a sophisticated integrated PV-MCHP-TEG-PCM system aimed at optimizing both thermal management and energy conversion pathways. As illustrated in the comprehensive evaluation in Fig. 16, this architecture is underpinned by an empirical framework that facilitates a rigorous comparative analysis against baseline configurations (Fig. 16a). The system strategically employs single-crystal silicon cells and Peltier 12709 TEG modules to equilibrate high electrical yield with robust thermal resilience. A salient feature of this design is the multi-layered assembly where a Micro Heat Pipe (MCHP) array is interposed between the PV posterior and a Phase Change Material (PCM) buffer (Fig. 16b).

Sensitivity analyses concerning the internal PCM layer reveal that structural dimensions exert a decisive influence on thermal dynamics. Specifically, the PV operating temperature exhibits a non-monotonic "parabolic" trajectory as PCM thickness varies (Fig. 16c). This thermal behavior directly modulates electrical output; empirical evidence in Fig. 16d indicates that electrical efficiency reaches a nadir of 16.74% at a critical PCM thickness of 40 mm, underscoring a complex trade-off

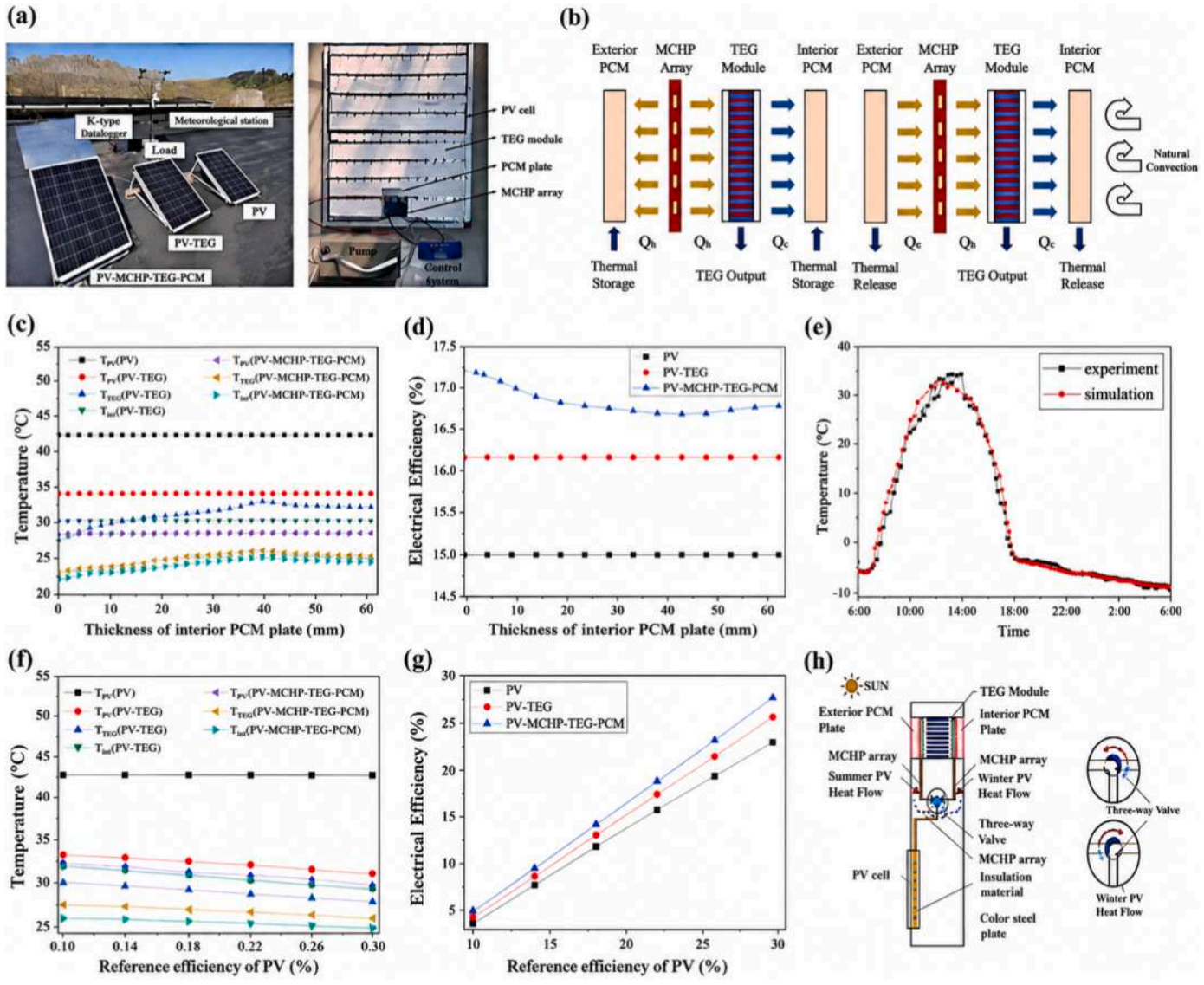


Fig. 16. (a) The experiment platform of PV-MCHP-TEG-PCM system. (b) The schematic of the PCM working principle. (c) Influence of internal PCM plate's thickness on Temperature efficiency. (d) Influence of internal PCM plate's thickness on Electrical efficiency. (e) Influence of PV reference efficiency on Temperature efficiency. (f) Influence of PV reference efficiency on Electrical efficiency [157]. (g) Controlled trial between numerical and experimental results of PV cell's temperature. (h) Structure of the PV-MCHP-TEG-PCM system [158].

between latent heat capacity and conductive thermal resistance. Beyond structural optimization, the intrinsic efficiency of the PV cells serves as a pivotal regulator of the system's aggregate thermal load. A robust negative correlation has been established, wherein the operating temperatures of both PV and TEG components subside as PV reference efficiency improves (Fig. 16e).

This mitigation of thermal stress enables a linear augmentation in holistic electrical performance. As depicted in Fig. 16f, the PV-MCHP-TEG-PCM system demonstrates superior efficiency gains-scaling from 10.9% to 31.9% as reference efficiency shifts from 10% to 30%-consistently eclipsing standalone PV and PV-TEG configurations. This superiority is attributed to the PCM's bifunctional role as a "thermal-seasonal buffer": it functions as a potent heat sink for the PV cells via the MCHPs during peak insolation and transitions into a primary heat source for the TEG modules during nocturnal periods, thereby sustaining the requisite temperature gradient. The methodological robustness of these findings is validated through a high degree of convergence between experimental data and mathematical simulations (Fig. 16g), with a maximum temperature deviation of 6.0 °C and a relative error confined to 9.3%. Ultimately, this synchronized thermal control is operationalized through an advanced architecture (Fig. 16h) incorporating three-way valves and inorganic working fluids, effectively resolving the spatiotemporal contradiction between solar cooling and thermoelectric heating requirements. Such optimized thermal-to-electric synergy provides a foundational theoretical framework for the comparative advancements documented in subsequent literature.

External studies corroborate these findings. A comparable hybrid system achieved peak parameters of 39 V and 4.9 An under 8 Ω load, yielding 191.1 W maximum power-a 15.38% improvement over standalone PV [158,159]. Similarly, Bi_2Te_3-based TEG-PV integration demonstrated 14.9% power gain under specific conditions [160]. Collectively, these results confirm that TEG integration enhances performance across diverse configurations. Nevertheless, constrained ΔT and high module costs persistently undermine economic viability, reinforcing the intrinsic trade-off between technical potential and commercial feasibility.

PV-TEG integration presents an inherent performance-cost dichotomy: while it improves power density and spatial efficiency, particularly in space-constrained settings, it is constrained by high LCOE and fundamental thermodynamic limitations. Future advancements must prioritize (i) thermal management strategies to maximize ΔT and (ii) concerted cost reduction in TEG manufacturing. Systematic exploration of novel materials and integrated system design will be essential to enable practical deployment of PV-TEG technology and achieve an optimal performance-cost equilibrium in renewable energy systems.

Based on the available techno-economic evidence, the conditions under which PV-TEG-PCM systems become economically viable can be summarized as follows.

(i) Temperature and irradiance conditions. Economic feasibility analyses in high-irradiation regions (GHI>2500 kWh/m^2/year) have demonstrated that hybrid PV-TEG systems can achieve profitability after approximately 5.5 years [153]. The economic viability is most favorable in hot climates with high solar irradiance where PV operating temperatures create substantial temperature differentials for TEG operation.

(ii) Performance enhancement threshold. The additional capital cost of PCM and TEG integration must be offset by sufficient performance gains. When performance gains fall below certain thresholds, the LCOE of hybrid systems can be substantially higher-up to 145% more than standalone PV systems [153]. Conversely, techno-economic assessments have shown LCOE reductions of 5-15% and payback period decreases of 10-20% for optimized configurations [155].

(iii) Scale and lifetime conditions. Economic feasibility improves significantly with system scale and extended operational lifetime. PV-PCM-TEG systems have demonstrated an LCOE of 0.076 USD/kWh, with energy payback times shorter than those of standalone PV systems [154]. The purchase of TEGs remains the dominant capital expenditure [153], indicating that future reductions in TEG manufacturing costs will be critical for broader economic viability.

(iv) System configuration optimization. The number of TEG modules, PCM thickness, PCM melting temperature, and system operating conditions are among the most influential parameters affecting both electrical efficiency and total life-cycle cost [157]. Multi-objective optimization approaches have been developed to balance thermal management objectives against power generation efficiency, identifying optimal configurations that maximize economic performance [157].

7. Photovoltaic-thermoelectric generator-phase change material (PV-TEG-PCM) systems

7.1. Structural design and operating mechanism of PV-TEG-PCM systems

The photovoltaic-thermoelectric generator-phase change material (PV-TEG-PCM) system represents an innovative energy conversion architecture engineered to resolve critical thermal management limitations inherent in conventional photovoltaic-thermoelectric (PV-TEG) systems. Traditional configurations suffer from inefficient heat transfer between the photovoltaic (PV) module and thermoelectric generator (TEG), resulting in elevated PV operating temperatures and insufficient temperature gradients across the TEG, thereby compromising overall energy conversion efficiency. By strategically integrating phase change material (PCM) as an intermediary thermal regulator, the PV-TEG-PCM system achieves bidirectional thermal control: The PV-TEG-PCM system integrates latent-heat storage into the thermal pathway of a conventional PV-TEG device. Depending on the selected layout, the PCM may be placed downstream of the TEG in a PV-TEG-PCM configuration or between the PV and TEG in a PV-PCM-TEG configuration. The former arrangement allows heat to reach the TEG hot side more directly, whereas the latter provides stronger buffering of PV-temperature fluctuations but introduces additional thermal resistance and thermal inertia. Therefore, PCM integration does not necessarily increase the instantaneous temperature difference across the TEG; its effect depends on PCM melting temperature, thickness, thermal conductivity, interfacial resistance and cold-side heat-rejection conditions [118]. The resultant 'sandwich' configuration optimizes internal heat flow pathways and leverages latent heat storage to ensure stable, high-efficiency operation during extended periods under variable solar irradiance.

The position of PCM within PV-TEG-PCM systems strongly determines the transient heat-transfer pathway and the effective temperature difference across the TEG. However, placing PCM between the PV module and the TEG does not necessarily increase the TEG temperature difference. Its actual effect depends on the PCM melting range, thickness, latent heat, thermal conductivity, interfacial thermal resistance, and the temporal matching between PCM charging/discharging and TEG operation.

In bottom PCM position, namely the PV-TEG-PCM configuration, heat generated by the PV module can be transferred more directly to the TEG hot side, while the PCM acts as a downstream thermal buffer. This arrangement may provide a faster thermal response and a larger instantaneous temperature difference when the PCM remains relatively cool or is undergoing phase transition. Nevertheless, once the PCM approaches thermal saturation, its heat-sink function weakens, the TEG cold-side temperature rises, and the effective temperature difference decreases.

In intermediate PCM position, namely the PV-PCM-TEG configuration, the PCM directly buffers the PV temperature and smooths thermal fluctuations before heat reaches the TEG. This can be beneficial for PV temperature regulation and extended heat release after solar irradiance

decreases. However, the intermediate PCM layer also introduces additional thermal resistance and thermal inertia. If PCM has low thermal conductivity, excessive thickness, poor interfacial contact, or an unsuitable melting range, it may damp or delay the heat flux reaching the TEG hot side and reduce instantaneous thermoelectric output.

Therefore, PCM spatial position should be treated as a system-level thermal-resistance and transient heat-transfer optimization problem. Bottom PCM position is more suitable when rapid heat transfer and higher instantaneous TEG output are prioritized, whereas intermediate PCM position may be preferable when PV temperature stabilization and prolonged heat release are the main design objectives. Neither configuration is universally superior, and the optimal design must be determined by coupling PCM properties, module geometry, interfacial resistance, and cold-side heat-rejection conditions.

Structurally, a representative PV-TEG-PCM configuration consists of a monocrystalline-silicon PV module, a thermoelectric module and a downstream PCM layer. In the numerical system developed by Lv et al. [118], the PV module measured 100 mm × 100 mm × 4.5 mm and was directly coupled to the TEG, while the PCM was positioned beneath the TEG as a transient thermal buffer. The corresponding three-dimensional transient model accounted for fluctuating solar irradiation and examined the effects of PCM melting temperature, layer thickness and thermal conductivity. For the specific geometry and boundary conditions considered, a PCM thickness of 30 mm and a melting temperature of 42 °C produced the best thermal performance [118]. The PV module consists of a 3.2 mm glass protective layer, a 0.46 mm ethylene-vinyl acetate (EVA) adhesive layer, a 0.18 mm monocrystalline silicon active layer, and a polyester backing, collectively forming a unit with a standard efficiency of 17% and a temperature coefficient of $-0.0045\ K^{-1}$. The TEG module integrates 0.8 mm ceramic substrates with series-connected bismuth telluride (Bi_2Te_3) semiconductor couples (1.5 mm thick), interconnected via 0.3 mm copper electrodes.

Fig. 17 comprehensively evaluates the performance advantages of the PV-TEG-PCM hybrid system through physical model construction, energy model analysis, grid independence verification, and comparison of temperature and efficiency. As shown in Fig. 17a, representative implementation employs a monocrystalline silicon PV panel (100 mm × 100 mm × 4.5 mm) integrated with TEG and PCM modules, underscoring the scalability of the layered design. The PCM heat sink, situated beneath the PV and TEG layers, absorbs substantial latent heat during phase transition, effectively mitigating thermal accumulation. System optimization balances thermal performance and economic viability through tailored PCM selection-varying melting points and fill heights-to maximize energy recovery. The operational mechanism hinges on dynamic energy flow and thermal equilibrium. Solar radiation constitutes the sole energy input, driving simultaneous electricity generation in the PV module and thermal energy harvesting in the TEG module. Photons incident on the PV layer convert partially to electrical energy, with residual energy dissipated as waste heat; the TEG harnesses heat conducted from the PV backplane and the temperature gradient sustained by the PCM on its opposing side. Environmental energy losses occur primarily via radiation and convection. To facilitate modeling, the system assumes adiabatic external boundaries, an initial temperature of 298.15 K, and negligible interfacial thermal resistance. The energy model in Fig. 17b assumes a uniform initial PCM temperature equal to ambient conditions, isotropic and homogeneous phases, and constant thermophysical properties within each phase. The liquid PCM is modelled as Newtonian, incompressible laminar flow, with negligible lateral heat transfer due to high aspect ratios. Uniform solar irradiation and homogenized TEG properties are also assumed. Governing equations quantify the energy balance [161], with PV and TEG electrical outputs derived from empirical relationships [161] that explicitly define the inverse correlation between PV efficiency and temperature rise. Crucially, the PCM functions as a thermal buffer: its phase transition absorbs waste heat, minimizing active cooling requirements, while natural convection within the molten PCM enhances internal heat distribution, thereby stabilizing operational temperatures and elevating net energy utilization. Numerical convergence and physical-model validation should be distinguished. The grid-independence test reported in Ref. [118] demonstrates that the numerical solution is insensitive to further mesh refinement, but it does not independently establish the physical accuracy of the complete PV-TEG-PCM model. Component-level validation may be conducted separately: published finite-element PV-temperature data can be used to assess the PV thermal sub-model [162], while experimental PCM melting data can be used to evaluate the phase-change formulation [163]. For integrated-system

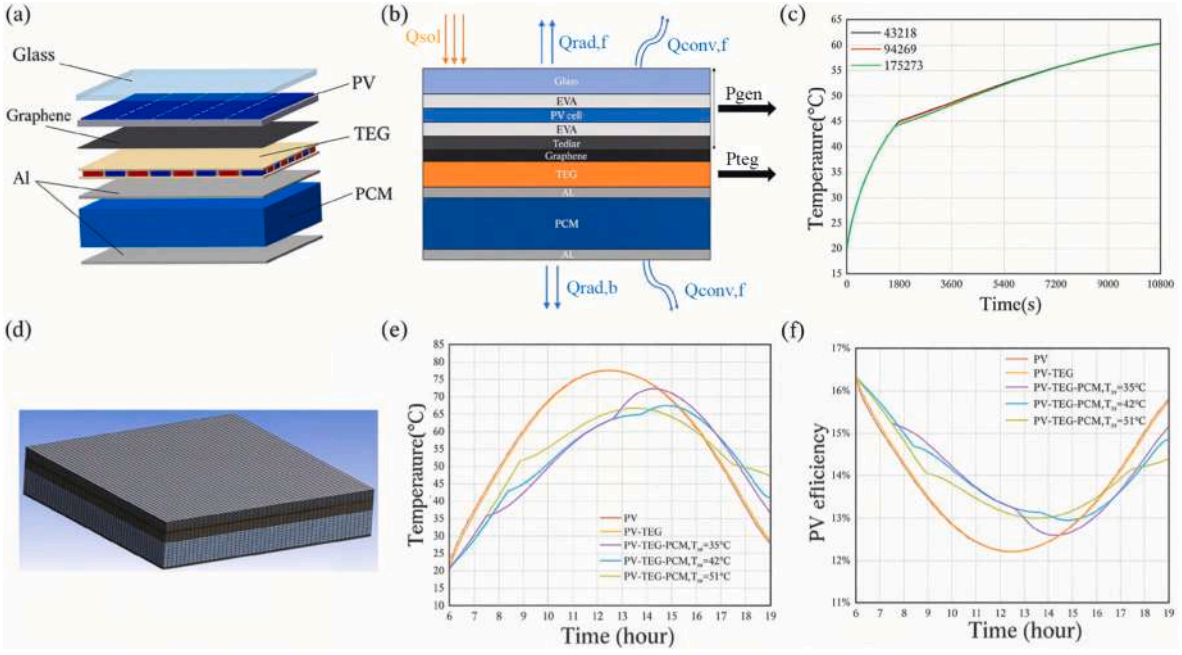


Fig. 17. (a) Physical model of the PV-TEG-PCM system; (b) energy-flow model; (c) grid-independence analysis; (d) computational mesh; (e) PV-temperature comparison; and (f) PV-efficiency comparison for different system configurations [163].

validation, Darkwa et al. [116] developed a coupled PV/TEG/PCM numerical model and compared its thermal predictions with experimental measurements. Nevertheless, the available validation remains mainly laboratory-scale and short-duration and should not be described as proof of long-term outdoor predictive accuracy. The modeling workflow encompasses parameter initialization, geometric construction, physics interface configuration, and rigorous mesh convergence analysis. As shown in Fig. 17c, mesh independence was rigorously verified using three resolutions (43,218; 94,269; and 175,273 elements), with PV back temperature variations below 0.5% confirming model robustness. A mesh density of 175,273 elements was selected for subsequent simulations, with geometric and mesh details illustrated in Fig. 17d. Model accuracy was cross-validated against literature data; as shown in Fig. 17e, the average PV cell temperature profiles for standalone PV, PV-TEG, and integrated PV-TEG-PCM systems demonstrate superior thermal regulation in the latter configuration [118]. Fig. 17f further quantifies diurnal PV efficiency trends, revealing that the PV-TEG-PCM system maintains higher efficiency for over 85% of operational time compared to alternatives, with peak enhancements of 1.04% achieved using a 42 °C melting-point PCM relative to the PV-TEG baseline. This comprehensive validation underscores the model's predictive reliability and the system's operational stability.

Synergistically, the structural design and thermal management mechanisms of the PV-TEG-PCM system enable exceptional performance under fluctuating irradiance. The PCM-integrated sandwich architecture facilitates passive thermal regulation, significantly boosting PV electrical output and TEG power generation. Combined energy balance analysis and numerical verification confirm the system's adaptability to dynamic environmental conditions and its resilience during prolonged operation. This integrated approach not only overcomes fundamental thermal barriers in conventional PV-TEG systems but also establishes a scalable framework for next-generation renewable energy technologies, offering a viable pathway toward high-efficiency, sustainable power generation.

7.2. PV-TEG-PCM thermal management performance optimization

In PV-TEG-PCM hybrid systems, thermal management performance is a pivotal determinant of energy conversion efficiency and operational stability. These systems integrate optical and thermal conversion units

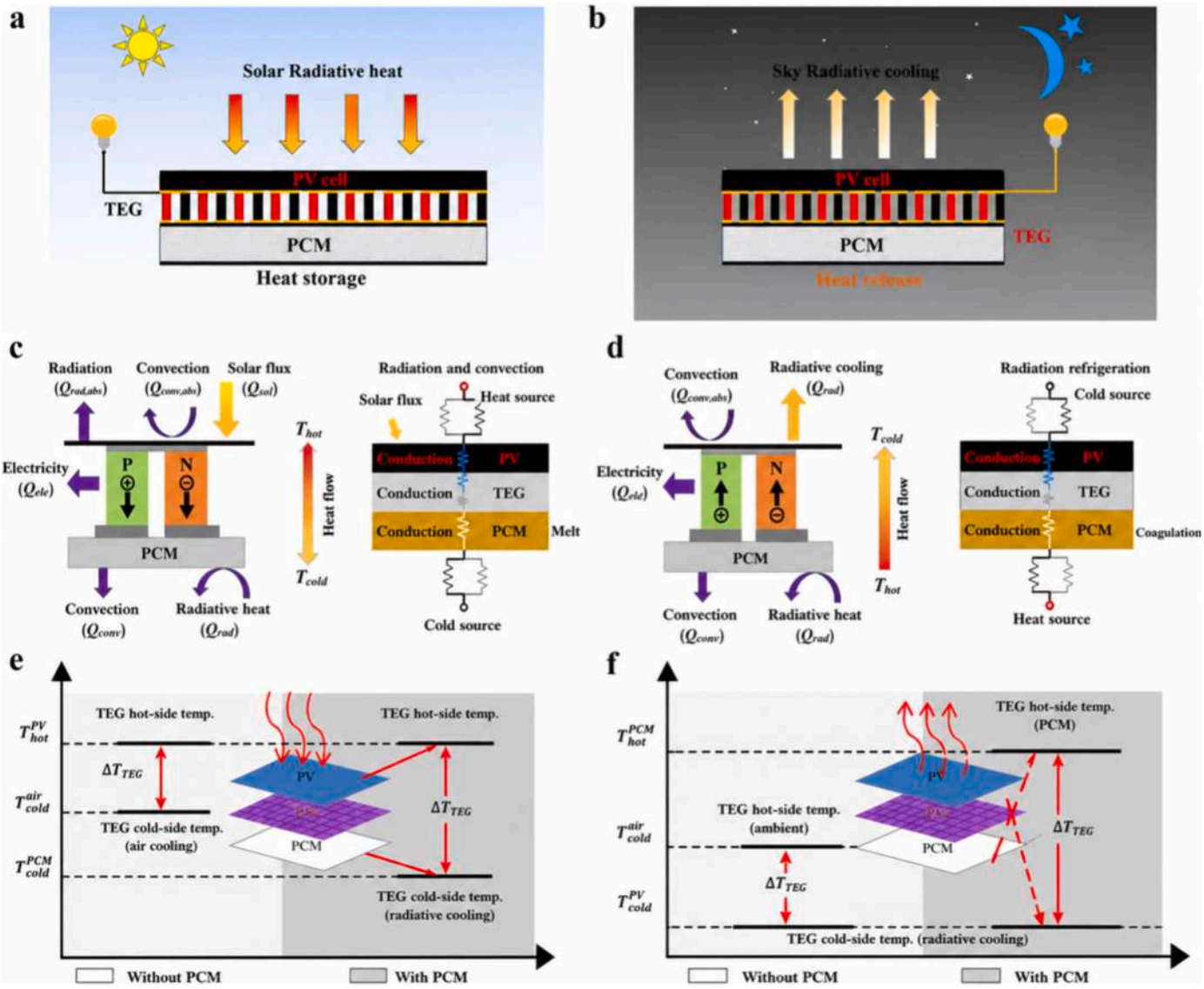


Fig. 18. Synergistic day–night energy harvesting in a PV-TEG-PCM system: (a) daytime solar conversion and PCM charging; (b) nighttime PCM heat release and sky radiative cooling; (c) daytime energy balance; (d) nighttime energy balance; (e) daytime temperature-gradient formation; and (f) enhancement of the nighttime temperature difference through radiative cooling [120].

to overcome the inherent limitation of daytime-only power generation in conventional photovoltaic setups. Nevertheless, dynamic internal heat distribution and unavoidable thermal losses present significant bottlenecks to performance enhancement. Specifically, diurnal cycles introduce dual challenges: fluctuating temperature gradients and energy supply-demand imbalances, exemplified by TEG module efficiency degradation due to daytime overheating or nighttime power generation interruptions from insufficient thermal storage. The structure and operating mechanism of a typical optimized system addressing these issues are systematically presented in Fig. 18. An innovative composite system incorporating glass cover, PV cells, TEG panels, and phase change materials (as shown in Fig. 18a) addresses these issues by directly converting solar thermal energy into electricity, diverging from traditional PV/T systems designed for hot water production. Consequently, optimizing thermal management necessitates not only precise temperature regulation but also coordinated energy flow control, forming the foundation for enhanced system reliability and solar energy utilization.

The core thermal management mechanism relies on the dynamic heat storage and release behavior of phase change materials (PCMs) and their synergistic interaction with PV and TEG components [161]. During the day, solar radiation strikes the photovoltaic cells to generate electricity, while simultaneously transferring waste heat radiation to the thermoelectric generator (TEG) panel, causing its temperature to rise. This establishes a temperature gradient within the TEG, with the photovoltaic cells acting as the hot side and the phase-change material (PCM) serving as the cold side (Fig. 18b). Concurrently, PCM absorbs excess heat to maintain the TEG within its operating temperature range (Fig. 18c), thereby enabling thermoelectric power generation. At night, solar absence triggers a mode shift: radiative cooling of the PV cell toward the sky (Fig. 18d) combines with thermal energy release from the PCM, reversing the gradient such that the PCM becomes the hot end and the PV cell the cold end. As shown in Fig. 18e, this reversed gradient sustains TEG power generation through the thermoelectric effect. The PCM's phase transition process dynamically balances heat flow distribution, mitigates abrupt temperature fluctuations, and reduces thermal stress, thereby enhancing environmental adaptability and long-term durability.

Optimization strategies for thermal management span multiple dimensions, including material selection, structural configuration, and control methodologies. High-latent-heat PCMs with tailored phase transition temperatures improve thermal storage density and response kinetics. Optimized integration layouts for TEG and PV components minimize interfacial thermal resistance and parasitic energy losses. Additionally, intelligent strategies-such as environment-responsive heat flow regulation-enable precise thermal control. These approaches collectively target diurnal cycle limitations, such as excessive thermal inertia or inadequate driving temperature differences, to maximize energy capture and conversion efficiency.

Sustained power generation during variable conditions, including nighttime or cloudy periods via stored thermal energy (Fig. 18f), exemplifies the efficacy of optimized thermal management. This capability yields multidimensional benefits: enhanced power generation efficiency, operational stability, and application feasibility. Reduced thermal losses and extended generation periods ensure consistent energy output across diverse environmental scenarios. Simultaneously, optimized thermal regulation mitigates thermal fatigue-induced component failures, prolongs system lifespan, and bolsters reliability for large-scale deployment. These advancements strengthen economic viability and expand applicability in off-grid power supply and building-integrated renewable energy systems.

Ultimately, thermal-management optimization should therefore consider the complete transient thermal-resistance network rather than PCM heat capacity alone. Relevant variables include PCM melting range, layer thickness, thermal conductivity, PV emissivity, TEG thermal resistance, interfacial contact quality and nighttime sky conditions. Although this configuration demonstrates the technical feasibility of full-day energy harvesting, longer outdoor tests are still required to evaluate seasonal repeatability, PCM cycling stability, degradation and practically useable nighttime energy output.

7.3. Integrated system reconfiguration and optimization algorithm

Previous studies have confirmed that photovoltaic-thermoelectric-phase change material (PV-TEG-PCM) hybrid systems possess significant advantages in thermal management and energy conversion. However, the performance of such systems is highly dependent on their parameter configuration and structural design, and fixed topological structures struggle to adapt to dynamically changing environments, thereby limiting the full realization of their potential. Consequently, the introduction of comprehensive system reconfiguration and optimization algorithms has emerged as a key pathway to enhance system adaptability and stability. These methods can proactively respond to external disturbances such as temperature fluctuations and irradiance variations by dynamically adjusting system parameters and structures, enabling more efficient energy utilization.

Need for dynamic optimization based on parameter sensitivity analyses of PCM phase change temperature and thickness in prior research indicate that system performance is highly dependent on parameter selection [161]. An inappropriate phase change of temperature may lead to failure of the thermal regulation function, while improper thickness can disrupt the delicate balance between heat capacity and thermal resistance. These findings highlight the necessity for dynamic parameter optimization. Algorithms such as multi-objective optimization or Bayesian inference can dynamically adjust these key parameters based on real-time operating conditions. For instance, by using measured PV module temperatures and PCM phase change states, they can automatically search for optimal material properties, thereby minimizing efficiency losses and maximizing the thermal buffering effect to ensure peak performance under variable conditions.

Impact of structural reconfiguration on thermal management and energy distribution dynamic reconfiguration of the system structure has been experimentally proven to significantly impact heat loss control and energy distribution. For example, adding insulation layers effectively reduces ineffective heat dissipation, whereas their removal leads to rapid temperature increases in components. This provides a basis for algorithm-driven structural reconfiguration strategies: during operation, the system can algorithmically evaluate the merits of different structural schemes (such as dynamically adjusting insulation layout or component arrangement) to optimize heat flow paths and energy transfer efficiency. This strategy not only relies on experimental data but can also integrate predictive models to respond in real-time to environmental changes, thereby enhancing system reliability and efficiency and avoiding performance degradation caused by structural rigidity.

To achieve synergy between parameter and structural optimization, constructing an integrated multi-objective optimization framework is crucial. This framework can utilize advanced algorithms such as model predictive control (MPC) or machine learning to process multi-source input data (e.g., ambient temperature, PV output, PCM phase change state) in real-time and output optimal reconfiguration decisions. For instance, the algorithm can balance thermal management objectives against power generation efficiency, simulating system performance under different configurations to select the optimal solution. Building on experimental observations, this approach can predict potential performance gains during long-term operation, such as higher photovoltaic conversion efficiency or more stable TEG output.

Comprehensive system reconfiguration and optimization algorithms show broad prospects in enhancing the performance of PV-TEG-PCM hybrid systems. Their core value lies in endowing systems with exceptional adaptability, efficiency, and stability through dynamic adjustments. However, implementing this method still faces challenges such as algorithm complexity, real-time computational overhead, and model

accuracy. Future research should focus on simplifying algorithm structures and integrating artificial intelligence technologies to promote the widespread application of such intelligent systems in the renewable energy sector. Through continuous algorithmic innovation, truly intelligent photovoltaic-thermoelectric systems can be realized, providing solid technical support for sustainable energy development.

7.4. Experimental validation and efficiency comparative analysis

Reliable validation is essential for determining whether the predicted benefits of PV-TEG-PCM systems can be reproduced under practical conditions. Existing studies employ numerical simulations, component-level benchmark validation, prototype experiments, and outdoor comparative tests, which provide different levels of evidence. The eight studies summarized in Table 4 were purposively selected from the available PV-TEG-PCM literature to provide a comparative representation of the principal ternary-system configurations and validation approaches. The selection was based on four criteria: (i) explicit integration of PV, TEG, and PCM components within the investigated system; (ii) availability of sufficient quantitative information for comparison across the key parameters summarized in Table 4; (iii) diversity in system configuration, operating conditions, and validation method; and (iv) coverage of different evidence types, including numerical modelling, laboratory-scale experiments, outdoor testing, and day-night prototype validation. The selected studies are compared in terms of PV technology, TEG characteristics, PCM type and melting temperature, PCM size, system configuration, operating conditions, validation method, electrical performance, TEG output, thermal performance, test duration, and principal limitation [99,120,121,119,118,116,117,122]. Among the validation approaches represented in Table 4, the numerical investigation by Gao et al. [117] illustrates the important distinction between numerical verification and physical validation.

Grid-independence analysis controls numerical discretization errors but does not establish the physical accuracy of a coupled model. In the numerical study shown in Fig. 19, the average PV temperature and TEG output became insensitive to further mesh refinement at approximately 309,913 elements, which were therefore used in the subsequent simulations [117]. Physical validation was then performed separately for the PCM, PV, and TEG sub-models.

The configurations and corresponding validation procedures are summarized in Fig. 19. Fig. 19a–b illustrates the two investigated arrangements: Configuration 1, in which the PCM is placed beneath the TEG in a PV-TEG-PCM sequence, and Configuration 2, in which the PCM is inserted between the PV and TEG in a PV-PCM-TEG sequence. The detailed assembly of Configuration 1, including an RT28HC-filled copper container with internal vertical fins, is shown in Fig. 19c. For the PCM sub-model, the simulated melting fronts were compared with experimental observations at 15 and 60 min, as shown in Fig. 19d–e,

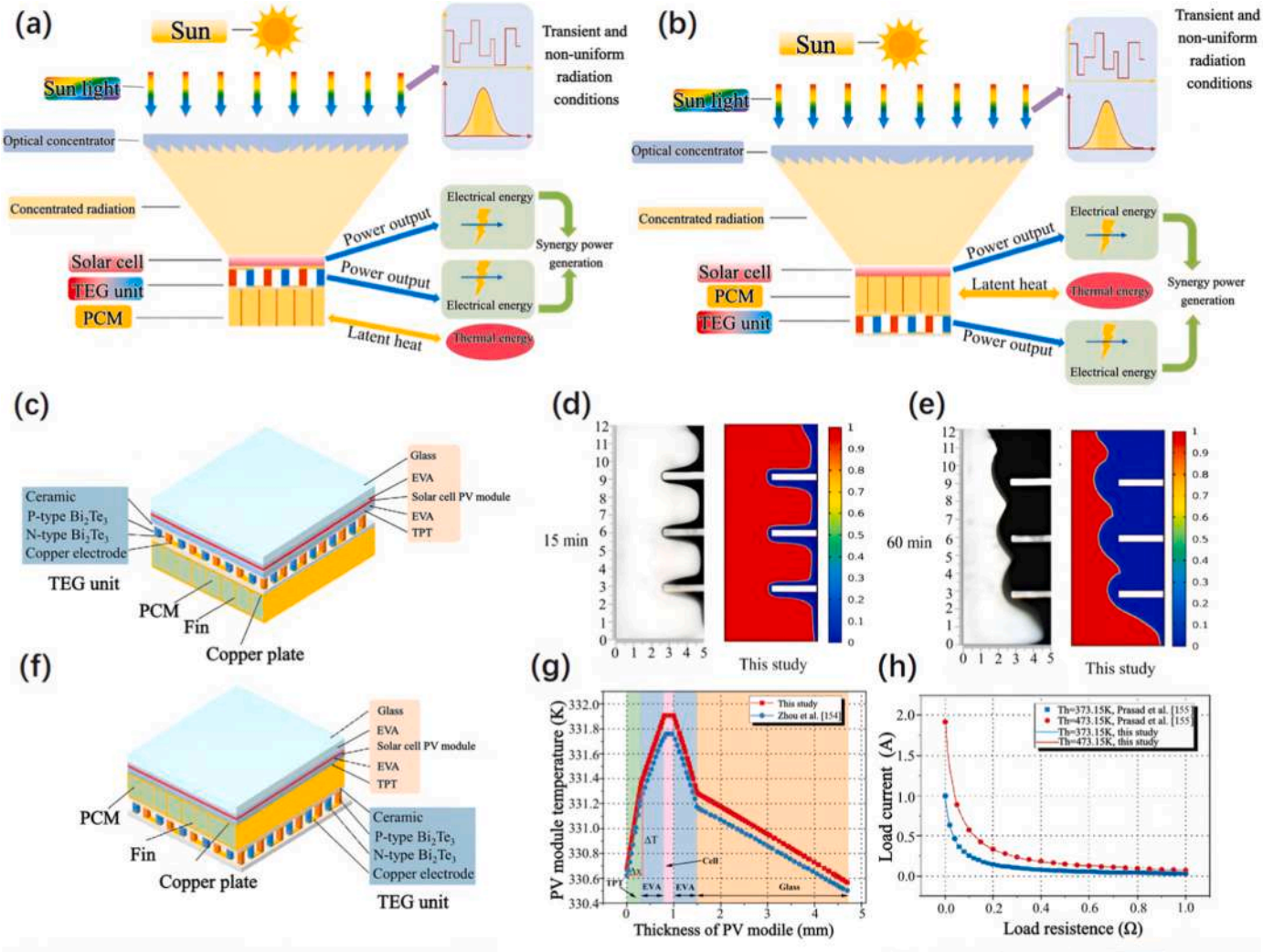


Fig. 19. Configuration and component-level model validation of the PCM-integrated PV-TEG system: (a) bottom PCM position; (b) intermediate PCM position; (c) assembly of Configuration 1; (d-e) PCM-model validation; (f) assembly of Configuration 2; (g) PV-model validation; and (h) TEG-model validation [117].

respectively. The observed agreement indicates that the model captured the principal transient phase-change behavior under the prescribed heating conditions. Fig. 19f presents the assembly of Configuration 2, for which the PV and TEG sub-models were evaluated separately. The through-thickness PV temperature gradients were compared with published benchmark results in Fig. 19g, while the TEG current-load characteristics were assessed in Fig. 19h at a fixed cold-side temperature of 293.15 K and hot-side temperatures ranging from 373.15 to 573.15 K [117].

While Fig. 19 provides detailed component-level validation for a representative PV-TEG-PCM system, evaluating the broader development and practical performance of this technology requires a systematic comparison across different system configurations, materials, operating conditions, and validation approaches.

As summarized in Table 4, the reviewed systems differ considerably in architecture and thermal-management strategy. The basic PV-TEG-PCM and PV-PCM-TEG arrangements are extended by pulsing heat pipes [99], night-sky radiative cooling [120], dual-loop active cooling [121], serpentine water circulation [119], graphene heat spreading [118], and fined PCM containers [117]. Commercial paraffins with nominal transition temperatures ranging from approximately 28 to 55 °C are commonly used, while the reported PCM dimensions vary from 5 to 25 mm layers to a 4 kg storage unit. These differences affect PV cooling, heat delivery to the TEG, cold-side heat rejection, system mass, and thermal response time.

PCM size and position must balance heat-storage capacity against thermal resistance. In Ref. [117], increasing the RT28HC thickness from 5 to 25 mm improved PV thermal regulation and increased the predicted peak TEG output from 0.0156 to 0.0292 W. However, a thicker PCM layer may delay heat transfer and weaken the temperature gradient across the TEG. Similarly, placing PCM between the PV and TEG can stabilize PV temperature but may reduce the heat flux reaching the TEG. By contrast, placing the TEG directly behind the PV improves hot-side thermal coupling, but useful TEG output can only be maintained if the PCM or cooling unit removes heat effectively from its cold side.

The reported electrical improvements should be interpreted cautiously because the reviewed studies adopted different performance indicators and system boundaries. The graphene-assisted system in LV S et al. [118] increased the peak PV efficiency by approximately 0.87-1.04 percentage points, whereas Lv et al. [122] increased the PV efficiency from 12.5% to 14.2%, corresponding to a gain of 1.7 percentage points, using a heat-accumulating prototype. Darkwa et al. [116] reported an approximately 9.5% increase in cumulative PV output under the specified indoor test conditions. Larger gains were reported for extended PV/T configurations. Gürbüz et al. [119] achieved a 10.29-14.22% increase in total power using a serpentine-tube system, while Li et al. [121] obtained a mean electrical efficiency of 12.97%, representing a 45.08% improvement over the reference PV module. However, Li et al. [121] excluded compressor power consumption; therefore, the reported improvement does not represent the net system gain. In addition, Kılkış [99] reported total exergy rather than a separate PV electrical gain, while Wei et al. [120] did not provide several key component-level performance parameters.

The independent contribution of the TEG was generally much smaller and strongly dependent on the attainable temperature difference. Darkwa et al. [116] reported a peak areal TEG output of approximately 0.55 W/m^2 at a maximum temperature difference of only about 1.6 K. The actively cooled system investigated by Li et al. [121] produced approximately 0.67 W at a maximum temperature difference of 9.6 °C, whereas the serpentine-tube system developed by Gürbüz et al. [119] achieved a peak output of 1.43-1.82 W. In contrast, the heat-accumulating system studied by Lv et al. [122] generated only approximately 0.112 mW. These results indicate that TEG output is governed more strongly by the attainable temperature difference, cold-side heat rejection, interfacial thermal resistance, and the duration of the temperature difference than by the number of TEG modules alone.

Thermal benefits were demonstrated more consistently than thermoelectric electricity generation. Maximum PV-temperature reductions of approximately 10.8 °C, 12.04-14.33 °C, and 19.22 K were reported by Li et al. [121], Lv et al. [118], and Gao et al. [117], respectively. Gürbüz et al. [119] further reported PV peak temperatures of approximately 71-75 °C, compared with 85-87 °C for the reference PV module. Nevertheless, these results were obtained under different irradiance levels, ambient conditions, PCM properties, test durations, and reference configurations. They therefore demonstrate the cooling potential of PCM integration but should not be interpreted as a direct performance ranking.

Test duration and validation quality remain major limitations. The investigations in Lv et al. [118] and by Gao et al. [117] were numerical or indirectly validated, whereas the experimental studies by Li et al. [121], Gürbüz et al. [119], Darkwa et al. [116], and Lv et al. [122] generally covered only several hours or a single day. Kilkiş [99] and Wei et al. [120] demonstrated extended thermal or day-night operation over 24 h but did not report sufficient component-level data for a comprehensive comparison. Moreover, Gürbüz et al. [119] conducted the comparative tests on different days, while Li et al. [121] excluded the auxiliary power required for active cooling. Quantitative nighttime TEG output and cumulative post-sunset electricity generation therefore remain insufficiently reported.

Based on the validation approaches and test durations summarized above, the available evidence can be broadly classified into four categories. Numerical studies provide preliminary evidence of system feasibility and support parametric optimization, but their predictions remain dependent on model assumptions and boundary conditions. Short-duration laboratory experiments provide proof-of-concept validation under controlled conditions, whereas outdoor experiments provide more practically relevant evidence under realistic and variable environmental conditions. Full day-night prototype studies provide the most direct evidence currently available for post-sunset power generation by demonstrating system operation across changing solar and thermal conditions.

Overall, the available evidence confirms that PCM integration can reduce PV temperature and improve PV electrical performance. However, TEG output is usually limited by the small temperature difference achievable in passive systems. Comparisons across studies are further complicated by inconsistent system boundaries and reporting metrics. Future studies should therefore separately report PV output, TEG output, auxiliary consumption, and net system gain, together with PCM thermophysical properties, TEG parameters, thermal contact resistance, measurement uncertainty, and normalized performance per unit area. Simultaneous testing against a reference PV module over repeated full day-night cycles is also required to establish the long-term practical value of PV-TEG-PCM systems.

8. Applications

8.1. Applications of PV/T–PCM systems in thermal management

The operating temperature of a photovoltaic module is a major factor affecting its electrical performance. However, the temperature coefficient is not a universal constant and varies with cell technology, irradiance level and operating conditions. For conventional crystalline-silicon modules, the power loss is generally of the order of 0.4% per degree Celsius, although the exact value should be taken from the module specification or the corresponding experimental study [164].

Phase change materials can provide passive thermal buffering by absorbing latent heat when their temperature enters the phase-transition range. When integrated behind a PV module or within a PV/T collector, the PCM can delay the rise in cell temperature and reduce short-term temperature fluctuations. Nevertheless, this thermal-regulation effect is temporary. Once the PCM has largely melted, its ability to absorb additional latent heat decreases, and the molten layer

may introduce additional thermal resistance unless an effective heat-rejection or solidification pathway is provided [156].

Representative studies demonstrate that system performance is highly dependent on PCM selection and structural configuration. Li et al. [165] developed a validated numerical model for an air-cooled double-channel PV/T-PCM system and reported that RT22HC, with a melting range of 17-29 °C, provided the best performance among the investigated paraffins. Under specific model conditions, a 15 mm PCM layer achieved an overall performance of 70.5% [165]. This value represents a case-specific optimum and should not be interpreted as a universal PCM thickness.

Experimental comparisons have also shown that PV/T-PCM systems can outperform corresponding PV/T systems without thermal storage [107,166]. Arici et al. [167] reported a maximum PV-temperature reduction of 10.26 °C and an efficiency improvement of up to 3.73% for the investigated PV-PCM configurations. These results depended on PCM type, layer thickness and local climatic conditions.

The relatively low thermal conductivity of conventional paraffin remains an important limitation. Thermal-conductivity enhancement can be achieved through nanoparticles, metal foams or fins. Nada et al. [168] reported electrical-efficiency improvements of 5.7% for pure PCM and 13.2% for Al_2O_3-enhanced PCM relative to an uncooled reference module [168]. Yousef et al. [169] showed that incorporating aluminium foam into paraffin produced greater seasonal electrical-efficiency improvements than PCM alone. In a numerical PVT/NEPCM study, Khodadadi and Sheikholeslami [170] found that an 18-fin configuration increased thermal performance by 9.47% and PCM liquid fraction by 16.74% relative to the corresponding finless configuration.

Overall, PCM integration can improve short-term PV-temperature regulation and support combined electrical and thermal utilization. However, the reported gains are configuration- and climate-specific. Practical assessment should include PCM solidification during repeated cycles, auxiliary power consumption, leakage prevention, additional mass, material cost and long-term outdoor degradation. Therefore, PCM-based cooling should be regarded as a promising thermal-management option rather than an inherently low-cost or universally optimal solution.

8.2. Applications of PV-TEG-PCM systems for post-sunset power generation

PV-TEG-PCM systems combine photovoltaic conversion, thermoelectric generation and latent-heat storage to improve solar-energy utilization and potentially extend electricity generation beyond daylight hours. However, these systems should not generally be described as dispatchable or continuous power generation. Their post-sunset output depends on the amount of thermal energy stored in the PCM, the temperature difference maintained across the TEG, nighttime sky and ambient conditions, interfacial thermal resistance and cold-side heat rejection.

During daytime operation, the PV cell converts a portion of the incident solar irradiance into electricity, while the remaining absorbed energy contributes to cell heating. A TEG can convert part of the maintained temperature difference into supplementary electrical power. PCM absorbs part of the thermal load during melting and can reduce excessive PV-temperature rise. After sunset, the PCM may release stored heat while radiative heat exchange with the sky lowers the temperature of the opposite side of the TEG. Under suitable conditions, this process can temporarily maintain or reverse the temperature difference and produce measurable nighttime thermoelectric output [120].

Wei et al. [120] designed, manufactured and experimentally tested a PV-TEG-PCM prototype combining daytime photovoltaic/thermal conversion with nighttime sky radiative cooling. The reported PV electrical efficiency reached 19.6%, while the TEG voltage reached up to 0.29 V during daytime and 0.15 V at night. The maximum overall system efficiency reached 20.8%. These results provide direct proof of concept for day-night electricity harvesting, but the values are specific to the tested prototype, a PCM phase-transition temperature of 38 °C, a 30 mm PCM layer and the corresponding outdoor conditions. The relatively low nighttime voltage also indicates that power conditioning or energy storage would be required for most practical electrical loads.

Building-envelope integration represents another potential application pathway. Luo et al. [171] investigated a wall- and roof-integrated PV-PCM-TEG system containing a double-layer PCM plate and conducted a whole-life-cycle optimization of energy saving, cost and carbon reduction. For the optimized configurations, improvements were reported relative to the corresponding unoptimized systems. However, the study was model-based, and its results should not be interpreted as direct experimental evidence of stable nighttime electricity supply.

Consequently, PV-TEG-PCM systems are more appropriately regarded as complementary energy-harvesting and thermal-management technologies. They may improve PV temperature regulation, recover a small proportion of waste heat and extend the useful electricity-generation period under favorable conditions. Future work should prioritize long-term outdoor testing, repeated PCM cycling, net-energy assessment including auxiliary electronics, power-conditioning requirements, climatic adaptability and system-level economic evaluation.

9. Conclusions and future outlook

This review demonstrates that the primary contribution of PV-TEG-PCM systems currently lies in photovoltaic thermal regulation and improved solar-energy utilization rather than in substantial thermoelectric electricity generation. Experimental studies have consistently shown that PCM-based thermal management can effectively reduce PV operating temperature and improve electrical performance. For example, a PV/dual-thermal-PCM-TEG configuration reduced the average PV-cell temperature by 10.8 °C and achieved an electrical efficiency of 12.97% under outdoor conditions [121]. Similarly, PCM and heat-transfer enhancement strategies increased power output by 10.29-14.22%, with the maximum overall efficiency reaching approximately 35.2% for a fin-enhanced configuration [119]. Although PV-TEG-PCM systems have demonstrated the feasibility of extending thermoelectric operation beyond daylight hours, the contribution of TEGs remains strongly restricted by limited temperature differences, thermal contact resistance, and cold-side heat dissipation conditions [120,143,116]. Improved thermal interfaces can significantly enhance heat transfer, with reported increases of at least 14% in PV output and 60% in TEG output through reduced interfacial thermal resistance [143]. Overall, current evidence supports PV-TEG-PCM as a hybrid thermal-management and supplementary energy-harvesting approach.

Future research should focus on resolving the key uncertainties that limit practical deployment of PV-TEG-PCM systems. Long-term PCM cycling stability requires systematic evaluation under realistic operating conditions, including retained latent heat, phase-transition-temperature variation, leakage, encapsulation degradation, corrosion, and changes in interfacial thermal resistance, as existing studies are mainly based on laboratory tests, short-term experiments, or numerical predictions [76,80,112,120,121,119,118,116]. Further improvements in TEG integration should consider the limited temperature differences available in unconcentrated PV systems and optimize thermal interfaces, cold-side heat rejection, and load matching rather than relying solely on additional TEG installation [120,143,116]. Moreover, techno-economic and lifecycle assessments should adopt consistent system boundaries and include the costs and environmental impacts associated with TEG modules, PCM materials, encapsulation structures, heat-transfer components, maintenance, replacement, and end-of-life treatment. Existing assessments indicate that economic feasibility remains highly dependent on climatic conditions, TEG cost reduction, and system assumptions; for example, PV-TEG systems in high-irradiance regions achieved an estimated LCOE of approximately 0.071 USD kWh^{-1} but were still less

competitive than conventional PV under current TEG conditions [153]. Future validation should therefore prioritize long-duration outdoor experiments reporting cumulative daytime and nighttime energy production, seasonal performance variation, useful operating duration, and net energy output after auxiliary consumption. Such standardized evaluations are essential to determine whether PV-TEG-PCM systems can provide meaningful advantages beyond conventional PV and PV-PCM configurations.

CRediT authorship contribution statement

Ning Yang: Writing – original draft. **Zishuo Zhao:** Writing – original draft. **Ze Wang:** Formal analysis. **Shipeng Yang:** Formal analysis. **Bowen Wang:** Investigation. **Zhiming Bao:** Formal analysis. **Yuhang Zhang:** Data curation. **Yaoxuan Shang:** Validation. **Binjian Nie:** Writing – review & editing. **Hongwei Li:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization. **Lei Xing:** Writing – review & editing, Validation, Supervision, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by the Science and Technology Development Plan Project of Jilin Province (Grant No. 20250205047GH), the National Natural Science Foundation of China (Grant No. 52006029), the Promotion Foundation for Young Science and Technology Talents in Jilin Province (Grant No. QT202113), the Special Foundation of Industrial Innovation in Jilin Province (Grant No. 2019C056-2) and the Special Foundation for Outstanding Young Talents Training in Jilin (Grant No. 20200104107).

Data availability

Data will be made available on request.

References

[1] Ali AO, Elgohr AT, El-Mahdy MH, et al. Advancements in photovoltaic technology: a comprehensive review of recent advances and future prospects. Energy Convers Manag X 2025;26:100952.
[2] Wei Z, Lu W, Li Z, Feng M, Yan B, Hu J-S, Xue D-J. Low-cost and high-performance selenium indoor photovoltaics. J Mater Chem A 2023;11:23837–43.
[3] Ghadikolaei SSC. Solar photovoltaic cells performance improvement by cooling technology: an overall review. Int J Hydrogen Energy 2021;46:10939–72.
[4] Sharma R, Sharma A, Agarwal S, Dhaka MS. Stability and efficiency issues, solutions and advancements in perovskite solar cells: a review,. Sol Energy 2023;244:516–35.
[5] Roy P, Sinha NK, Tiwari S, Khare A. A review on perovskite solar cells: evolution of architecture, fabrication techniques, commercialization issues and status. Sol Energy 2020;198:665–88.
[6] Firoozi AA, Firoozi AA, Maghami MR. Harnessing photovoltaic innovation: advancements, challenges, and strategic pathways for sustainable global development. Energy Convers Manag X 2025;27:101058.
[7] Bermudez-Garcia A, Voarino P, Raccurt O. Environments, needs and opportunities for future space photovoltaic power generation: a review,. Appl Energy 2021;290:116728.
[8] Fritzsche H. Amorphous silicon and related materials: in 2 parts. Singapore: World Scientific Publishing Company; 1989.
[9] O'Regan B, Grätzel M. A low-cost, high-efficiency solar cell based on dye-sensitized colloidal TiO2 films,. Nature 1991;353:737–40.
[10] Narayanan S, Wohlgemuth JH, Bennett K, et al. Large-area multicrystalline silicon solar cells in high-volume production. Sol Energy Mater Sol Cell 2002;74:125–34.
[11] Zhang CF, Zhang JC, Ma XH, et al. CdTe solar cells. In: Zhang CF, Zhang JC, Ma XH, et al., editors. Semiconductor photovoltaic cells. Singapore: Springer Singapore; 2021. p. 293–324.
[12] Kim HS, Lee CR, Im JH, et al. Lead iodide perovskite sensitized all-solid-state submicron thin-film mesoscopic solar cell with efficiency exceeding 9%. Sci Rep 2012;2:591.
[13] Werner J, Barraud L, Walter A, et al. Efficient near-infrared-transparent perovskite solar cells enabling direct comparison of 4-terminal and monolithic perovskite/silicon tandem cells. ACS Energy Lett 2016;1:474–80.
[14] Schindler F, Michl B, Krenckel P, et al. How to achieve efficiencies exceeding 22% with multicrystalline n-type silicon solar cells. Energy Proc 2017;124:777–80.
[15] Lan D, Green MA. Pathways towards a 50% efficiency spectrum-splitting photovoltaic system: application of built-in filters and generalization of concept. Energy Proc 2018;150:83–6.
[16] Liu D, Zheng H, Ji L, et al. Improved crystallinity of perovskite via molecularly tailored surface modification of SnO_2. J Power Sources 2019;441:227161.
[17] Al-Ashouri A, Köhnen E, Li B, et al. Monolithic perovskite/silicon tandem solar cell with more than 29% efficiency by enhanced hole extraction. Science 2020;370:1300–9.
[18] Wang G, Zheng J, Duan W, et al. Molecular engineering of hole-selective layer for high-band-gap perovskites for highly efficient and stable perovskite-silicon tandem solar cells. Joule 2023;7:2583–94.
[19] Kirk AP. Comment on CdTe solar cell efficiency. Journal of Alloys and Metallurgical Systems 2024;5:100050.
[20] Wang X, Zheng J, Ying Z, et al. Ultrathin approximately 30 μm flexible monolithic perovskite/silicon tandem solar cell. Sci Bull 2024;69:1887–94.
[21] Artuk K, Oranskaia A, Turkay D, et al. 60 cm^2 perovskite-silicon tandem solar cells with an efficiency of 28.9% by homogeneous passivation. Nat Commun 2025;16:8672.
[22] Zhang C, Chen C, Ran C. Research progress and future prospects of perovskite tandem photovoltaics. Sustainable Chemistry for Energy Materials 2026;3:100032.
[23] Altinkök S, Altinkök A. Overheating issues in photovoltaic systems and mitigation strategies: a comprehensive review,. Energy 2026;12:100637.
[24] Ma T, Yang H, Zhang Y, et al. Using phase change materials in photovoltaic systems for thermal regulation and electrical efficiency improvement: a review and outlook,. Renew Sustain Energy Rev 2015;43:1273–84.
[25] Madurai Elavarasan R, Pugazhendhi R, Shafiq S, et al. Efficiency enhancement of PV panels with passive thermal management using PCM: an exhaustive review on materials, designs and effective techniques. Appl Energy 2025;382:125217.
[26] Siecker J, Kusakana K, Numbi BP. A review of solar photovoltaic systems cooling technologies,. Renew Sustain Energy Rev 2017;79:192–203.
[27] Al-Waeli AH, Sopian K, Kazem HA, et al. Photovoltaic/Thermal systems: status and future prospects. Renew Sustain Energy Rev 2017;77:109–30.
[28] Yu Q, Chen X, Yang H. Research progress on utilization of phase change materials in photovoltaic/thermal systems: a critical review,. Renew Sustain Energy Rev 2021;149:111313.
[29] Van Sark W. Feasibility of photovoltaic-thermoelectric hybrid modules. Appl Energy 2011;88:2785–90.
[30] Utomo B, Darkwa J, Du D, et al. Solar photovoltaic cooling and power enhancement systems: a review,. Renew Sustain Energy Rev 2025;216:115644.
[31] Paudyal BR, Imenes AG. Investigation of temperature coefficients of PV modules through field measured data. Sol Energy 2021;224:425–39.
[32] Peng Q, Li D, Sun X, Li J, Zhou Y, Zeng L. Experimental and numerical investigation on photovoltaic/thermal characteristics of an integrated PV/T-PCM system with metal fin and heat exchange pipe. Renew Energy 2025;239:122113.
[33] Scheuermann U, Schmidt R, Newman P. Power cycling testing with different load pulse durations, in: 7Th IET international conference on power electronics. Machines and Drives (PEMD 2014), IET 2014:1–6.
[34] Huang H, Mawby PA. A lifetime estimation technique for voltage source inverters,. IEEE Trans Power Electron 2012;28:4113–9.
[35] Teo H, Lee P, Hawlader M. An active cooling system for photovoltaic modules. Appl Energy 2012;90:309–15.
[36] Bahaidarah H, Subhan A, Gandhidasan P, Rehman S. Performance evaluation of a PV (Photovoltaic) module by back surface water cooling for hot climatic conditions. Energy 2013;59:445–53.
[37] Yin H, Yang D, Kelly G, Garant J. Design and performance of a novel building integrated PV/thermal system for energy efficiency of buildings. Sol Energy 2013;87:184–95.
[38] Robles-Ocampo B, Ruíz-Vasquez E, Canseco-Sánchez H, Cornejo-Meza R, Trápaga-Martínez G, García-Rodriguez F, et al. Photovoltaic/Thermal solar hybrid system with bifacial PV module and transparent plane collector. Sol Energy Mater Sol Cell 2007;91:1966–71.
[39] Al-Mahmodi AF, Munusamy Y, Atta MR, Muniyadi M, Leong SS. Optimization of synthesis parameters for polyamide 610: strategic tailoring for superior latent heat performance. Surf Interfaces 2024;51:104692.
[40] Royo P, Acevedo L, Ferreira VJ, García-Armingol T, López-Sabirón AM, Ferreira G. High-temperature PCM-based thermal energy storage for industrial furnaces installed in energy-intensive industries. Energy 2019;173:1030–40.
[41] Al-Mahmodi AF, Munusamy Y. A comparative analysis of nylon 610 production: thermal efficiency and environmental considerations,. Mater Lett 2025;398:138904.
[42] Sulaiman NS, Mohamad Amini MH. Review on the phase change materials in wood for thermal regulative wood-based products. Forests 2022;13:1622.
[43] Liu Q, Zhang J, Liu J, Sun W, Xu H, Liu C. Self-healed inorganic phase change materials for thermal energy harvesting and management. Appl Therm Eng 2023;219:119423.

[44] Zeng C, Yuan Y, Cao H, et al. Stability and durability of microencapsulated phase change materials (MePCMs) in building applications: a state of the review,. J Energy Storage 2024;80:110249.
[45] Rashid MK, Abas A, Awad AT. Macro-encapsulated phase change materials for latent heat storage: design innovations, performance enhancement, and application insights. J Energy Storage 2026;160:121573.
[46] Mjbel HMM, Hamzah HK, Alkhazaleh R, et al. A roadmap for high-performance phase change materials in thermal energy applications,. Int Commun Heat Mass Tran 2025;169:109921.
[47] Gao Y, Zhao Y, Wang X, et al. Review of supercooling suppression of phase change materials based on nanoparticles. Thermochim Acta 2025;745:179936.
[48] Yu K, Liu Y, Yang Y. Review on form-stable inorganic hydrated salt phase change materials: preparation, characterization and effect on the thermophysical properties. Appl Energy 2021;292:116845.
[49] Jang H, Han Y, Jeong J, et al. Fire-retardant phase-change material composites: a review of organic and inorganic systems for thermal safety, composites part B: Engineering 2026;314:113448.
[50] Elbir A. Dynamic phase change materials for sustainable energy storage: long-term performance trends and enhancement strategies. J Build Eng 2025;113:114030.
[51] Jayabal R, Sekar S, Leo GML, et al. Advanced phase change material-thermal energy storage for low-carbon heat: materials, reliability, and AI-driven system integration for industry, buildings, and data centers. Appl Energy 2026;409:127479.
[52] Kalidasan B, Pandey A, Saidur R, Tyagi V. Energizing organic phase change materials using silver nanoparticles for thermal energy storage. J Energy Storage 2023;58:106361.
[53] Man X, Lu H, Xu Q, Wang C, Ling Z. Review on the thermal property enhancement of inorganic salt hydrate phase change materials. J Energy Storage 2023;72:108699.
[54] Junaid MF, Ur Rehman Z, Čekon M, Čurpek J, Farooq R, Cui H, et al. Inorganic phase change materials in thermal energy storage: a review on perspectives and technological advances in building applications,. Energy Build 2021;252:111443.
[55] Abdellatif HE, Belaadi A, Arshad A, Chai BX, Ghernaout D. Enhancing thermal energy storage system efficiency: geometric analysis of phase change material integrated wedge-shaped heat exchangers. Appl Therm Eng 2025;262:125268.
[56] Sun M, Liu T, Sha H, Li M, Liu T, Wang X, et al. A review on thermal energy storage with eutectic phase change materials: fundamentals and applications,. J Energy Storage 2023;68:107713.
[57] Chen C, Wang X, Ma F, Wang Y, Jiu S, Chen Y. Preparation and characterization of modified activated carbon-based shape stabilized eutectic phase change materials for gypsum composites application. Constr Build Mater 2023;369:130551.
[58] Kufel A, Kuciel S. Composites based on polypropylene modified with natural fillers to increase stiffness. Technical Transactions 2020;116:187–95.
[59] Lin X, Zhang X, Liu L, Liang J, Liu W. Polymer/Expanded graphite-based flexible phase change material with high thermal conductivity for battery thermal management. J Clean Prod 2022;331:130014.
[60] Jing D, Xing M, Zhang Z, Wang R. Solidification/Melting enhancement in ice thermal energy storage by synergistic effect of metal foam and carbon nanotube under magnetic field. Appl Therm Eng 2024;236:121472.
[61] Wu X, Wang C, Wang Y, Zhu Y. Experimental study of thermo-physical properties and application of paraffin-carbon nanotubes composite phase change materials. Int J Heat Mass Tran 2019;140:671–7.
[62] Mahdi JM, Mohammed HI, Hashim ET, Talebizadehsardari P, Nsofor EC. Solidification enhancement with multiple PCMs, cascaded metal foam and nanoparticles in the shell-and-tube energy storage system. Appl Energy 2020;257:113993.
[63] Esapour M, Hamzehnezhad A, Darzi AAR, Jourabian M. Melting and solidification of PCM embedded in porous metal foam in horizontal multi-tube heat storage system. Energy Convers Manag 2018;171:398–410.
[64] Ramakrishnan S, Wang X, Sanjayan J. Effects of various carbon additives on the thermal storage performance of form-stable PCM integrated cementitious composites. Appl Therm Eng 2019;148:491–501.
[65] Ramakrishnan S, Wang X, Sanjayan J, Petinakis E, Wilson J. Development of thermal energy storage cementitious composites (TESC) containing a novel paraffin/hydrophobic expanded perlite composite phase change material. Sol Energy 2017;158:626–35.
[66] Wang Z, Liu S, Ma G, Xie S, Du G, Sun J, et al. Preparation and properties of caprylic-nonanoic acid mixture/expanded graphite composite as phase change material for thermal energy storage. Int J Energy Res 2017;41:2555–64.
[67] He Z, Ma H, Lu S. Preparation and thermal performance study of a novel hydrated salt composite PCM for space heating. J Energy Storage 2024;90:111906.
[68] Fan LW, Fang X, Wang X, Zeng Y, Xiao YQ, Yu ZT, et al. Effects of various carbon nanofillers on the thermal conductivity and energy storage properties of paraffin-based nanocomposite phase change materials. Appl Energy 2013;110:163–72.
[69] Li Y, Wang S, Liu H, et al. Preparation and thermal properties of paraffin/multi-walled carbon nanotubes composite phase change materials. Sol Energy Mater Sol Cell 2014;126:51–7.
[70] Williams JD, Peterson G. A review of thermal property enhancements of low-temperature nano-enhanced phase change materials,. Nanomaterials 2021;11:2578.
[71] Aiswarya V, Das S. Magnetized graphene oxide-modified microencapsulated phase change material for enhanced heat transfer performance with reduced leakage. Therm Sci Eng Prog 2023;41:101807.
[72] Ghufran M, Huitink D. Advances in encapsulated phase change materials for integration in thermal management applications. Emerg Mater 2025;8:5355–86.
[73] Huo X, Li W, Wang Y, Han N, Wang J, Wang N, et al. Chitosan composite microencapsulated comb-like polymeric phase change material via coacervation microencapsulation. Carbohydr Polym 2018;200:602–10.
[74] Jamekhorshid M, Sadrameli SM, Farid M. A review of microencapsulation methods of phase change materials as a thermal energy storage medium,. Renew Sustain Energy Rev 2014;31:531–42.
[75] Kumar GN, Al-Aifan B, Parameshwaran R, Ram VV. Facile synthesis of microencapsulated 1-dodecanol/melamine-formaldehyde phase change material using in-situ polymerization for thermal energy storage. Colloids Surf A Physicochem Eng Asp 2021;610:125698.
[76] Al-mahmodi AF, Munusamy Y, Atta MR, Mutaafi AA, Sethupathi S. Advancements in polymer-based phase change materials for thermal energy storage: a systematic review. Energy Storage Sav 2026:1–10.
[77] Methaapanon R, Kornbongkotmas S, Ataboonwongse C, Soottitantawat A. Microencapsulation of n-octadecane and methyl palmitate phase change materials in silica by spray drying process. Powder Technol 2020;361:910–6.
[78] Methaapanon R, Perngyai T, Ataboonwongse C, Tang SY, Assabumrungrat S, Soottitantawat A. The production of microencapsulated methyl palmitate phase change materials via different scales of spray drying process,. Adv Powder Technol 2025;36:104853.
[79] Pasarkar NP, Yadav M, Mahanwar PA. A review on the micro-encapsulation of phase change materials: classification, study of synthesis technique and their applications,. J Polym Res 2023;30:13.
[80] Dixit P, Reddy VJ, Parvate S, Balwani A, Singh J, Maiti TK, et al. Salt hydrate phase change materials: current state of art and the road ahead. J Energy Storage 2022;51:104360.
[81] Pomianowski M, Heiselberg P, Zhang Y. Review of thermal energy storage technologies based on PCM application in buildings. Energy Build 2013;67:56–69.
[82] Zhu L, Zhong Z, Wang C, Song J, Guo J, Zheng W. Thermo-economic comparison of integrating compressed air energy storage and molten salt thermal energy storage in a combined heat and power plant. Appl Therm Eng 2025;260:124931.
[83] Cabeza LF, Castell A, Barreneche C, et al. Materials used as PCM in thermal energy storage in buildings: a review,. Renew Sustain Energy Rev 2011;15:1675–95.
[84] Saeed RM, Frick KL, Shigrekar A, Mikkelson D, Bragg-Sitton S. Mapping thermal energy storage technologies with advanced nuclear reactors. Energy Convers Manag 2022;267:115872.
[85] Liu M, Saman W, Bruno F. Review on storage materials and thermal performance enhancement techniques for high-temperature phase change thermal storage systems. Renew Sustain Energy Rev 2012;16:2118–32.
[86] Liu L, Liu W, Yao J, Jia T, Zhao Y, Dai Y. Life cycle energy, economic, and environmental analysis for the direct-expansion photovoltaic-thermal heat pump system in China. J Clean Prod 2024;434:139730.
[87] Browne MC, Norton B, McCormack SJ. Phase change materials for photovoltaic thermal management. Renew Sustain Energy Rev 2015;47:762–82.
[88] Bian CL, Song JZ, Jin X, et al. Thermal management in photovoltaic-thermal systems: advances, challenges, and cross-domain applications. Appl Therm Eng 2025;279:127791.
[89] Sommerfeldt N, Madani H. In-depth techno-economic analysis of PV/Thermal plus ground source heat pump systems for multi-family houses in a heating dominated climate. Sol Energy 2019;190:44–62.
[90] Ren H, Quan Z, Wang Z, Wang L, Jing H, Zhao Y. Performance simulation and analysis of a multi-energy complementary energy supply system for a novel BIPVT nearly zero energy building. Energy Convers Manag 2023;282:116879.
[91] Gallero FJG, Siles GG, Maestre IR, Blazquez JLF, Bottarelli M. Performance analysis of a novel façade-based building integrated photovoltaic-thermal system with phase change material for domestic hot water. Appl Therm Eng 2024;240:122321.
[92] Shahsavar A, Azimi N. Performance evaluation and multi-objective optimization of a hybrid earth-air heat exchanger and building-integrated photovoltaic/thermal system with phase change material and exhaust air heat recovery. J Build Eng 2024;90:109531.
[93] Bouadila S, Baddadi M, Ben Nasrallah S. Performance analysis of a solar heating and cooling system combining a photovoltaic/thermal collector, latent heat storage and an earth-air heat exchanger. Renew Energy 2021;168:1320–33.
[94] Ito R, Lee S. Development of adjustable solar photovoltaic system for integration with solar shading louvers on building façades. Appl Energy 2024;359:122711.
[95] Javadijam R, Dehbashi M, Shahverdian MH, Sohani A, Arici M, Sayyaadi H. Artificial intelligent based techno-economic-exergetic optimization of a thermoelectric enhanced building integrated photovoltaic thermal system. J Build Eng 2024;84:108526.
[96] Singh AP, Tiwari S, Sinhmar H. Modelling and analysis of photovoltaic-thermal-thermoelectric-cooler air collector integrated mixed-mode greenhouse dryer with heat storage material. J Energy Storage 2024;95:112369.
[97] Singh AP, Tiwari S, Sinhmar H. A novel photovoltaic thermal and thermoelectric converter air collector integrated with solar dryer having thermal energy storage-An experimental approach,. J Energy Storage 2025;108:115115.
[98] Yüksel C, Öztürk M, Çiftçi E. Analysis of a novel V-grooved double pass photovoltaic thermal solar dryer including thermal energy storage. Appl Therm Eng 2024;236:121697.
[99] Kilkiş B. Development of a composite PVT panel with PCM embodiment, TEG modules, flat-plate solar collector, and thermally pulsing heat pipes. Sol Energy 2020;200:89–107.

[100] Liu J, Wang J, Tang Y, Jin J, Li W. Solar photovoltaic-thermal hydrogen production system based on full-spectrum utilization. J Clean Prod 2023;430:139340.
[101] Fterich M, Chouikhi H, Bentaher H, et al. Experimental analysis of a solar dryer integrated with phase change material for drying of date fruits. Sol Energy 2021;225:877–89.
[102] Mehrpooya M, Hemmatabady H, Ahmadi MH. Techno-economic analysis of a hybrid system consisting of a concentrating photovoltaic, phase change material storage and a solid oxide electrolyzer for hydrogen production. Energy Convers Manag 2021;244:114515.
[103] Kasaeian A, Raziei S, Pourfayaz F, et al. Dynamic modeling and performance analysis of a PV/PCM system coupled with a solid oxide electrolyzer for hydrogen generation. Renew Energy 2023;217:119205.
[104] El Alami Y, Lamkaddem A, Bendaoud R, Talbi S, Baghaz E. Numerical study of a water-based photovoltaic-thermal (PVT) hybrid solar collector with a new heat exchanger, E-Prime-Advances in electrical engineering. Electronics and Energy 2024;9:100693.
[105] Hossain M, Pandey A, Selvaraj J, Abd Rahim N, Islam M, Tyagi V. Two side serpentine flow based photovoltaic-thermal-phase change materials (PVT-PCM) system: energy, exergy and economic analysis. Renew Energy 2019;136:1320–36.
[106] Nada SA, Rady MA, Al Derham A. Performance enhancement of a photovoltaic panel by phase change material for thermal management: an experimental retrofit study. J Energy Storage 2021;33:101879.
[107] Yang X, Sun L, Yuan Y, Zhao X, Cao X. Experimental investigation on performance comparison of PV/T-PCM system and PV/T system. Renew Energy 2018;119:152–9.
[108] Maatallah T, Zachariah R, Al-Amri FG. Exergo-economic analysis of a serpentine flow type water based photovoltaic thermal system with phase change material (PVT-PCM/water). Sol Energy 2019;193:195–204.
[109] Li Z, Ma T, Zhao J, Song A, Cheng Y. Experimental study and performance analysis on solar photovoltaic panel integrated with phase change material. Energy 2019;178:471–86.
[110] Carmona M, Bastos AP, García JD. Experimental evaluation of a hybrid photovoltaic and thermal solar energy collector with integrated phase change material (PVT-PCM) in comparison with a traditional photovoltaic (PV) module. Renew Energy 2021;172:680–96.
[111] Yu Q, Romagnoli A, Yang R, Xie D, Liu C, Ding Y, et al. Numerical study on energy and exergy performances of a microencapsulated phase change material slurry based photovoltaic/thermal module. Energy Convers Manag 2019;183:708–20.
[112] Anand A, Mansor M, Sharma K, et al. A comprehensive review on eutectic phase change materials: development, thermophysical properties, thermal stability, reliability, and applications,. Alex Eng J 2025;112:254–80.
[113] Hasan K, Yousuf SB, Tushar MSHK, Das BK, Das P, Islam MS. Effects of different environmental and operational factors on the PV performance: a comprehensive review,. Energy Sci Eng 2022;10:656–75.
[114] Burlafinger K, Strohm S, Joisten C, Woiton M, Classen A, Hepp J, et al. Accelerated lifetime testing of thin-film solar cells at high irradiances and controlled temperatures. Prog Photovoltaics Res Appl 2022;30:518–27.
[115] Ram JP, Rajasekar N. A novel flower pollination based global maximum power point method for solar maximum power point tracking,. IEEE Trans Power Electron 2016;32:8486–99.
[116] Darkwa J, Calautit J, Du D, Kokogianakis G. A numerical and experimental analysis of an integrated TEG-PCM power enhancement system for photovoltaic cells,. Appl Energy 2019;248:688–701.
[117] Gao Y, Dai Z, Wu D, Wang C, Chen B, Zhang X. Transient performance assessment of a hybrid PV-TEG system integrated with PCM under non-uniform radiation conditions: a numerical investigation. Renew Energy 2022;198:352–66.
[118] Lv S, Yang J, Ren J, Zhang B, Lai Y, Chang Z. Research and numerical analysis on performance optimization of photovoltaic-thermoelectric system incorporated with phase change materials. Energy 2023;263:125850.
[119] Gürbüz H, Demirtürk S, Akçay H. Ü. Topalci, thermal stabilization and energy harvesting in a solar PV/T-PCM-TEG hybrid system: a case study on the design of system components,. Energy Convers Manag 2023;294:117536.
[120] Wei W, Li N, Che L, Fan Y, Liu H, Ji J, et al. A continuous 24-hour power generated PV-TEG-PCM hybrid system enabled by solar diurnal photovoltaic/thermal conversion and nocturnal sky radiative cooling,. Energy Convers Manag 2024;321:119086.
[121] Li T, Yu J, Peng X, Zhou W, Xu C, Li G, et al. Comparison of the thermoelectric performance of different photovoltaic/thermal hybrid thermoelectric generation modules: an experimental study. Appl Energy 2025;378:124771.
[122] Lv S, Liu W, Yang J. Performance optimization of photovoltaic thermoelectric systems based on phase change materials. Renew Energy 2025;240:122203.
[123] Islam MN, Rahman MZ, Mominuzzaman SM. The effect of irradiation on different parameters of monocrystalline photovoltaic solar cell. In: 2014 3rd international conference on the developments in renewable energy technology (ICDRET). IEEE; 2014. p. 1–6.
[124] Kumar R, Singh S. Solar photovoltaic modeling and simulation: as a renewable energy solution. Energy Rep 2018;4:701–12.
[125] Hossain MS, Bashir MBA, Akter K. Design a low-cost solar PV data logger and assess the application and accuracy with off-grid monocrystalline panels. Measurement 258 (Part B) 2026:119176.
[126] Gebreabe ST, Bogale D, Hailu A, Worku A. Advances in solar energy technologies: a comprehensive review of photovoltaic, thermal, and hybrid systems for sustainable energy solutions,. Research 2026;3:101176.
[127] Pabon JJG, Wang J, Chamanehpour E, Salami D, Khosravi A. Dynamic integration of solar-powered hydrogen systems with fuel cells and district heating for green data centers. Int J Hydrogen Energy 2025;196:152557.
[128] Chegaar M, Hamzaoui A, Namoda A, Petit P, Aillerie M, Herguth A. Effect of illumination intensity on solar cells parameters. Energy Proc 2013;36:722–9.
[129] Xiao C, Yu X, Yang D, Que D. Impact of solar irradiance intensity and temperature on the performance of compensated crystalline silicon solar cells. Sol Energy Mater Sol Cell 2014;128:427–34.
[130] Wang Y, Lu B, Han Z. Rapid increase of the nighttime electricity demand in Beijing due to compound heatwaves. Urban Clim 2023;50:101595.
[131] Zhang B, Wakui T. A near-optimal solution method for year-round operational planning of energy supply-storage systems utilizing time-domain decomposition,. Energy 2025;335:137358.
[132] Shi L, Liu Y, Sun C, Guo Y, Wang D, Liu Y, et al. Energy solution for rural household in remote cold regions: an innovative photovoltaic-based thermal energy storage system. Energy Convers Manag 2025;343:120216.
[133] Bie Y, Qin W, Wu J. Optimal electric bus scheduling method under hybrid energy supply mode of photovoltaic-energy storage system-power grid. Appl Energy 2024;372:123774.
[134] Hou M, Chen H, Li S, Zhang X, Chen J, Jiang Z, et al. A tri-mode photothermal, phase-change, and radiative-cooling film for all-day thermoelectric generation,. Adv Mater 2025;37:2505601.
[135] Zhang J, Xuan Y. Performance improvement of a photovoltaic-thermoelectric hybrid system subjecting to fluctuant solar radiation. Renew Energy 2017;113:1551–8.
[136] Benghanem M, Al-Mashraqi AA, Daffallah KO. Performance of solar cells using thermoelectric module in hot sites. Renew Energy 2016;89:51–9.
[137] Harb H, Paprott JN, Matthes P, Schütz T, Streblow R, Müller D. Decentralized scheduling strategy of heating systems for balancing the residual load. Build Environ 2015;86:132–40.
[138] Wang Y, Lin Z, Wang B, Liu C, Cai G, Liu H, et al. Two-stage day-ahead and intraday low-carbon dispatch method based on enhancing the peak-load regulation capability of cogeneration units with a novel multi-stage Tesla valve thermal storage device. Energy 2025;316:134467.
[139] Wang P, Hu B, Vafai K. A data-driven stochastic model predictive control framework for optimal operation of CSP-PV hybrid system under grid coordination,. Energy Convers Manag 2026;348:120653.
[140] Vanderbeck F, Savelsbergh MW. A generic view of Dantzig-Wolfe decomposition in mixed integer programming,. Oper Res Lett 2006;34:296–306.
[141] Jans R. Classification of Dantzig-Wolfe reformulations for binary mixed integer programming problems. Eur J Oper Res 2010;204:251–4.
[142] Wakui T, Hashiguchi M, Yokoyama R. A near-optimal solution method for coordinated operation planning problem of power-and heat-interchange networks using column generation-based decomposition,. Energy 2020;197:117118.
[143] Zhang J, Zhai H, Wu Z, Wang Y, Xie H, Zhang M. Enhanced performance of photovoltaic-thermoelectric coupling devices with thermal interface materials. Energy Rep 2020;6:116–22.
[144] Moshwan R, Shi XL, Zhang M, et al. Advances and challenges in hybrid photovoltaic-thermoelectric systems for renewable energy. Appl Energy 2025;380:125032.
[145] Yang B, Zeng C, Li D, Guo Z, Chen Y, Shu H, Cao P, Li Z. Improved immune genetic algorithm based TEG system reconfiguration under non-uniform temperature distribution. Appl Energy 2022;325:119691.
[146] Yang B, Wu S, Li Q, et al. Jellyfish search algorithm based optimal thermoelectric generation array reconfiguration under non-uniform temperature distribution condition. Renew Energy 2023;204:197–217.
[147] Fathy A. Efficient energy valley optimization approach for reconfiguring thermoelectric generator system under non-uniform heat distribution. Renew Energy 2023;217:119177.
[148] Yousri D, Fathy A, Farag HEZ, El-Saadany EF. Optimal dynamic reconfiguration of thermoelectric generator array using RIME optimizer to maximize the generated power. Appl Therm Eng 2024;238:122174.
[149] Wang C, Zhang G, Wang Y, Song L, Liu B. Comprehensive optimization of photovoltaic-thermoelectric hybrid systems: experimental analysis of cooling methods, photovoltaic strategies, and their synergistic effects. Sol Energy 2025;298:113667.
[150] Müller E, Drašar Č, Schilz J, Kaysser WA. Functionally graded materials for sensor and energy applications. Mater Sci Eng, A 2003;362:17–39.
[151] Cramer CL, Wang H, Ma K. Continuous functionally graded material to improve the thermoelectric power generation performance under transient heat sources. J Eur Ceram Soc 2018;38:5127–35.
[152] Harb AEMA, Elsayed K, Sedrak M, et al. Enhancing the performance of thermoelectric generators using novel segmental arrangement of multifunctional gradient materials. Renew Energy 2024;226:120311.
[153] Montero FJ, Kumar R, Lamba R, Escobar RA, Vashishtha M, Upadhyaya S, Guzmán AM. Hybrid photovoltaic-thermoelectric system: economic feasibility analysis in the Atacama Desert, Chile. Energy 2022;239:122058.
[154] Li W, Kan J, Zhao W, Wang J, Zhang X, Zhao J. Economic and life cycle analysis of a photovoltaic thermal application system based on phase change thermal storage. Appl Therm Eng 2024;257:124261.
[155] Garud KS, Lee MY. Thermodynamic, environmental and economic analyses of photovoltaic/thermal-thermoelectric generator system using single and hybrid particle nanofluids. Energy 2022;255:124515.
[156] Mazzeo D, Colombo LPM, Leva S. A review of photovoltaic cooling with phase change materials: technical advances, modeling approaches, efficiency gains and economic/environmental impact,. Appl Therm Eng 2025;280:128222.

[157] Zhao R, Zhu N, Zhao X, Luo Z, Chang J. Multi-objective optimization of a novel photovoltaic-thermoelectric generator system based on hybrid enhanced algorithm. Energy 2025;319:135046.
[158] Zhang Y, Zhu N, Zhao X, Luo Z, Hu P, Lei F. Energy performance and enviroeconomic analysis of a novel PV-MCHP-TEG system. Energy 2023;274:127342.
[159] Saleh UA, Jumaat SA, Johar MA, Jamaludin WAW. Photovoltaic-thermoelectric generator monitoring system using Arduino-based data acquisition system technique. In: 2021 IEEE international conference on artificial intelligence in engineering and technology (IICAIET). IEEE; 2021. p. 1–6.
[160] Gao Y, Wu D, Dai Z, Wang C, Chen B, Zhang X. Performance analysis of a hybrid photovoltaic-thermoelectric generator system using heat pipe as heat sink for synergistic production of electricity. Energy Convers Manag 2021;249:114830.
[161] Praveenkumar S, Agyekum EB, Kumar A, Velkin VI. Thermo-enviro-economic analysis of solar photovoltaic/thermal system incorporated with u-shaped grid copper pipe, thermal electric generators and nanofluids: an experimental investigation. J Energy Storage 2023;60:106611.
[162] Zhou J, Yi Q, Wang Y, Ye Z. Temperature distribution of photovoltaic module based on finite element simulation. Sol Energy 2015;111:97–103.
[163] Kamkari B, Groulx D. Experimental investigation of melting behaviour of phase change material in finned rectangular enclosures under different inclination angles. Exp Therm Fluid Sci 2018;97:94–108.
[164] Skoplaki E, Palyvos JA. On the temperature dependence of photovoltaic module electrical performance: a review of efficiency/power correlations. Sol Energy 2009;83(5):614–24.
[165] Li M, Lu Z, Kang Z, Ban L, Cong H, Lu Y. Investigation of an air-cooled double-channel photovoltaic/thermal system with integrated thermal energy storage. J Build Eng 2023;77:107539.
[166] Rao VT, Sekhar YR, Pandey AK, Said Z, Prasad DMR, Hossain MS, Selvaraj J. Thermal analysis of hybrid photovoltaic-thermal water collector modified with latent heat thermal energy storage and two side serpentine absorber design. J Energy Storage 2022;56:105968.
[167] Arici M, Bilgin F, Nižetić S, Papadopoulos AM. Phase change material based cooling of photovoltaic panel: a simplified numerical model for the optimization of the phase change material layer and general economic evaluation,. J Clean Prod 2018;189:738–45.
[168] Nada S, El-Nagar D, Hussein H. Improving the thermal regulation and efficiency enhancement of PCM-Integrated PV modules using nano particles. Energy Convers Manag 2018;166:735–43.
[169] Yousef MS, Sharaf M, Huzayyin A. Energy, exergy, economic, and enviroeconomic assessment of a photovoltaic module incorporated with a paraffin-metal foam composite: an experimental study. Energy 2022;238:121807.
[170] Khodadadi M, Sheikholeslami M. Numerical simulation on the efficiency of PVT system integrated with PCM under the influence of using fins. Sol Energy Mater Sol Cell 2021;233:111402.
[171] Luo Z, Zhu N, Zhao X, Chang J. Whole life cycle energy, economy, and environment collaborative optimization on a building envelope integrated with electricity generator system. Renew Energy 2025;255:123832.